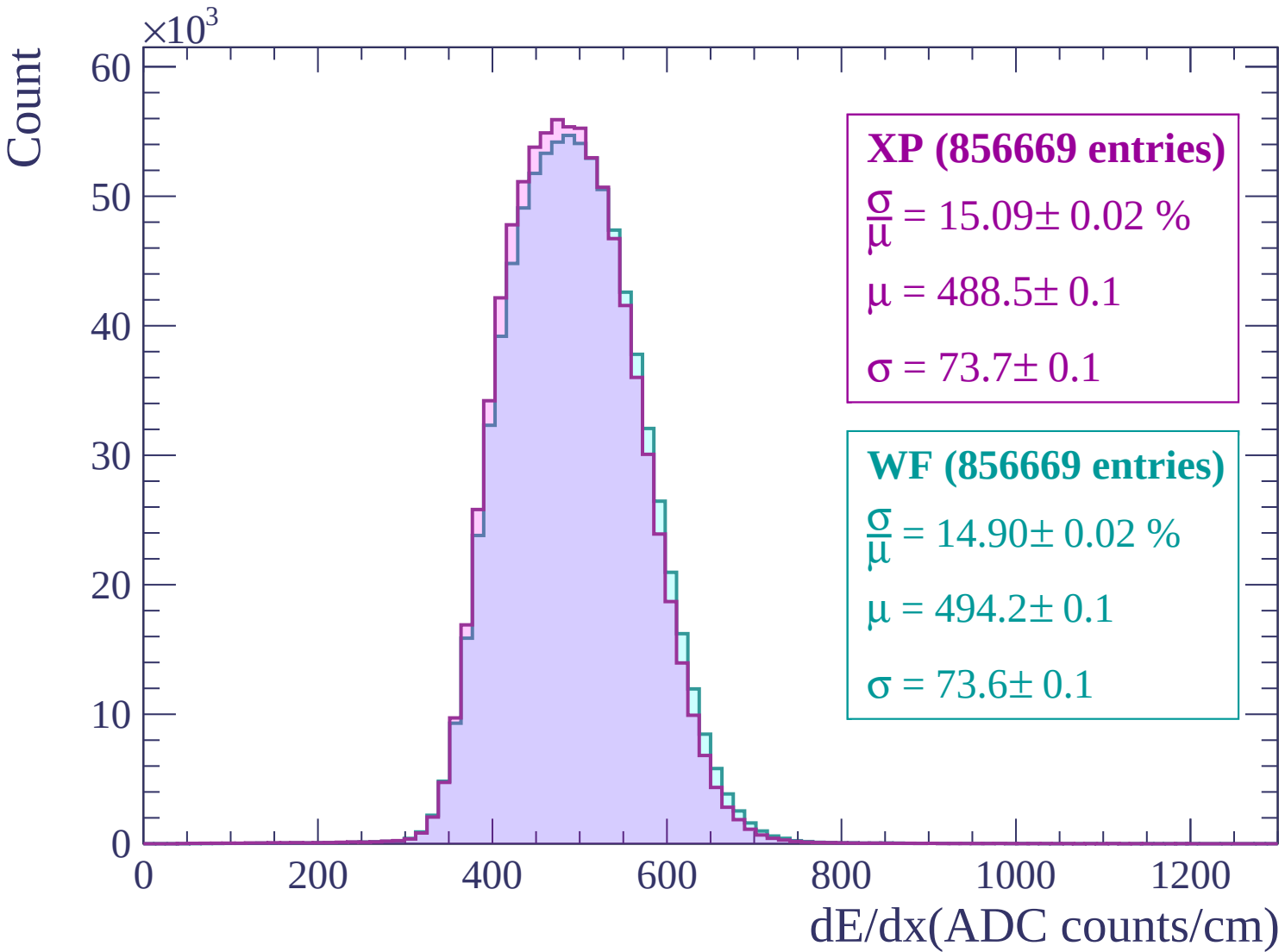
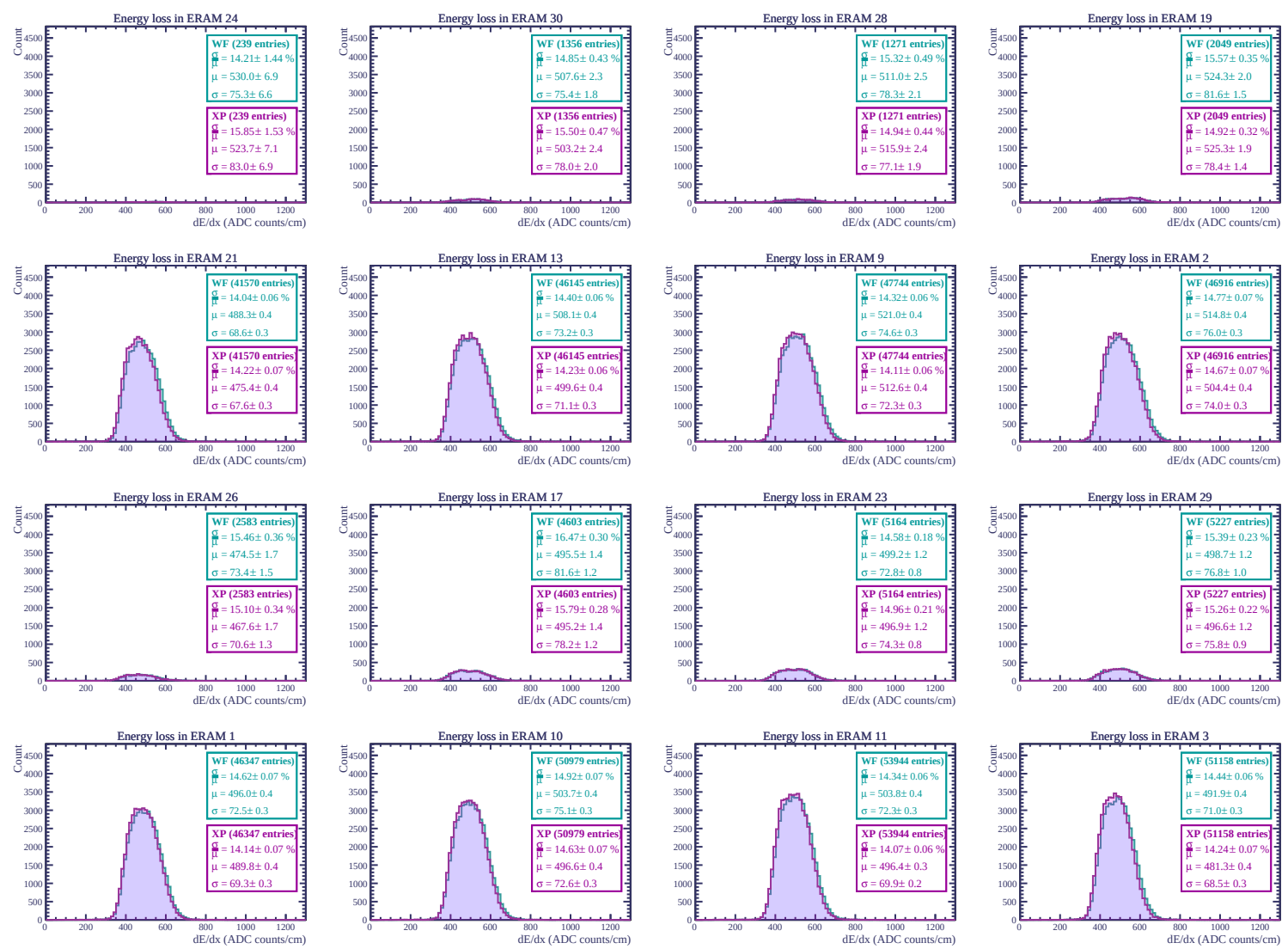
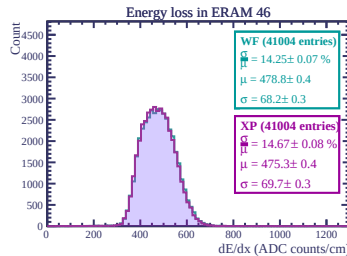
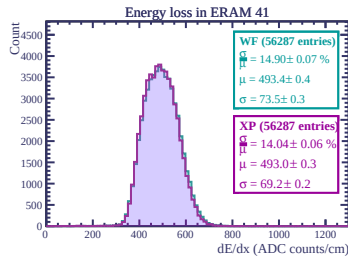
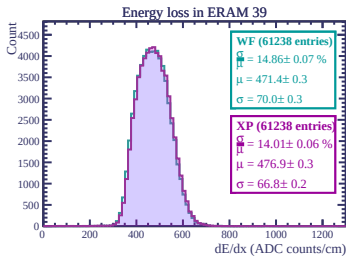
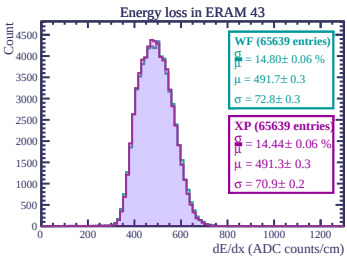
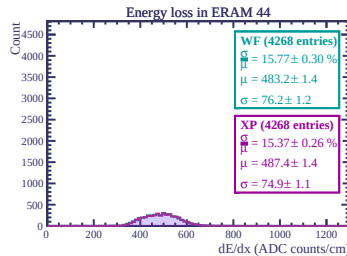
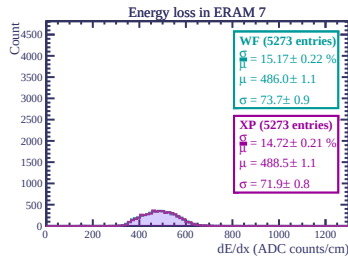
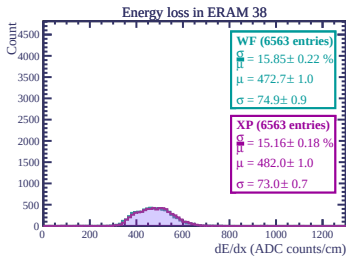
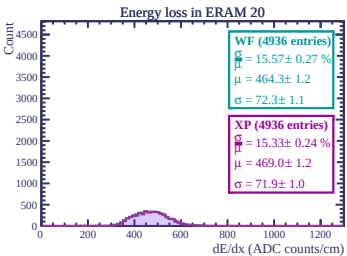
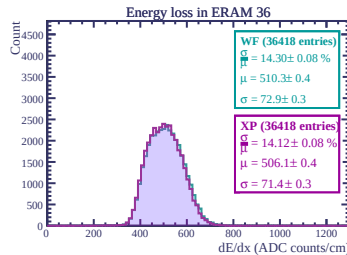
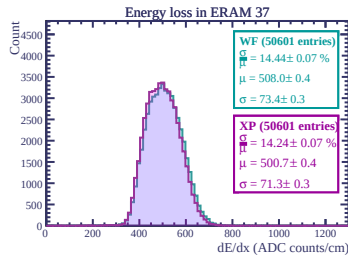
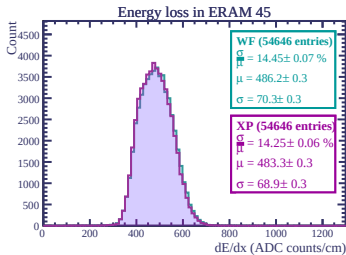
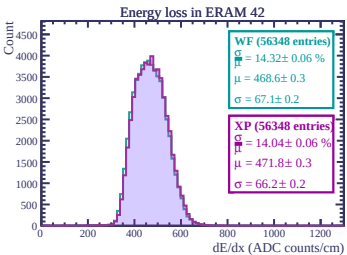
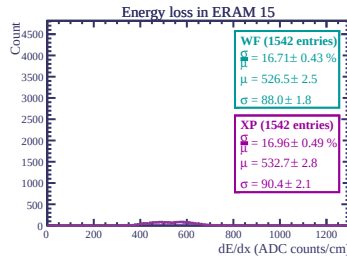
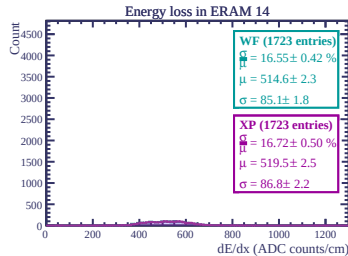
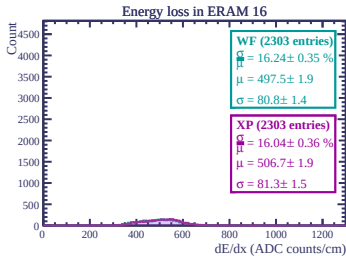
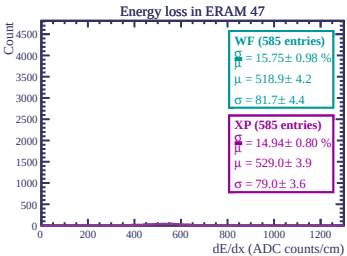
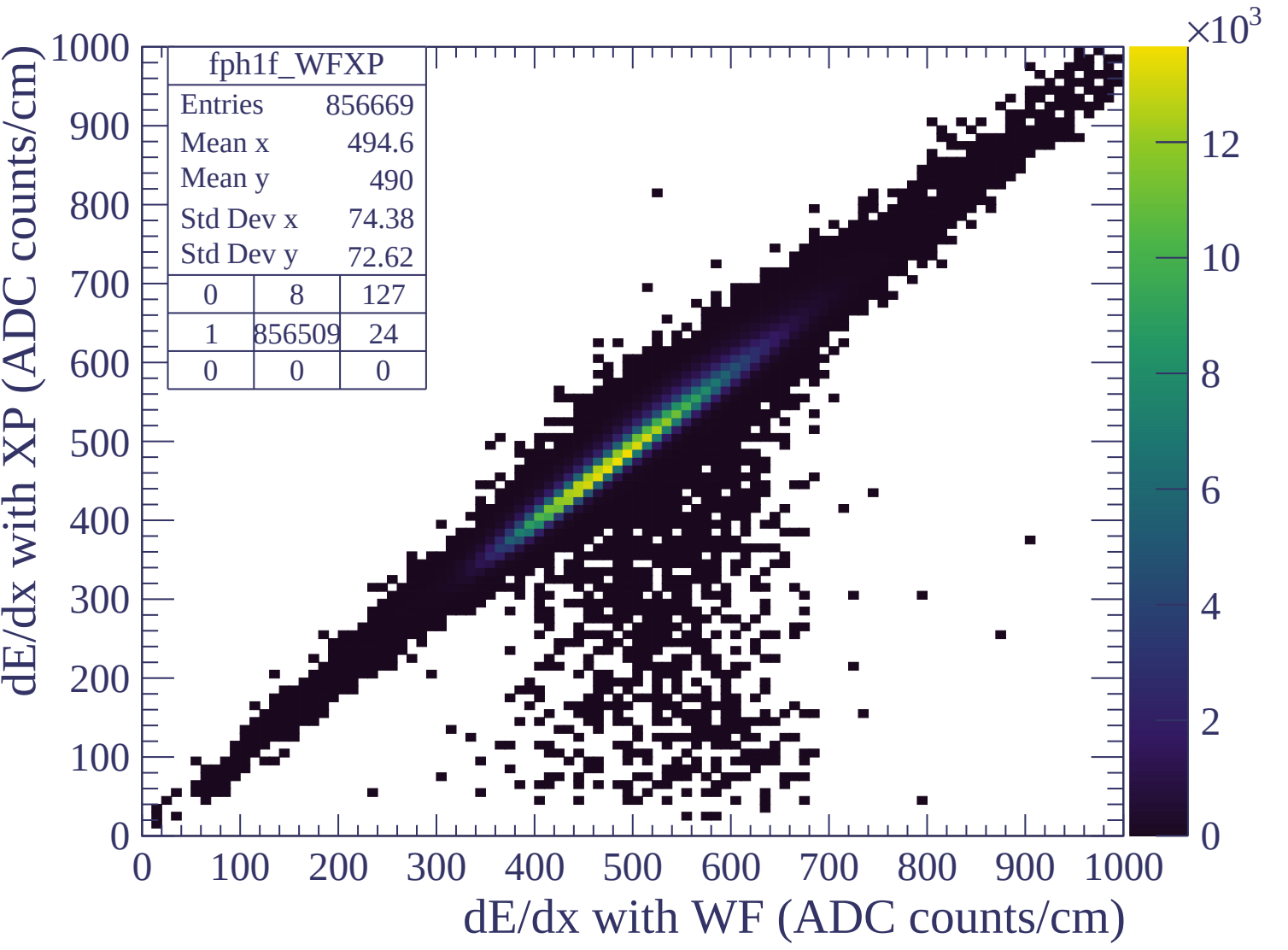
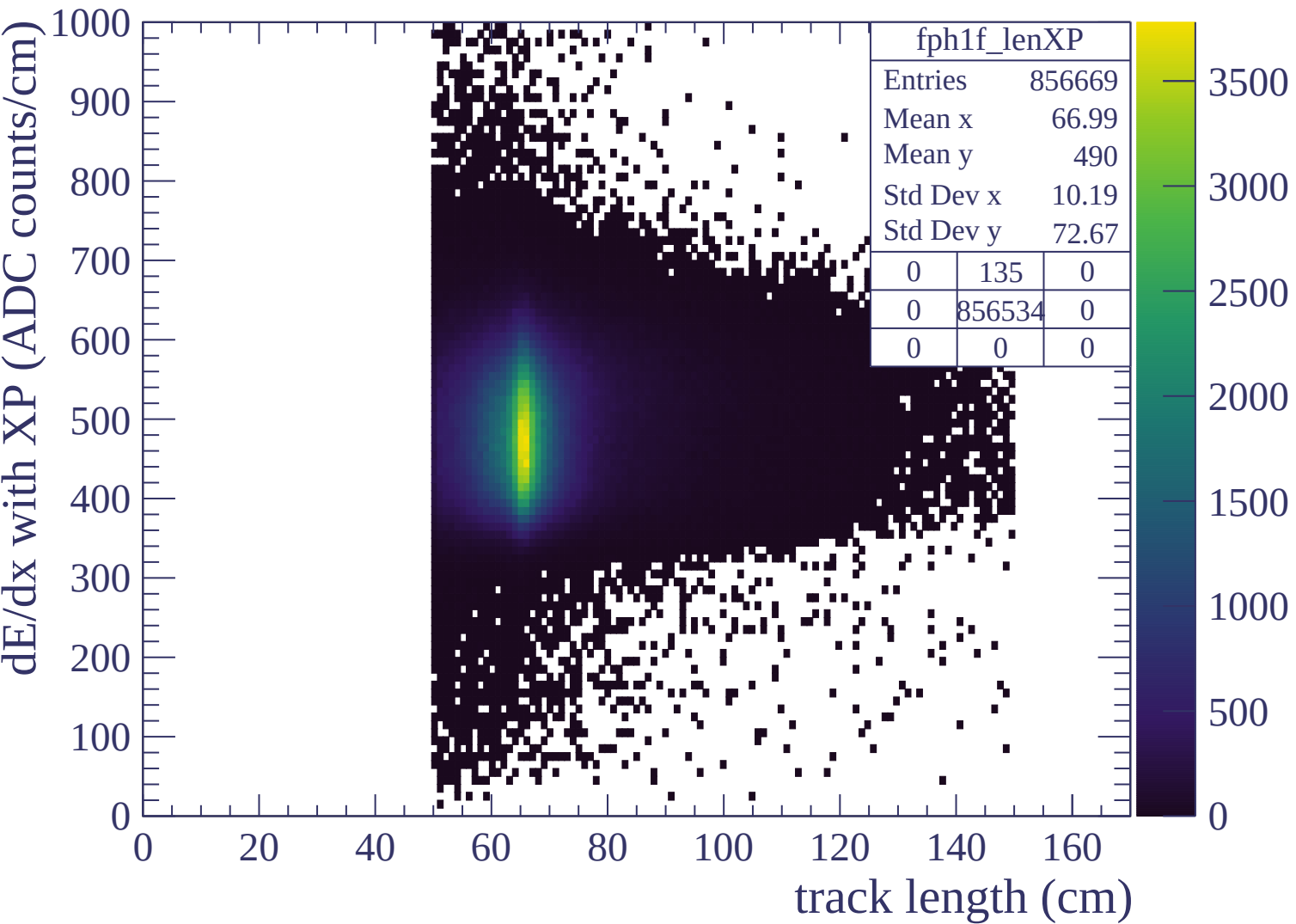


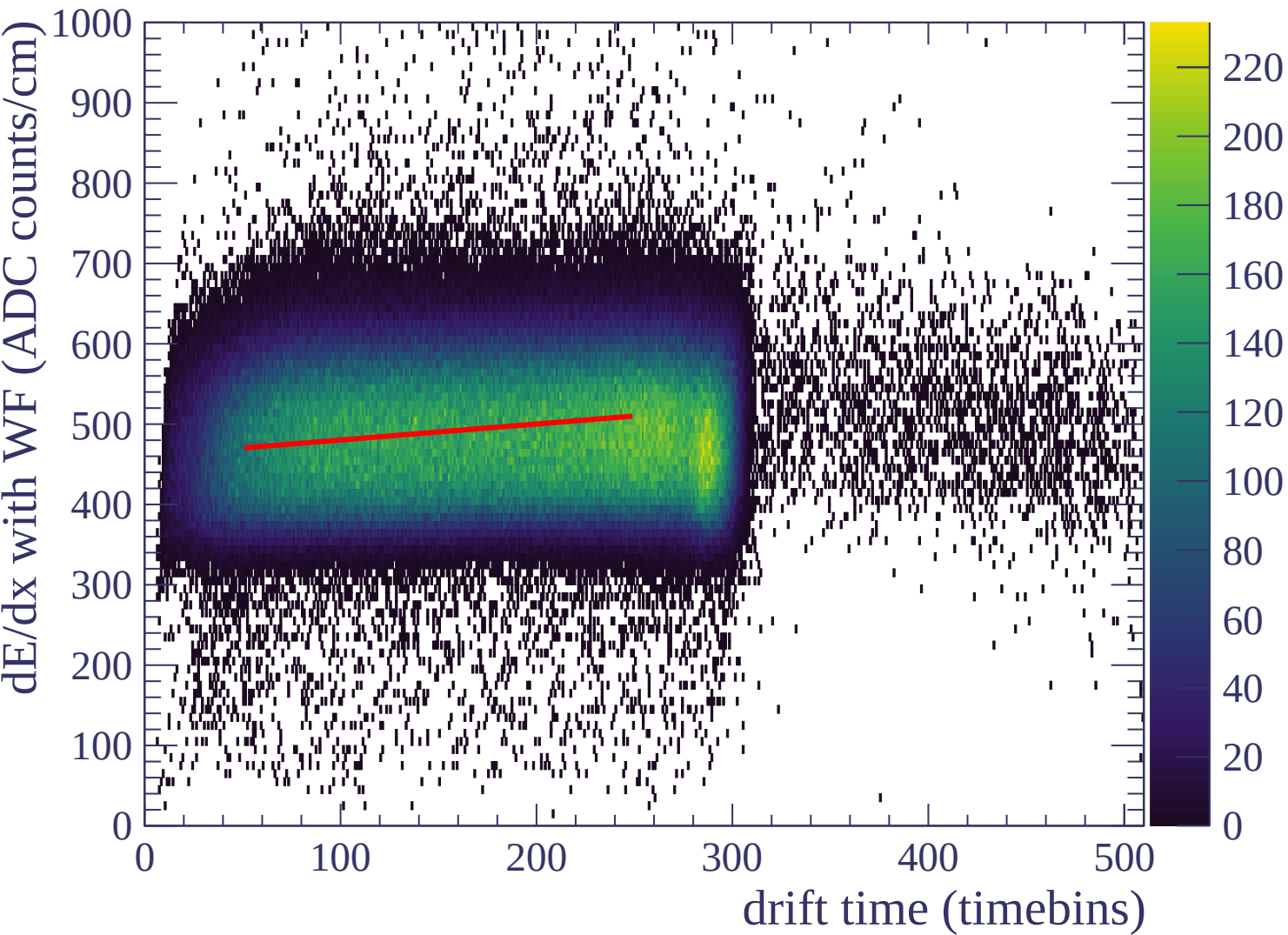


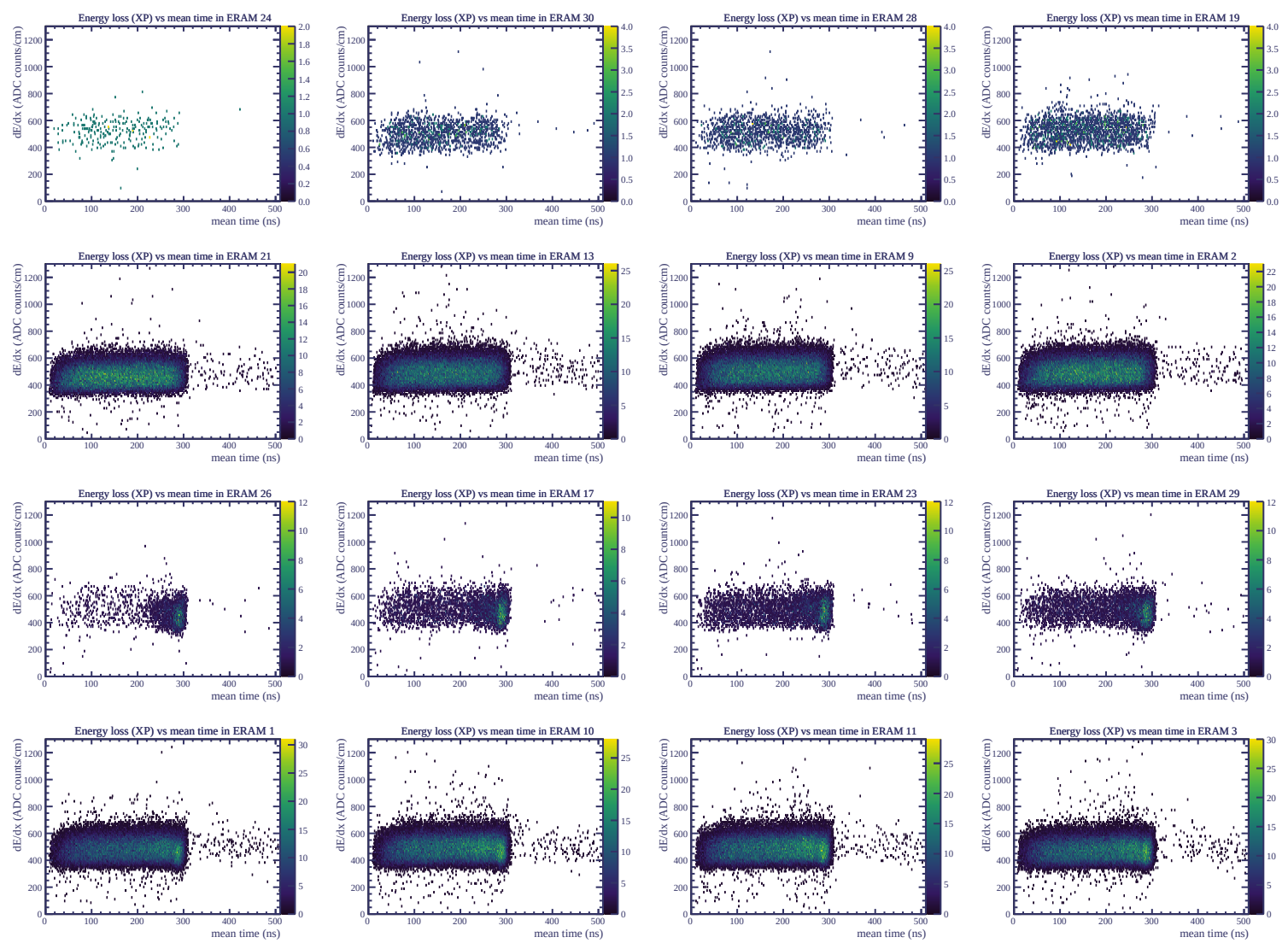


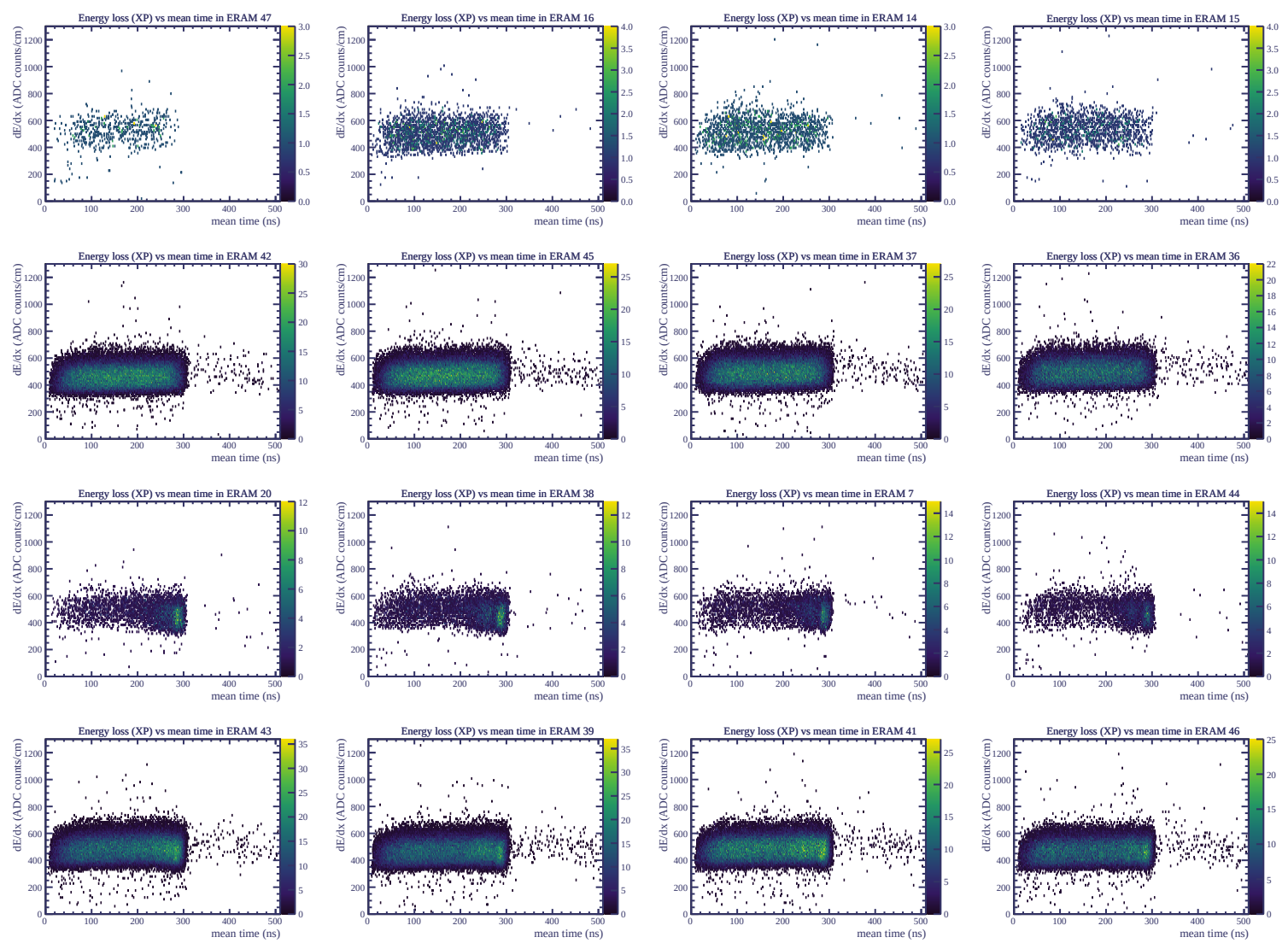




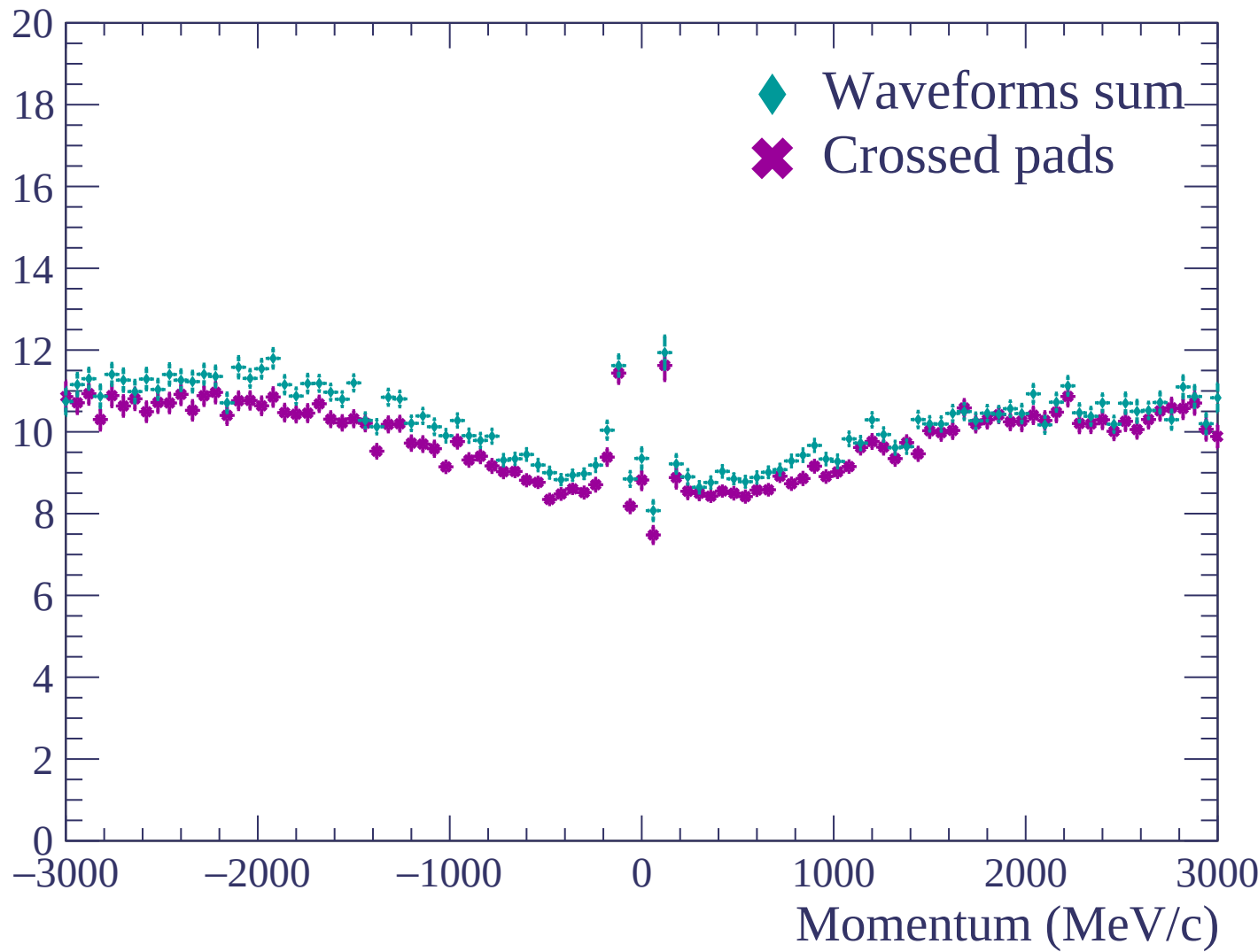


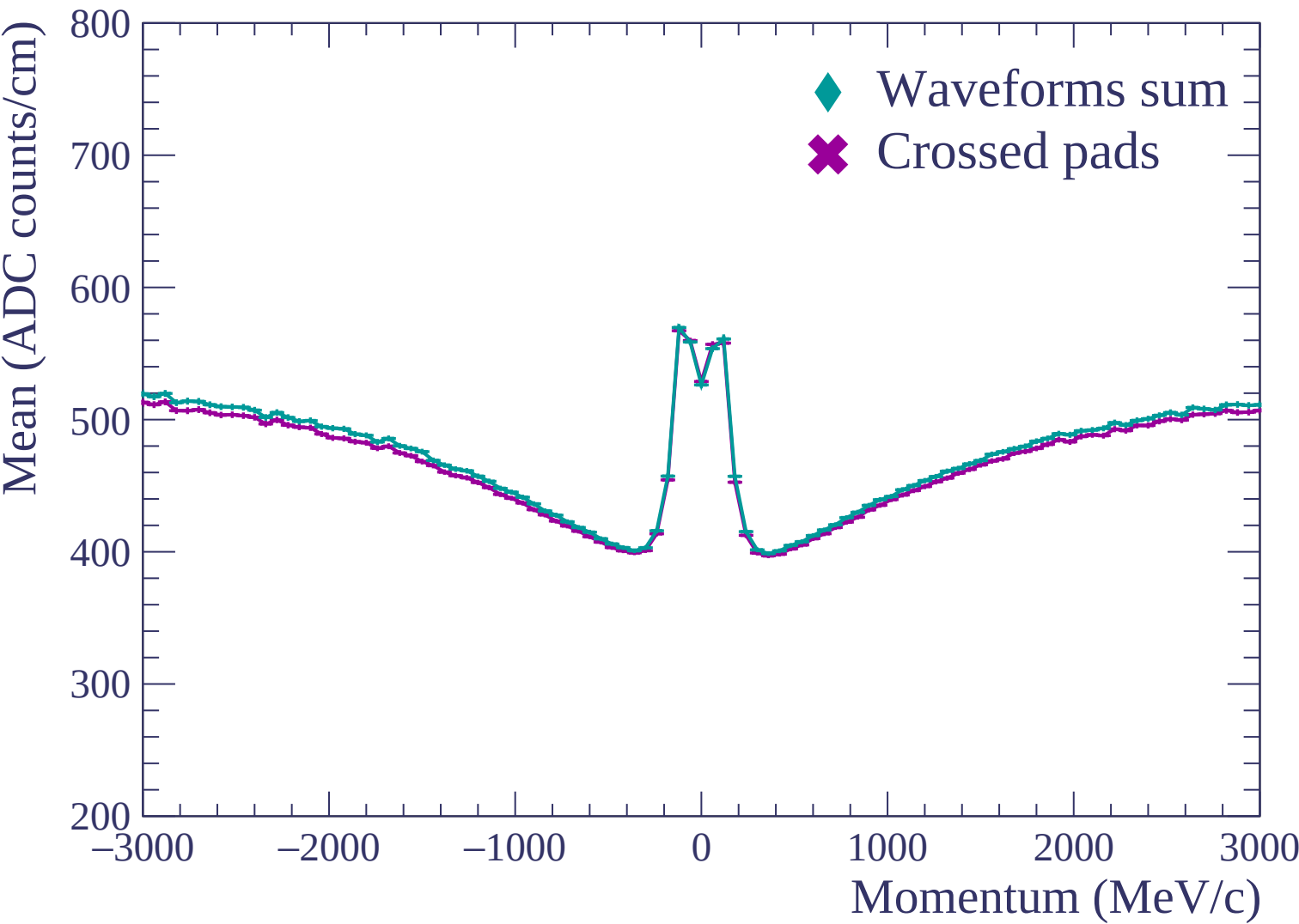


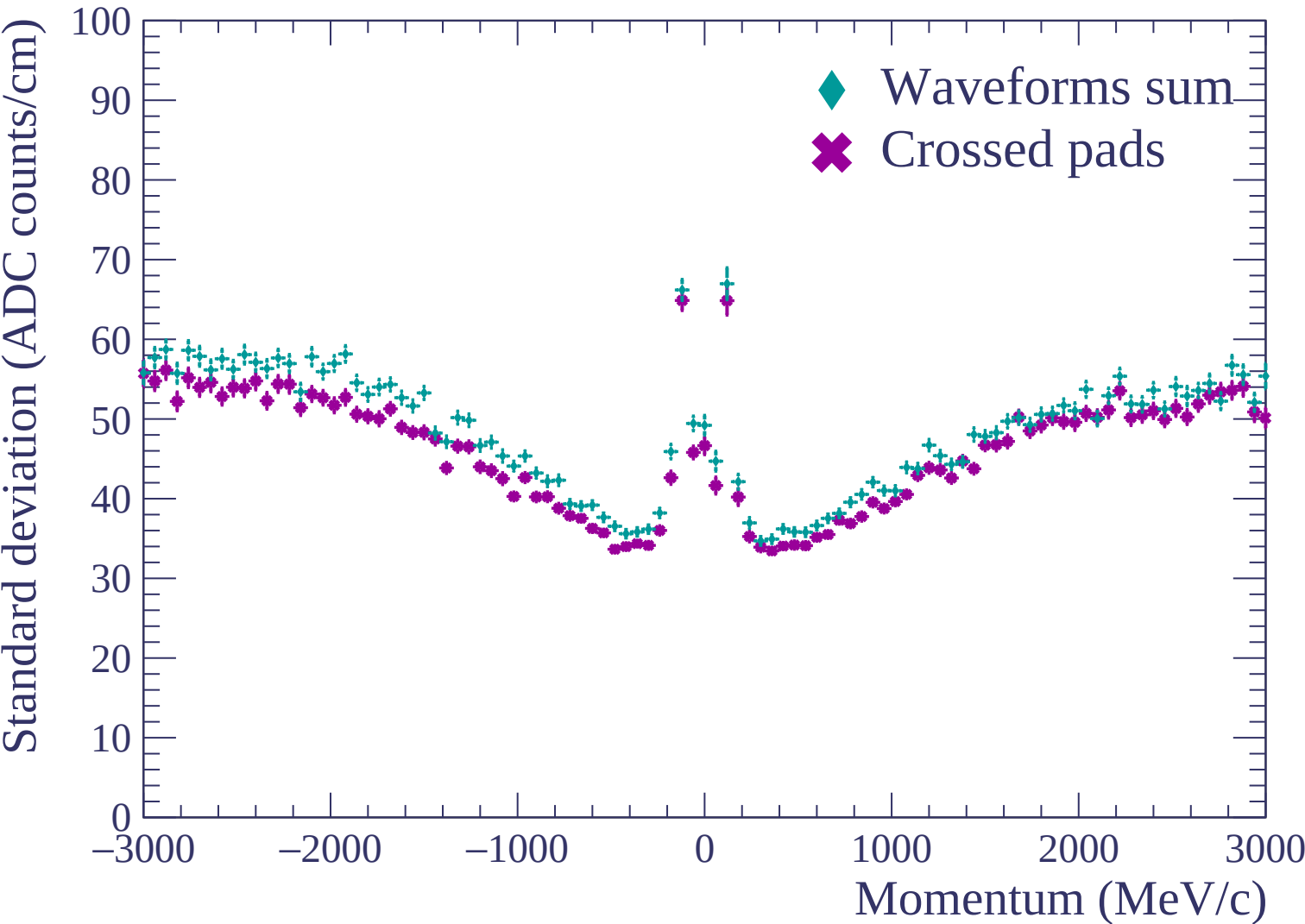


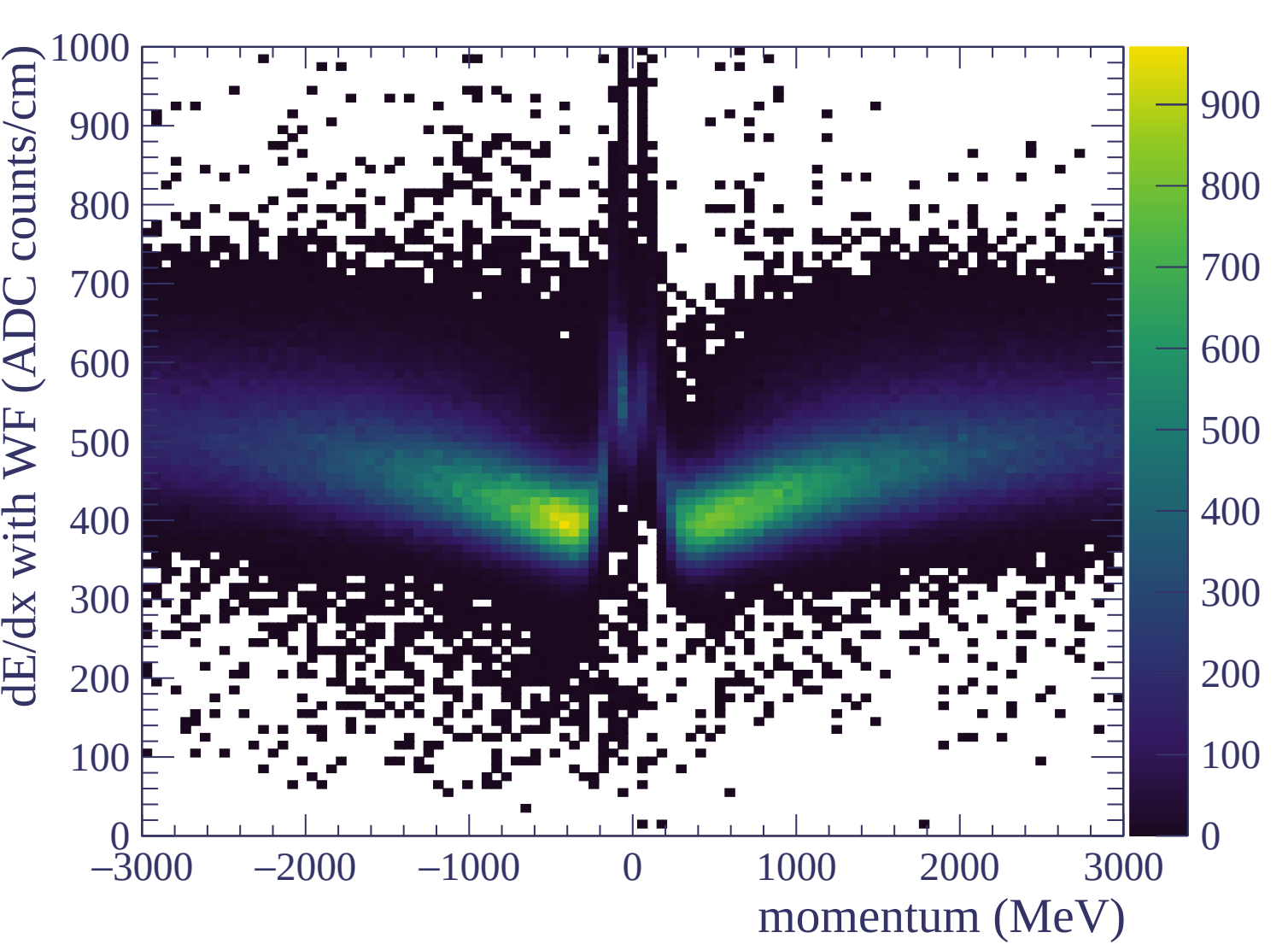


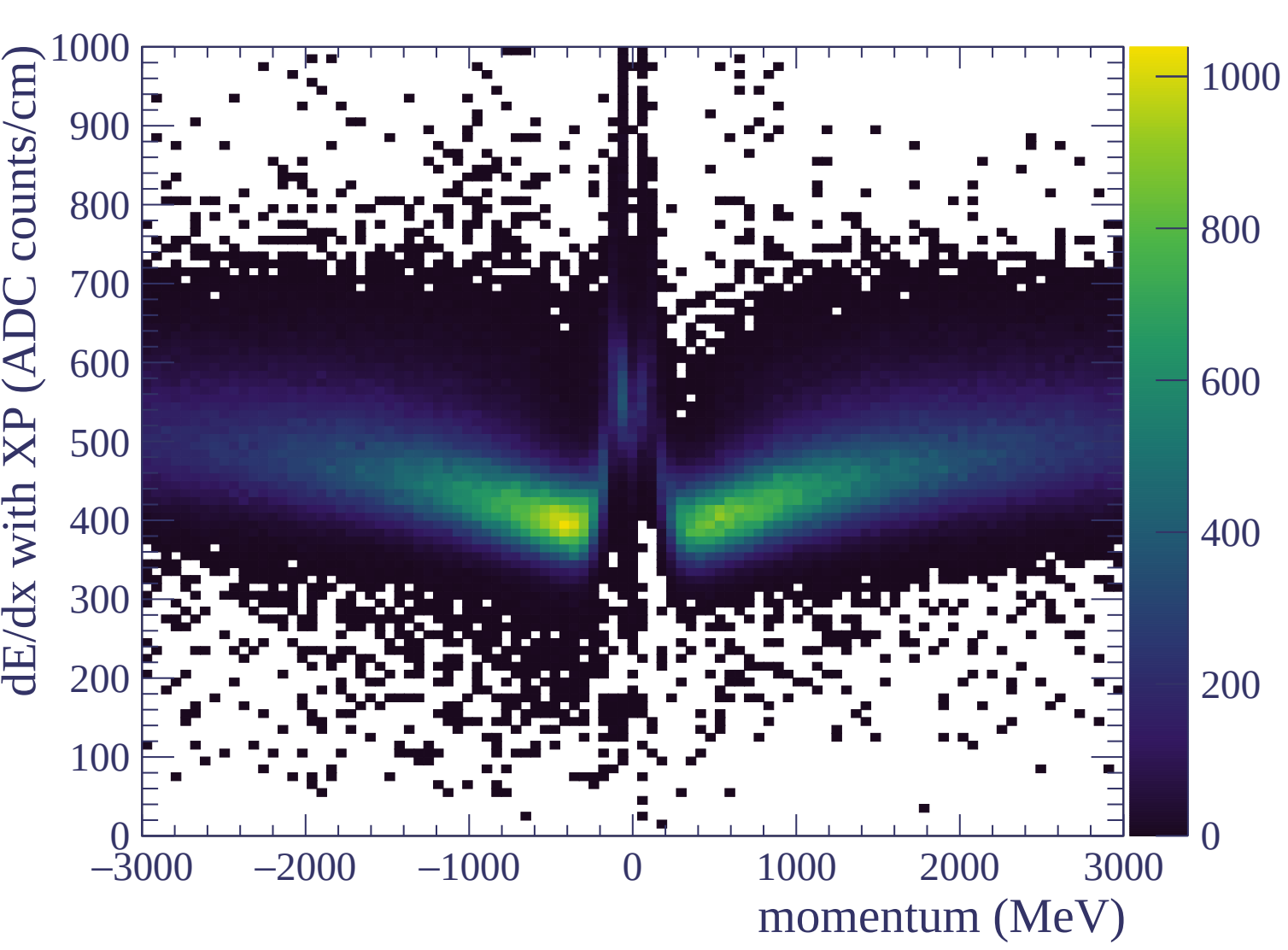
Resolution (%)

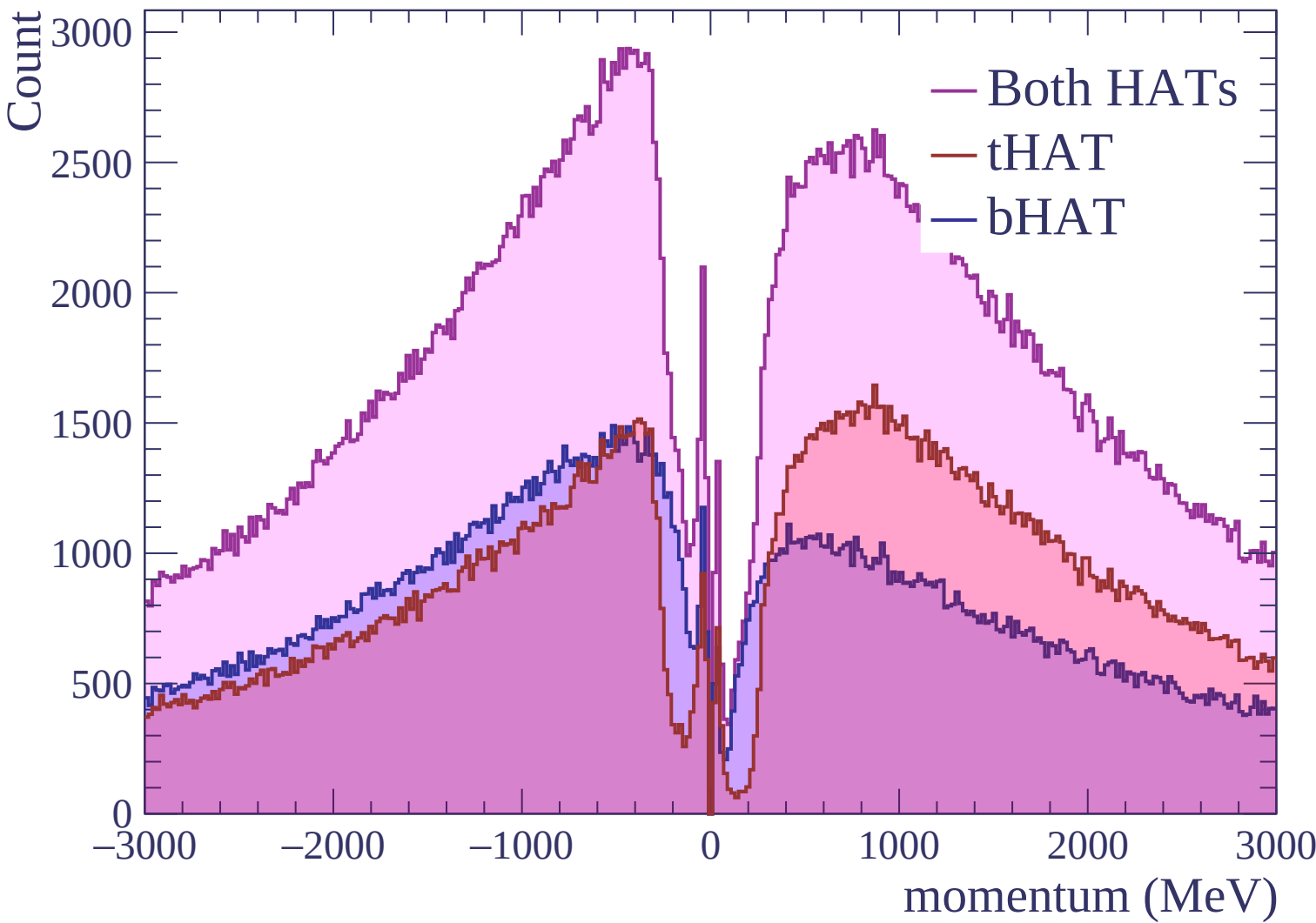




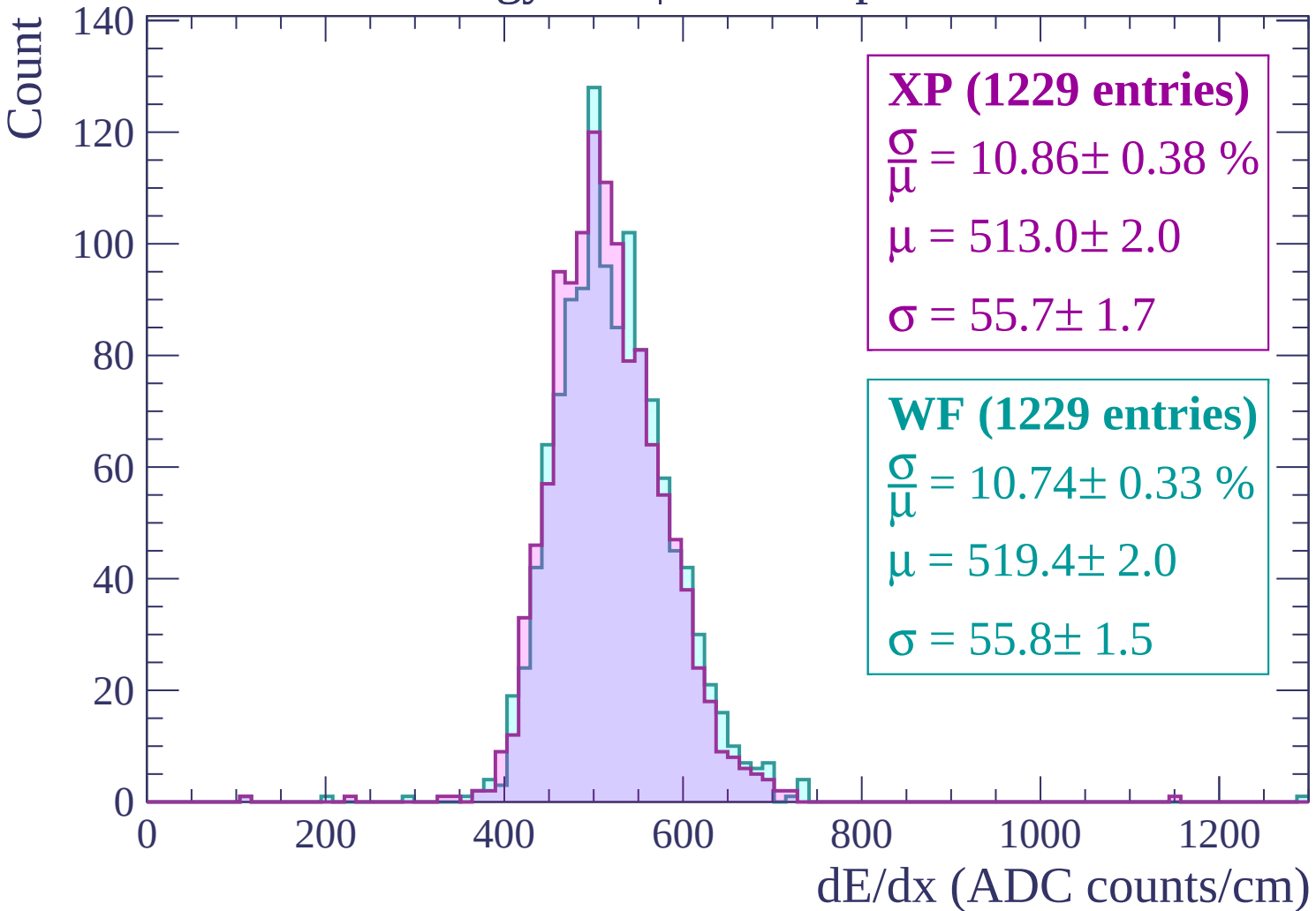






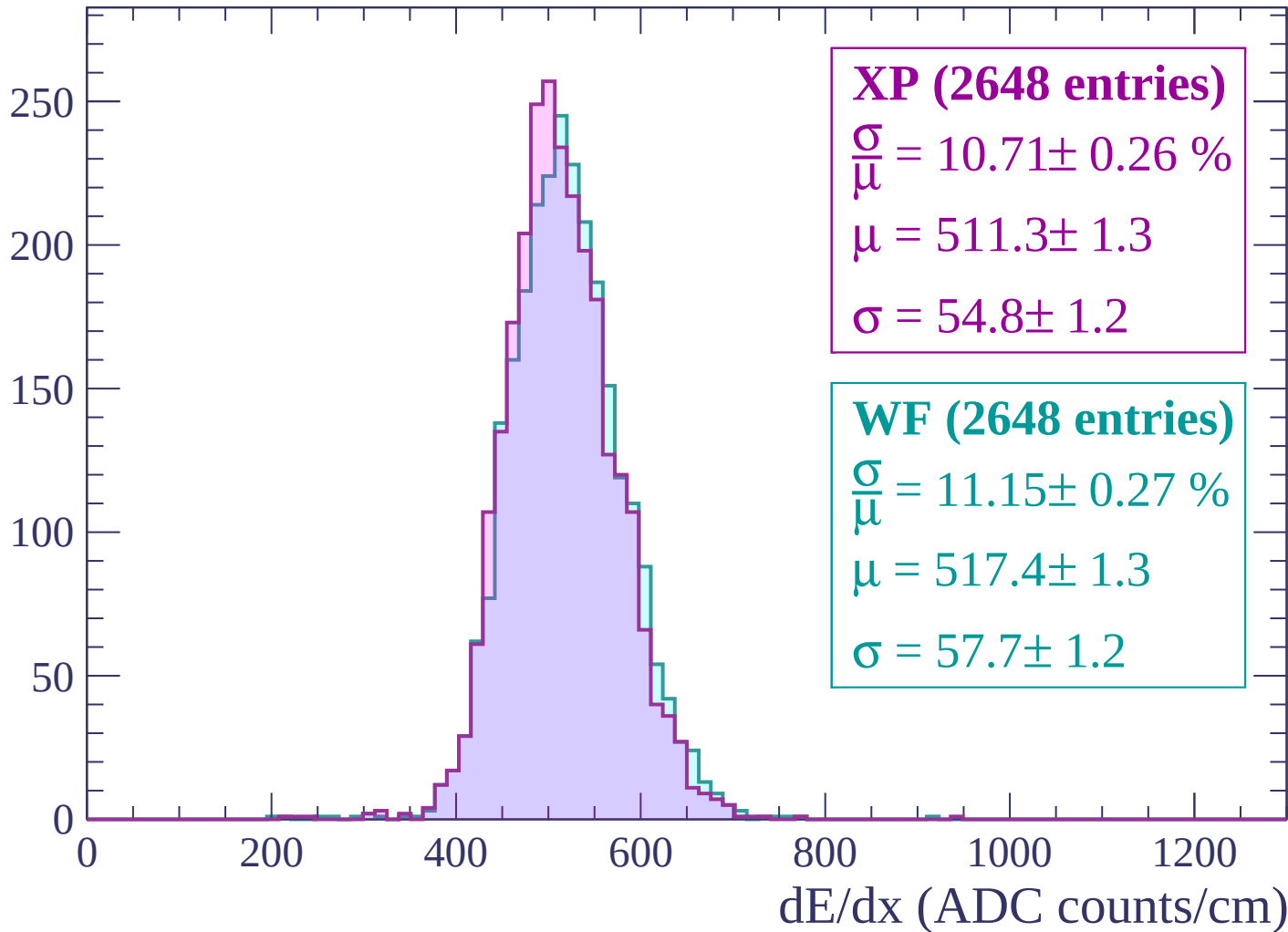


Energy loss | -3000 < p < -2940



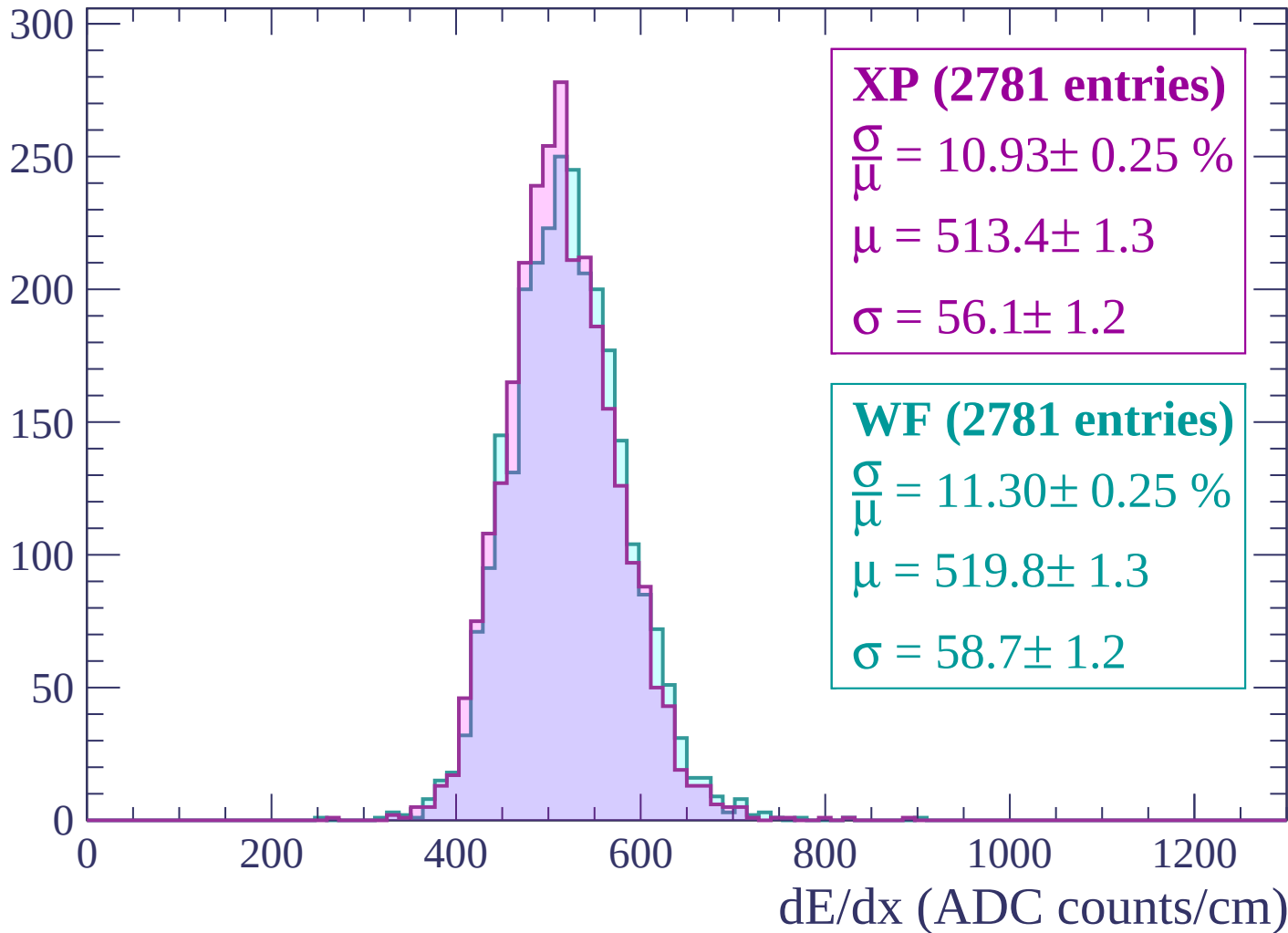
Energy loss | $-2940 < p < -2880$

Count



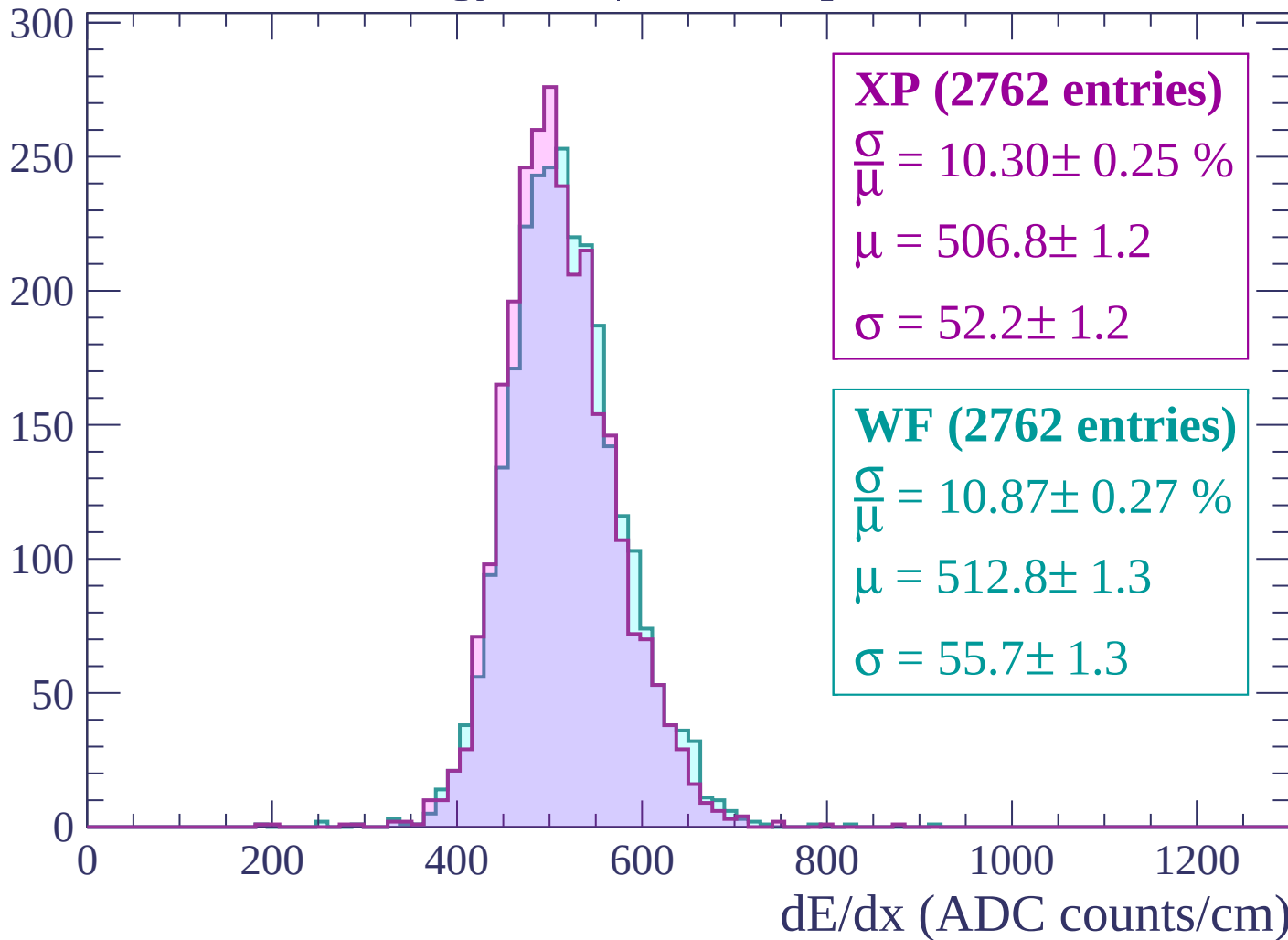
Energy loss | $-2880 < p < -2820$

Count



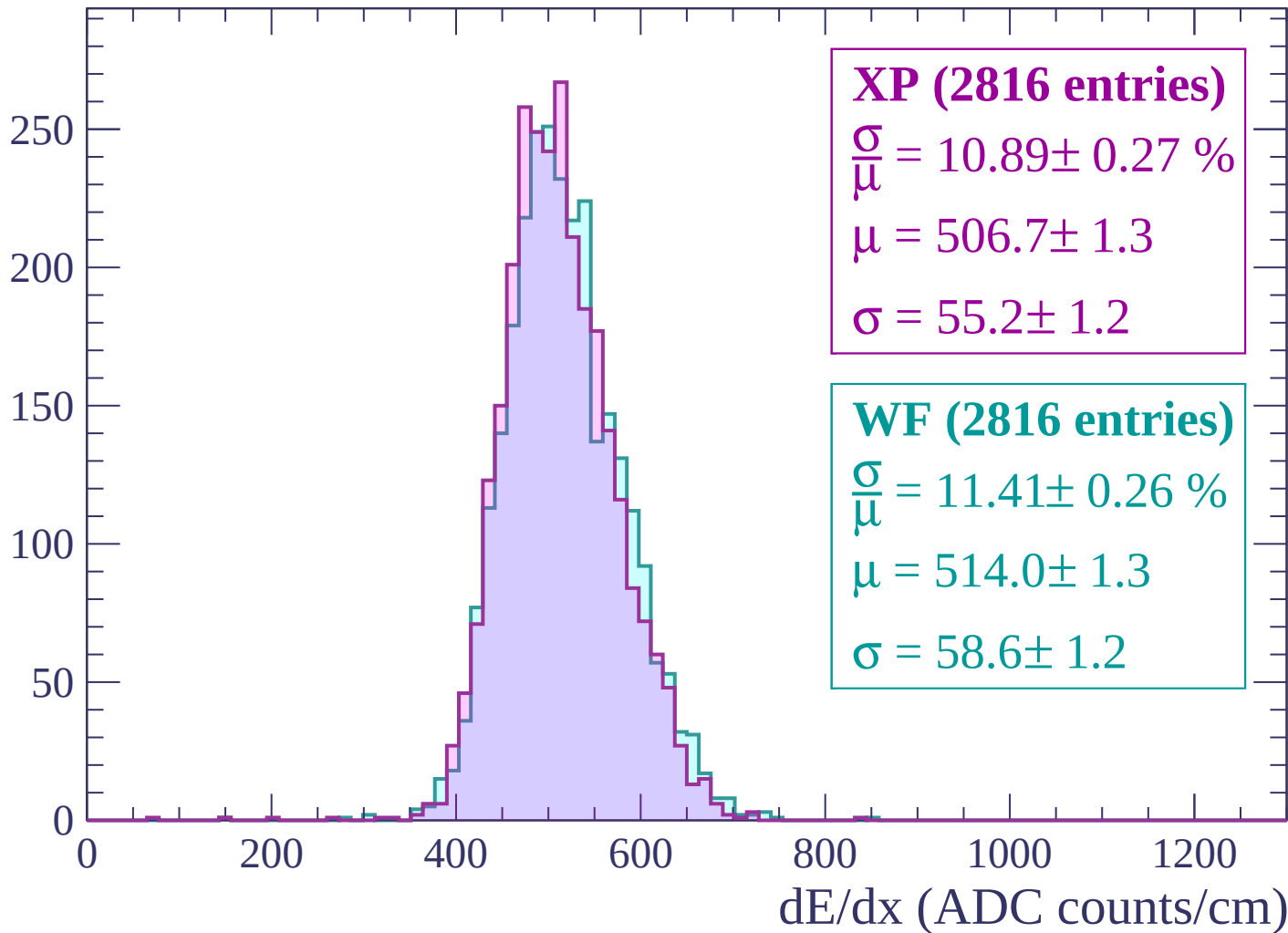
Energy loss | $-2820 < p < -2760$

Count



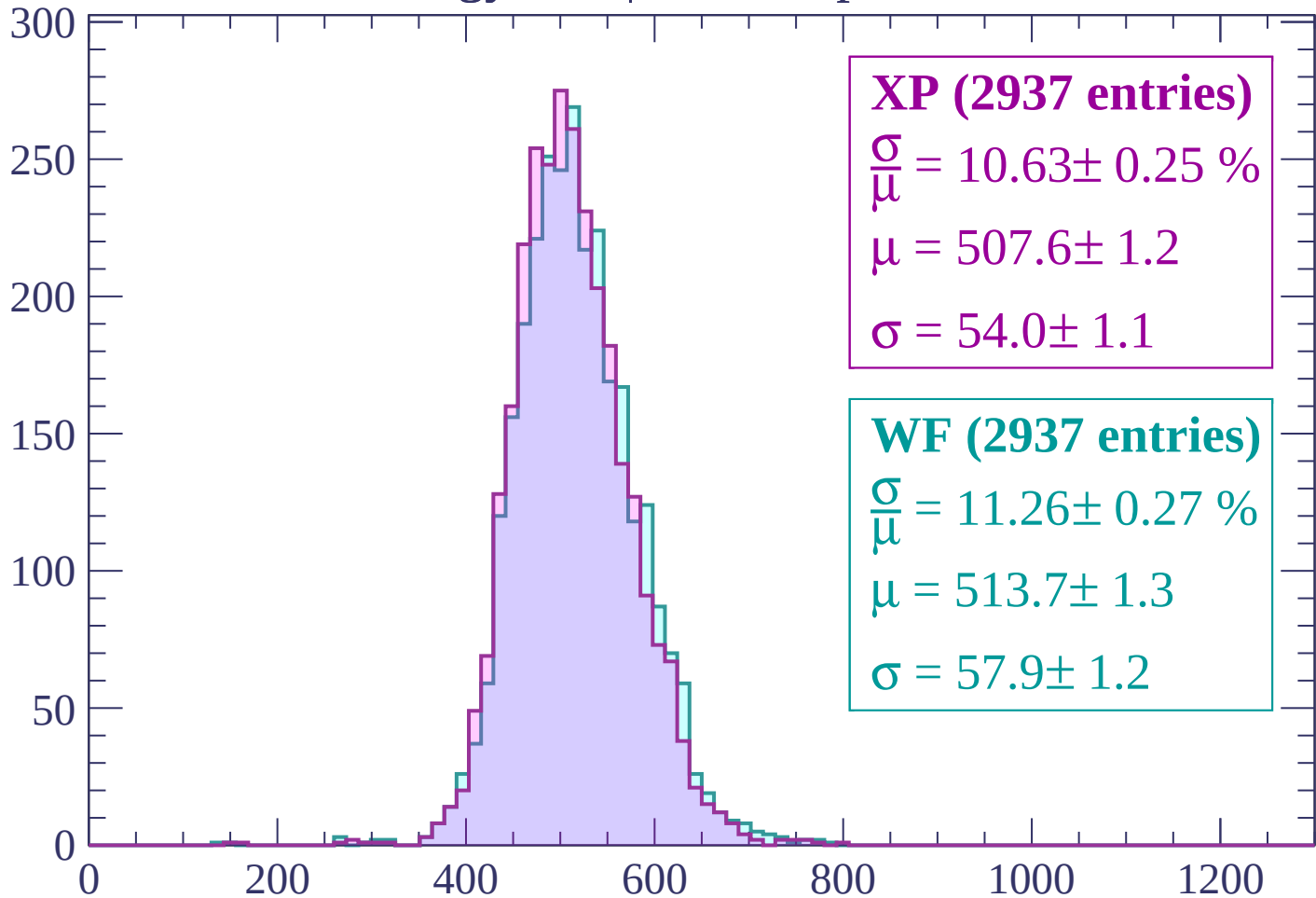
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (2937 entries)

$$\frac{\sigma}{\mu} = 10.63 \pm 0.25 \%$$

$$\mu = 507.6 \pm 1.2$$

$$\sigma = 54.0 \pm 1.1$$

WF (2937 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.27 \%$$

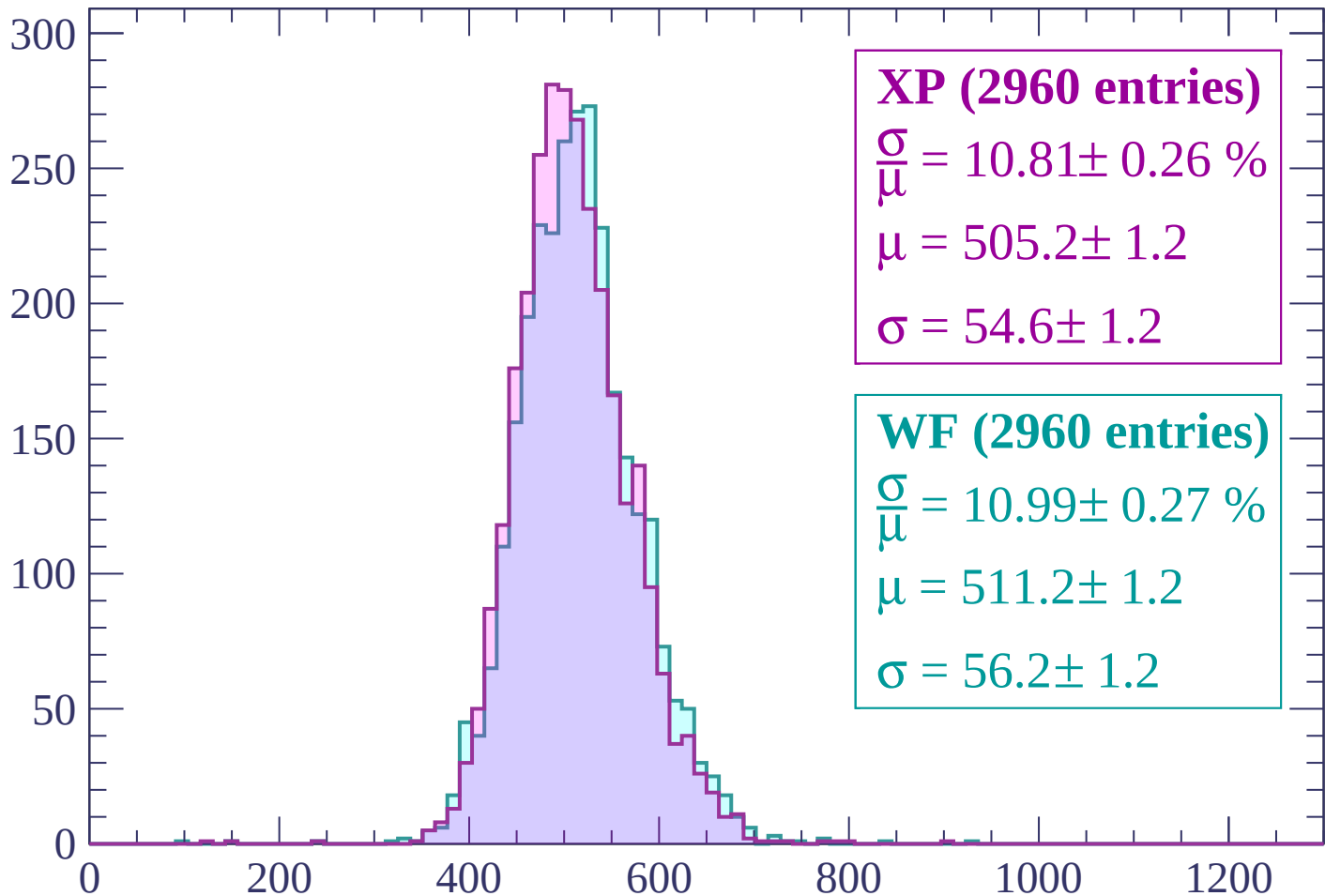
$$\mu = 513.7 \pm 1.3$$

$$\sigma = 57.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (2960 entries)

$\frac{\sigma}{\mu} = 10.81 \pm 0.26 \%$

$\mu = 505.2 \pm 1.2$

$\sigma = 54.6 \pm 1.2$

WF (2960 entries)

$\frac{\sigma}{\mu} = 10.99 \pm 0.27 \%$

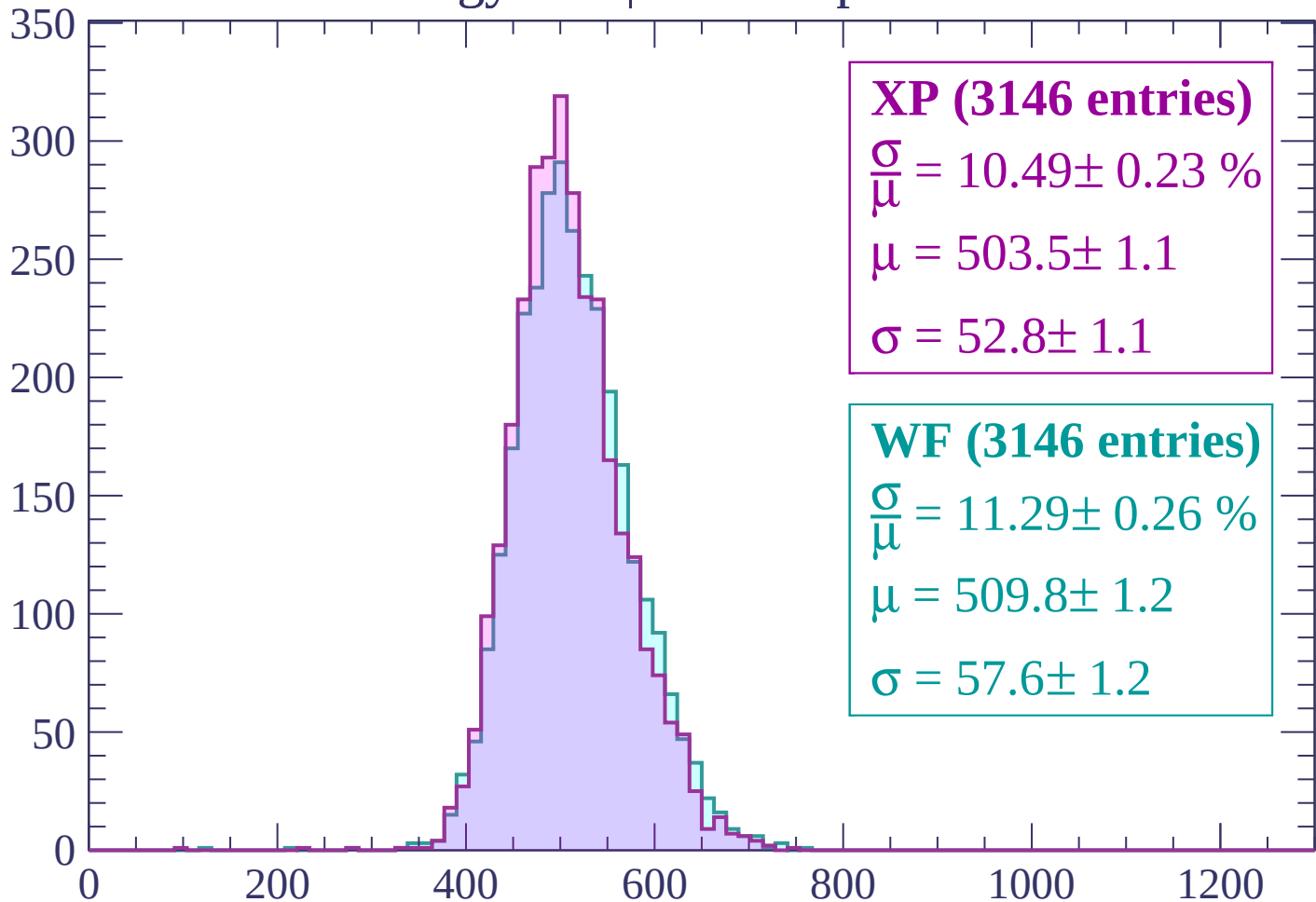
$\mu = 511.2 \pm 1.2$

$\sigma = 56.2 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $-2580 < p < -2520$

Count



XP (3146 entries)

$$\frac{\sigma}{\mu} = 10.49 \pm 0.23 \%$$

$$\mu = 503.5 \pm 1.1$$

$$\sigma = 52.8 \pm 1.1$$

WF (3146 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 0.26 \%$$

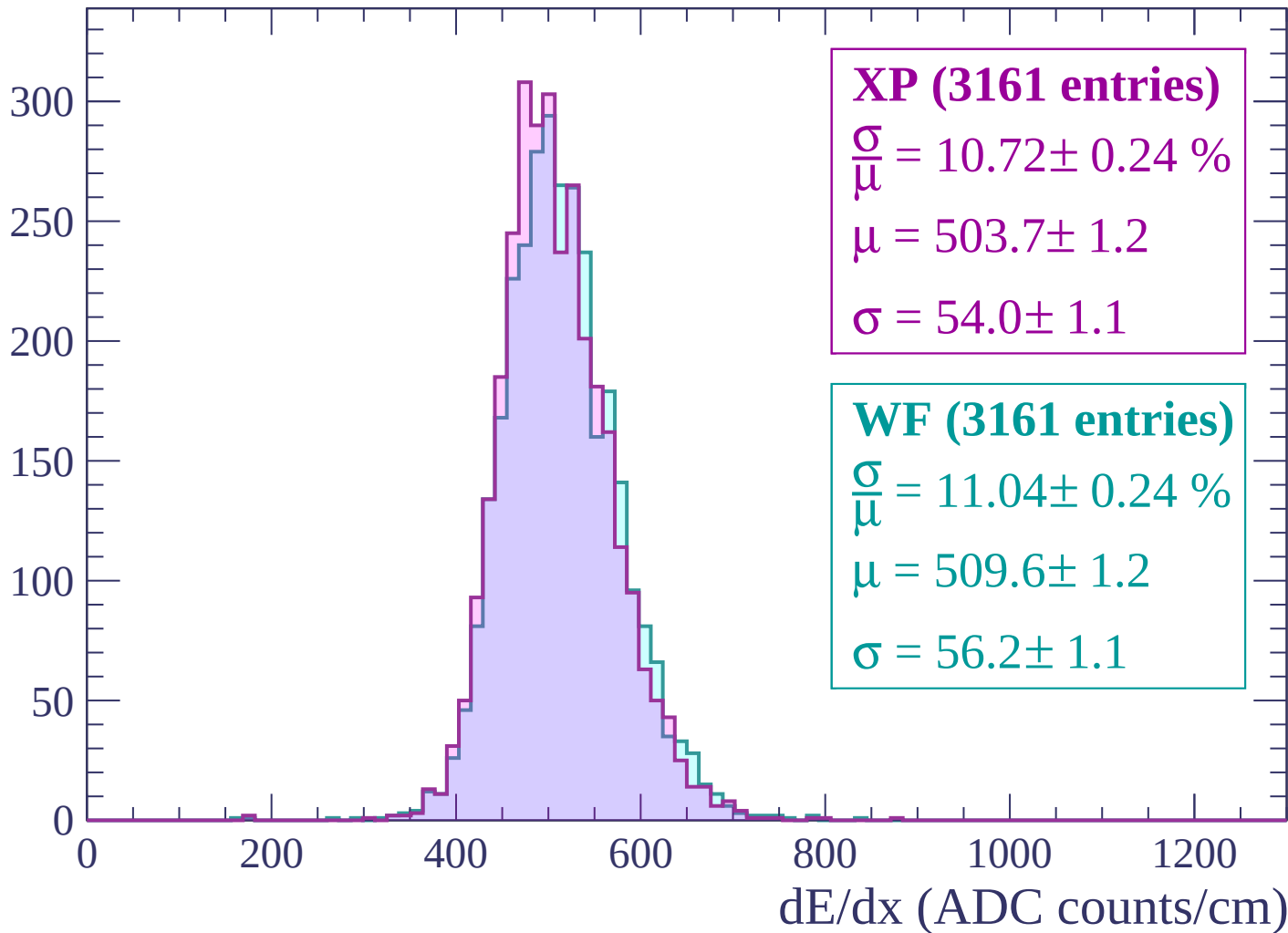
$$\mu = 509.8 \pm 1.2$$

$$\sigma = 57.6 \pm 1.2$$

dE/dx (ADC counts/cm)

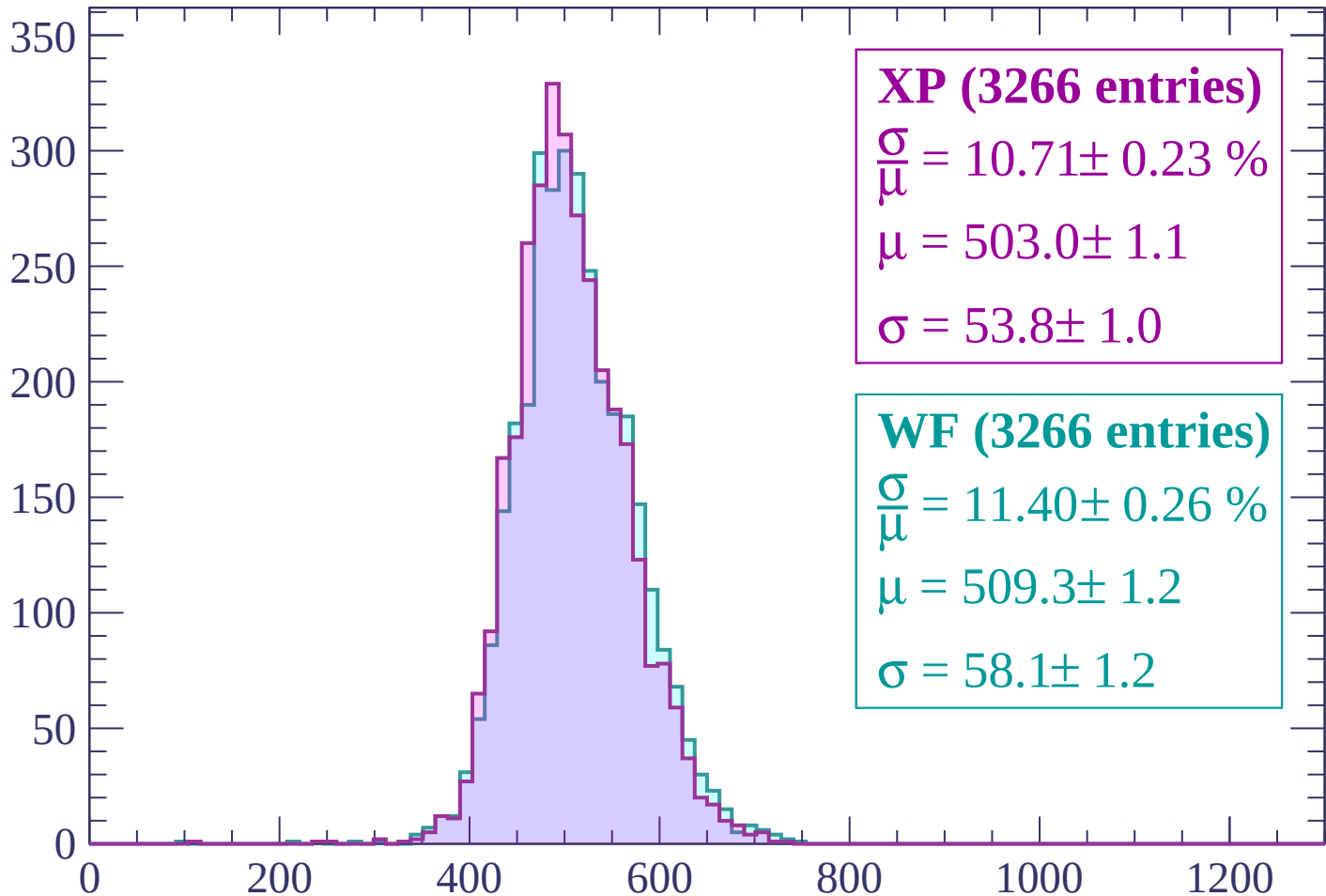
Energy loss | -2520 < p < -2460

Count



Energy loss | $-2460 < p < -2400$

Count



XP (3266 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$

$\mu = 503.0 \pm 1.1$

$\sigma = 53.8 \pm 1.0$

WF (3266 entries)

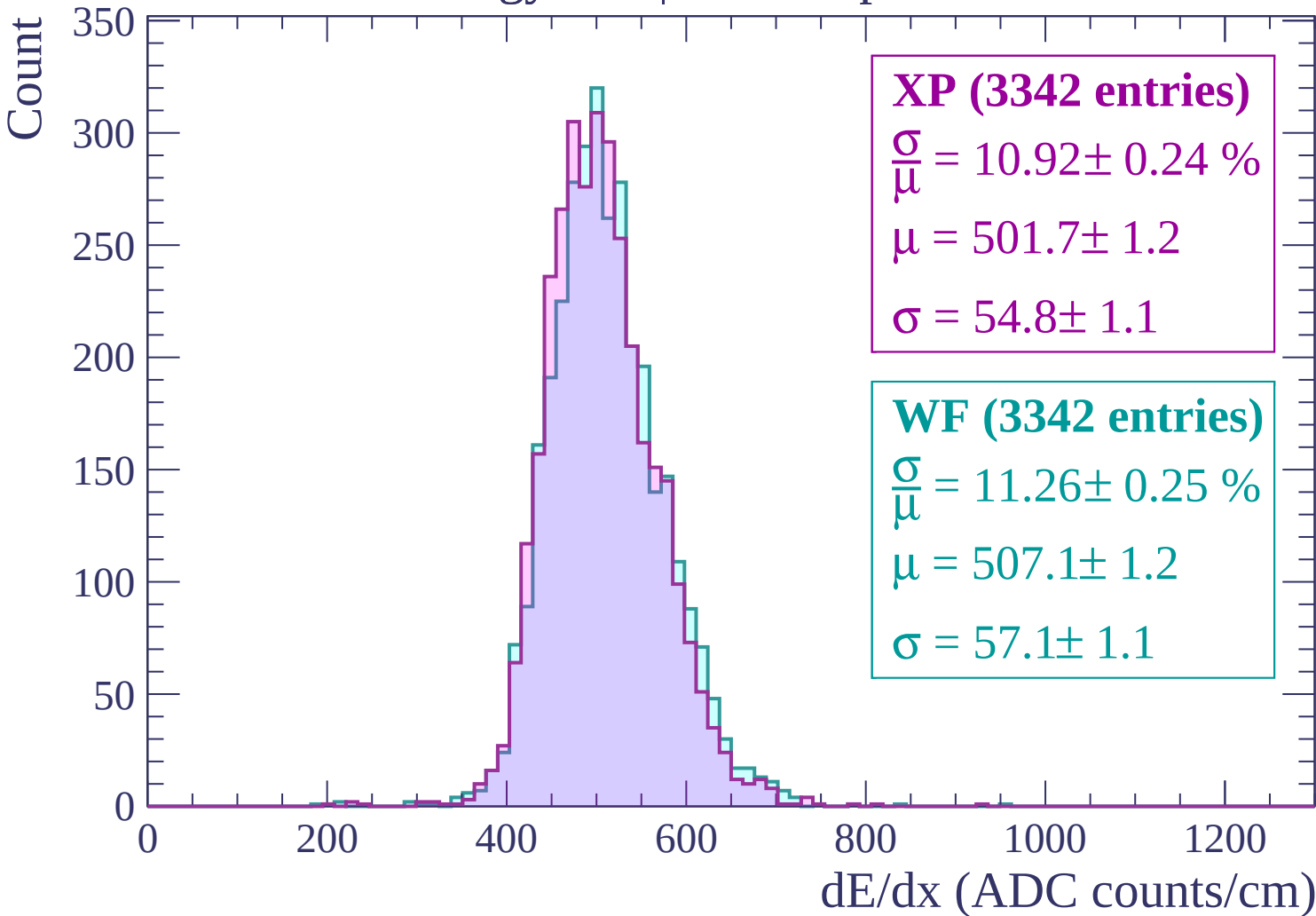
$\frac{\sigma}{\mu} = 11.40 \pm 0.26 \%$

$\mu = 509.3 \pm 1.2$

$\sigma = 58.1 \pm 1.2$

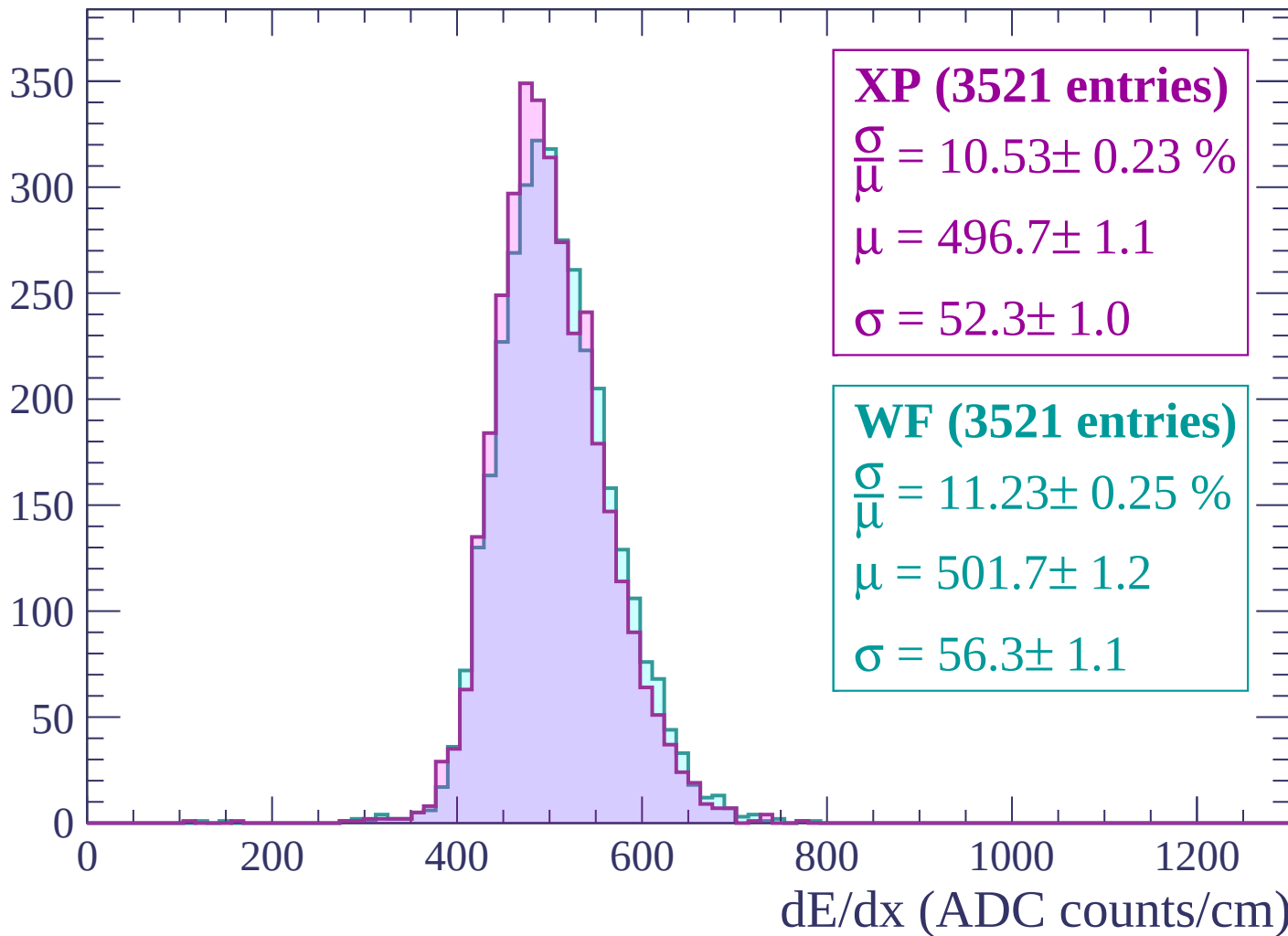
dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340



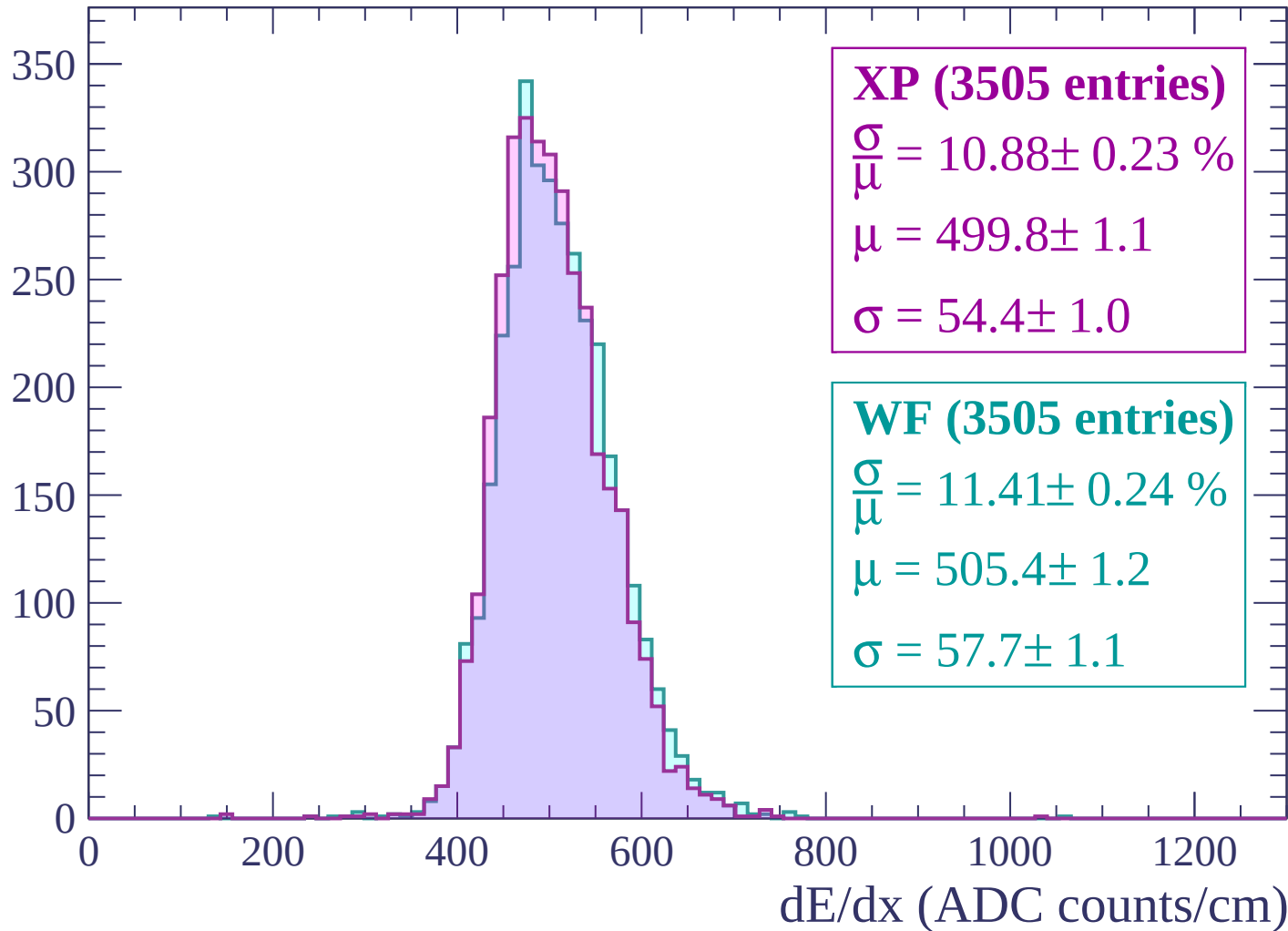
Energy loss | $-2340 < p < -2280$

Count



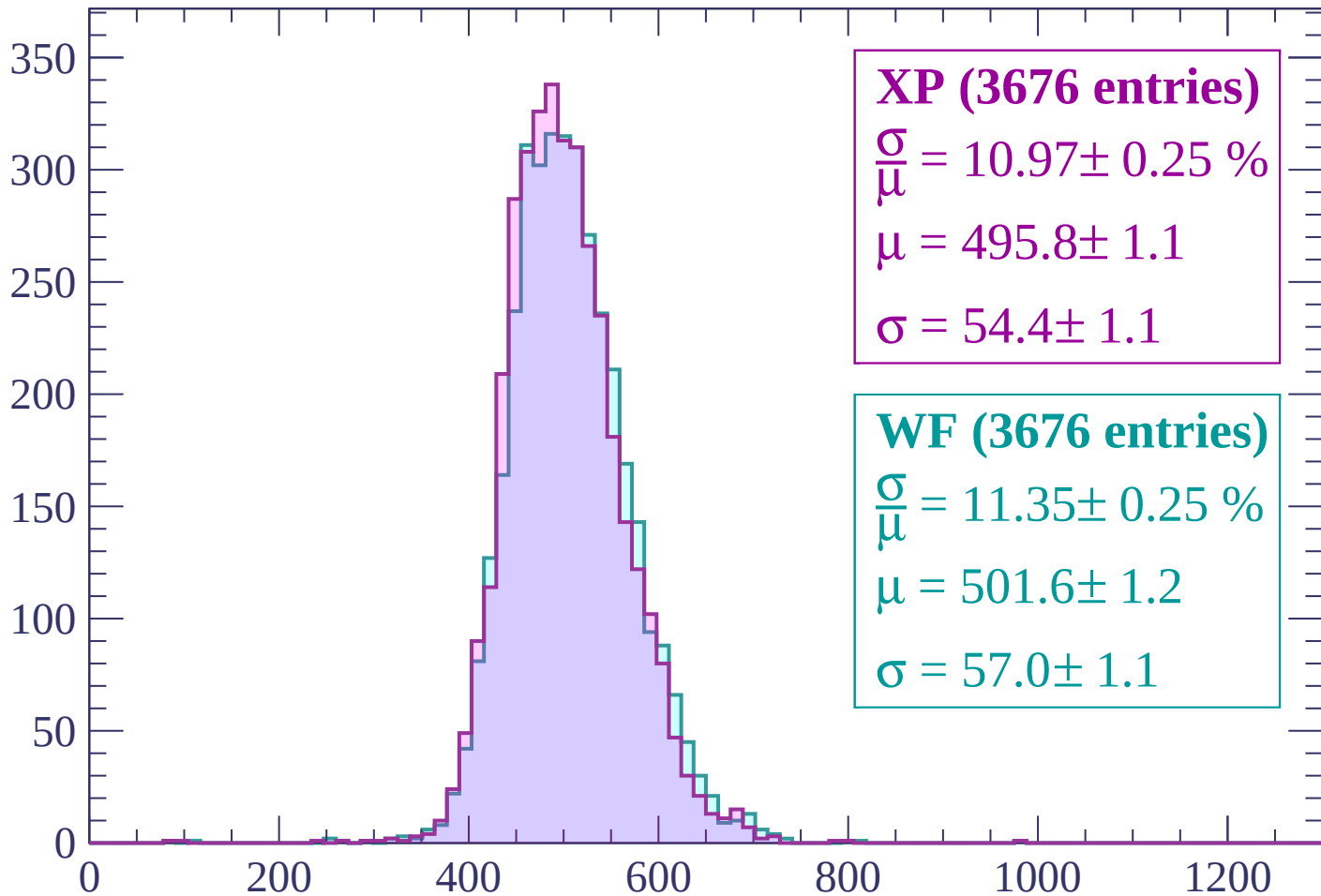
Energy loss | -2280 < p < -2220

Count



Energy loss | -2220 < p < -2160

Count



XP (3676 entries)

$\frac{\sigma}{\mu} = 10.97 \pm 0.25 \%$

$\mu = 495.8 \pm 1.1$

$\sigma = 54.4 \pm 1.1$

WF (3676 entries)

$\frac{\sigma}{\mu} = 11.35 \pm 0.25 \%$

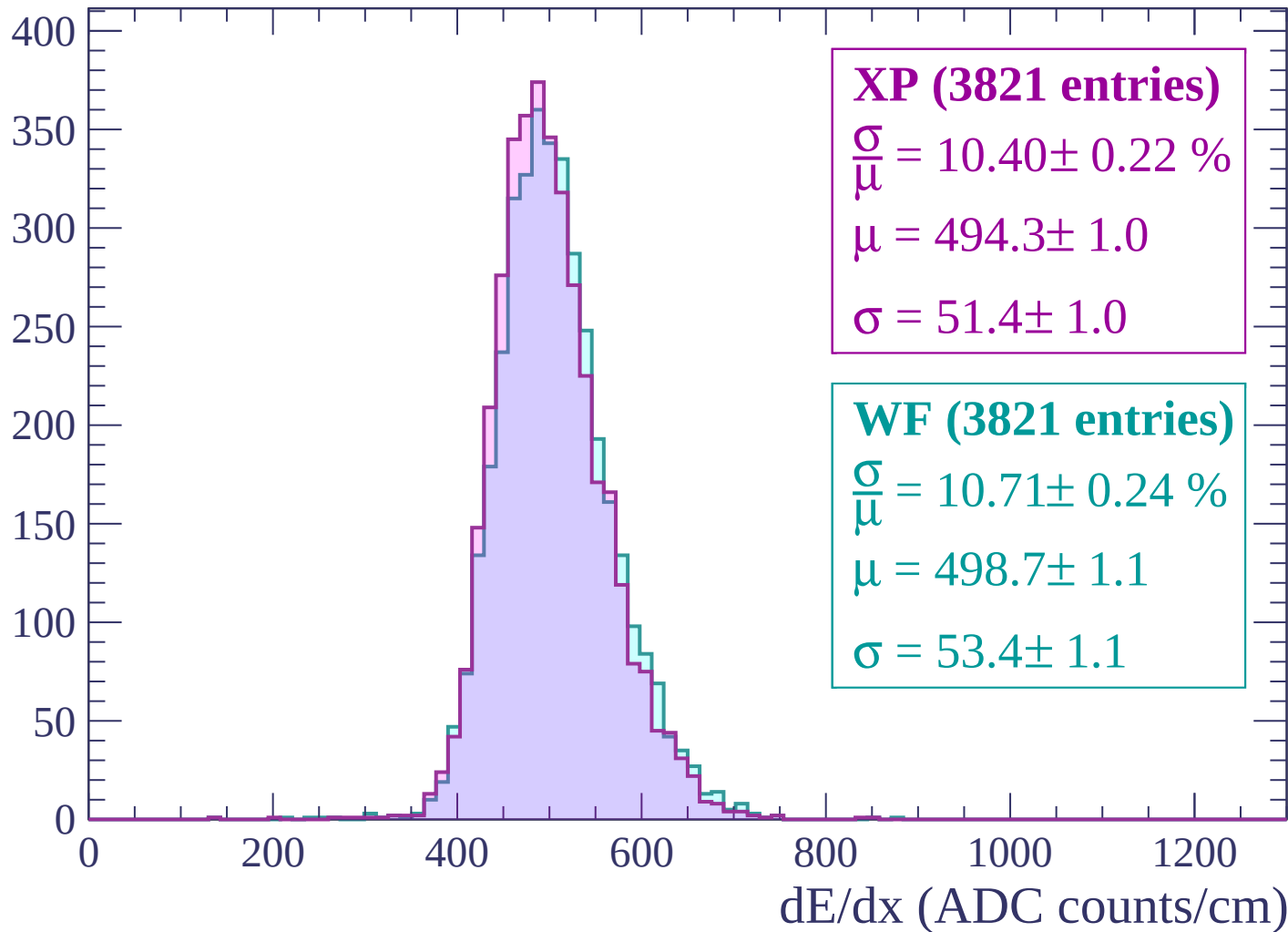
$\mu = 501.6 \pm 1.2$

$\sigma = 57.0 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

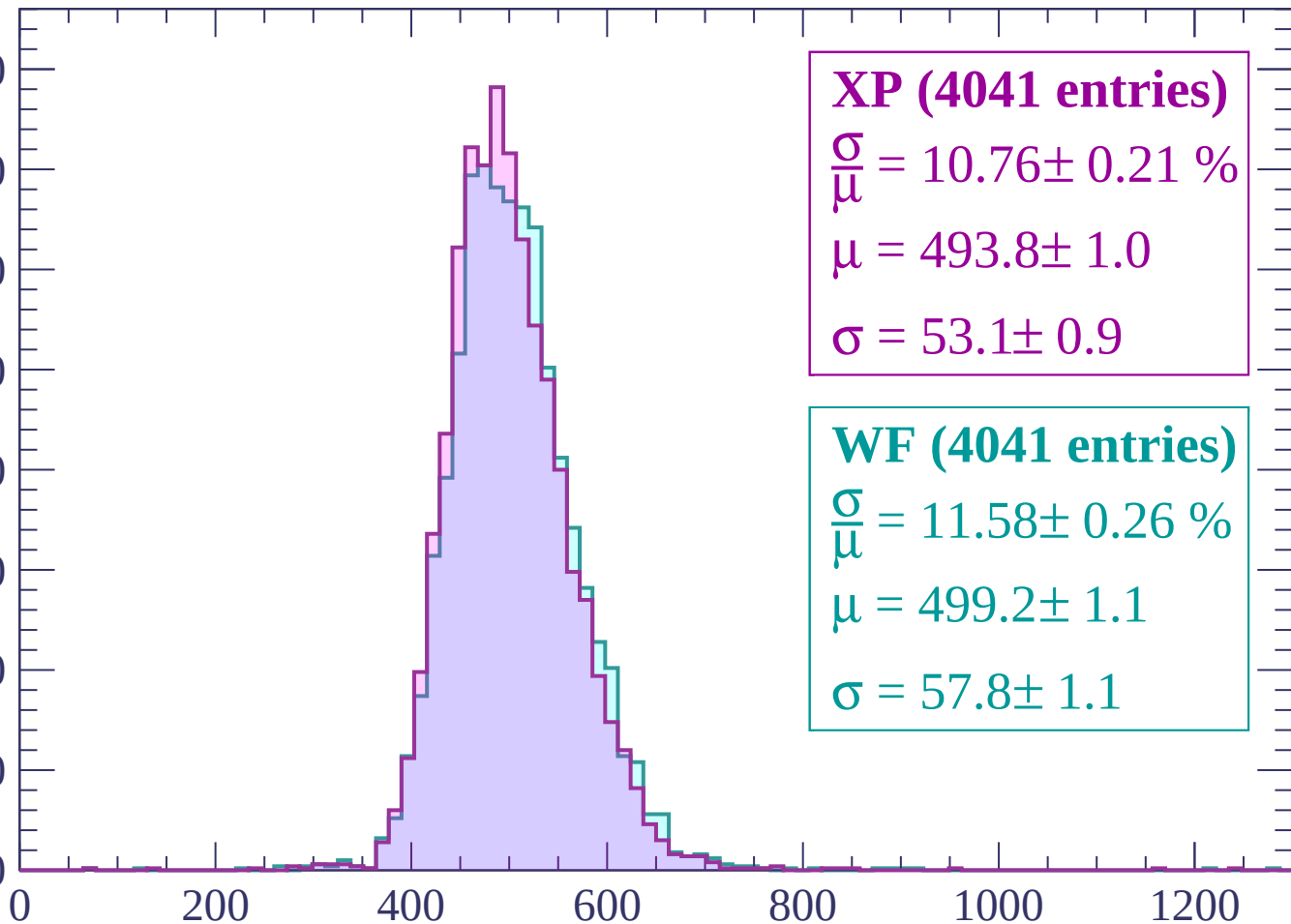
Count



Energy loss | -2100 < p < -2040

Count

400
350
300
250
200
150
100
50
0



XP (4041 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.21 \%$$

$$\mu = 493.8 \pm 1.0$$

$$\sigma = 53.1 \pm 0.9$$

WF (4041 entries)

$$\frac{\sigma}{\mu} = 11.58 \pm 0.26 \%$$

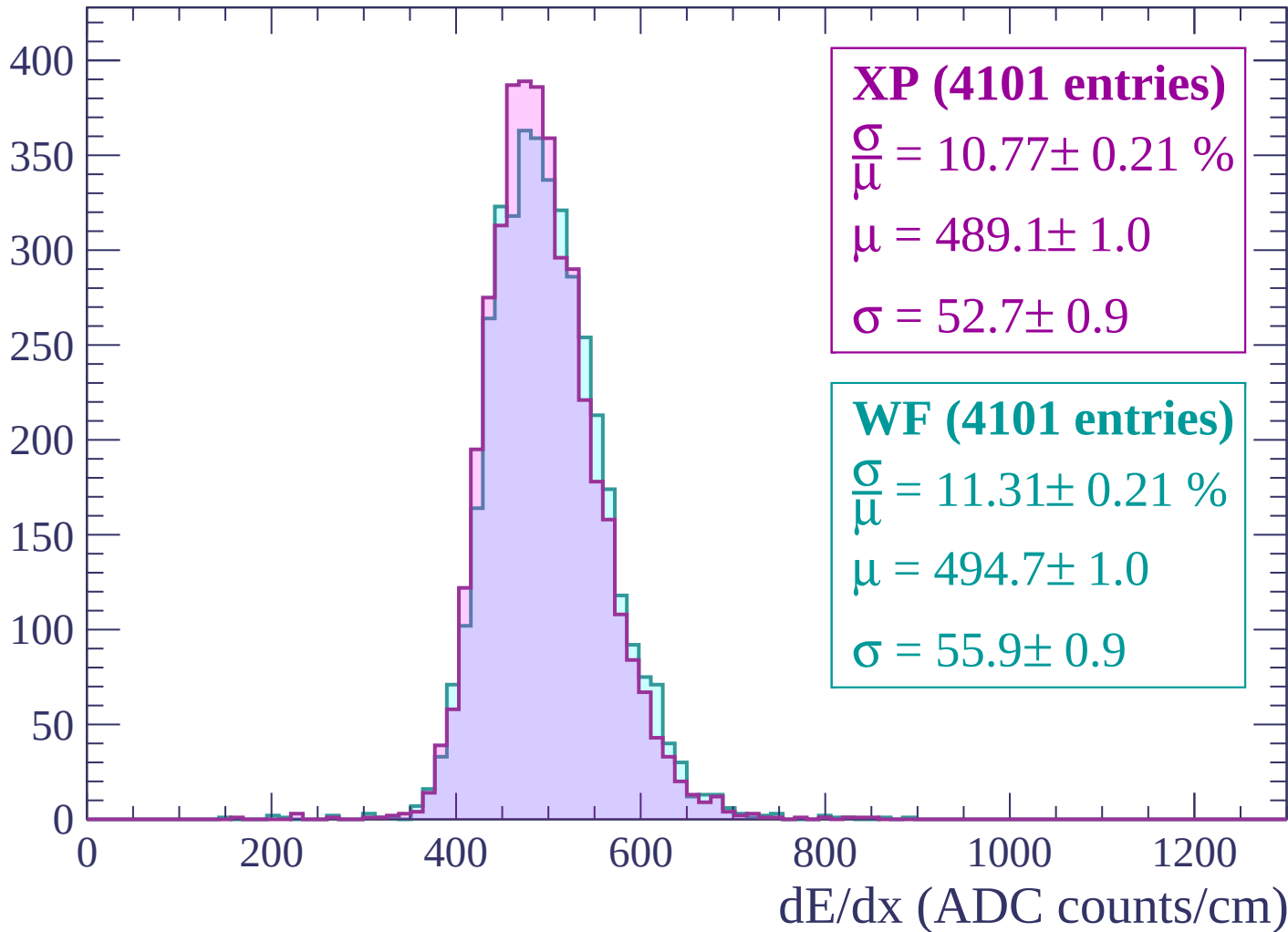
$$\mu = 499.2 \pm 1.1$$

$$\sigma = 57.8 \pm 1.1$$

dE/dx (ADC counts/cm)

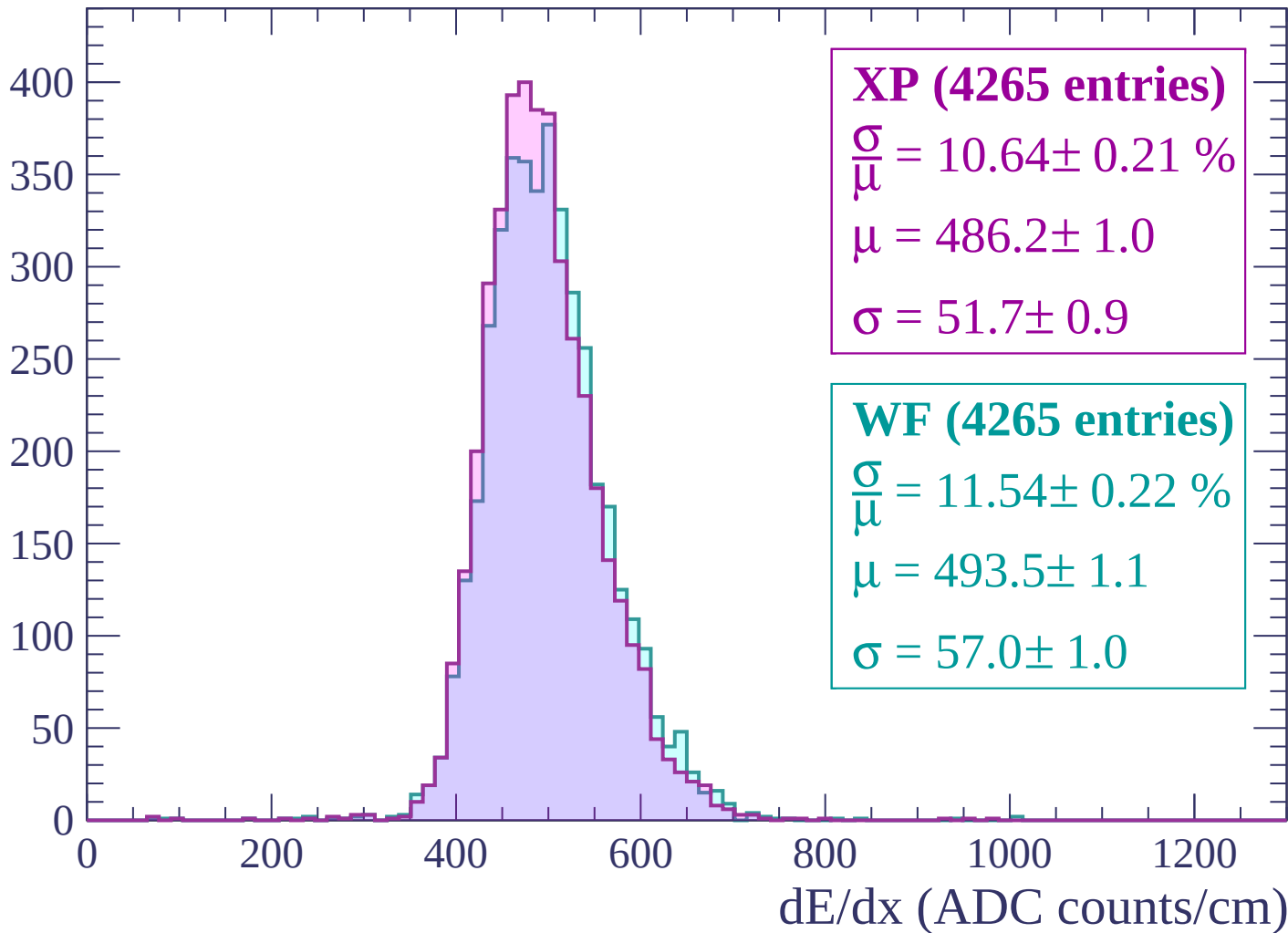
Energy loss | $-2040 < p < -1980$

Count



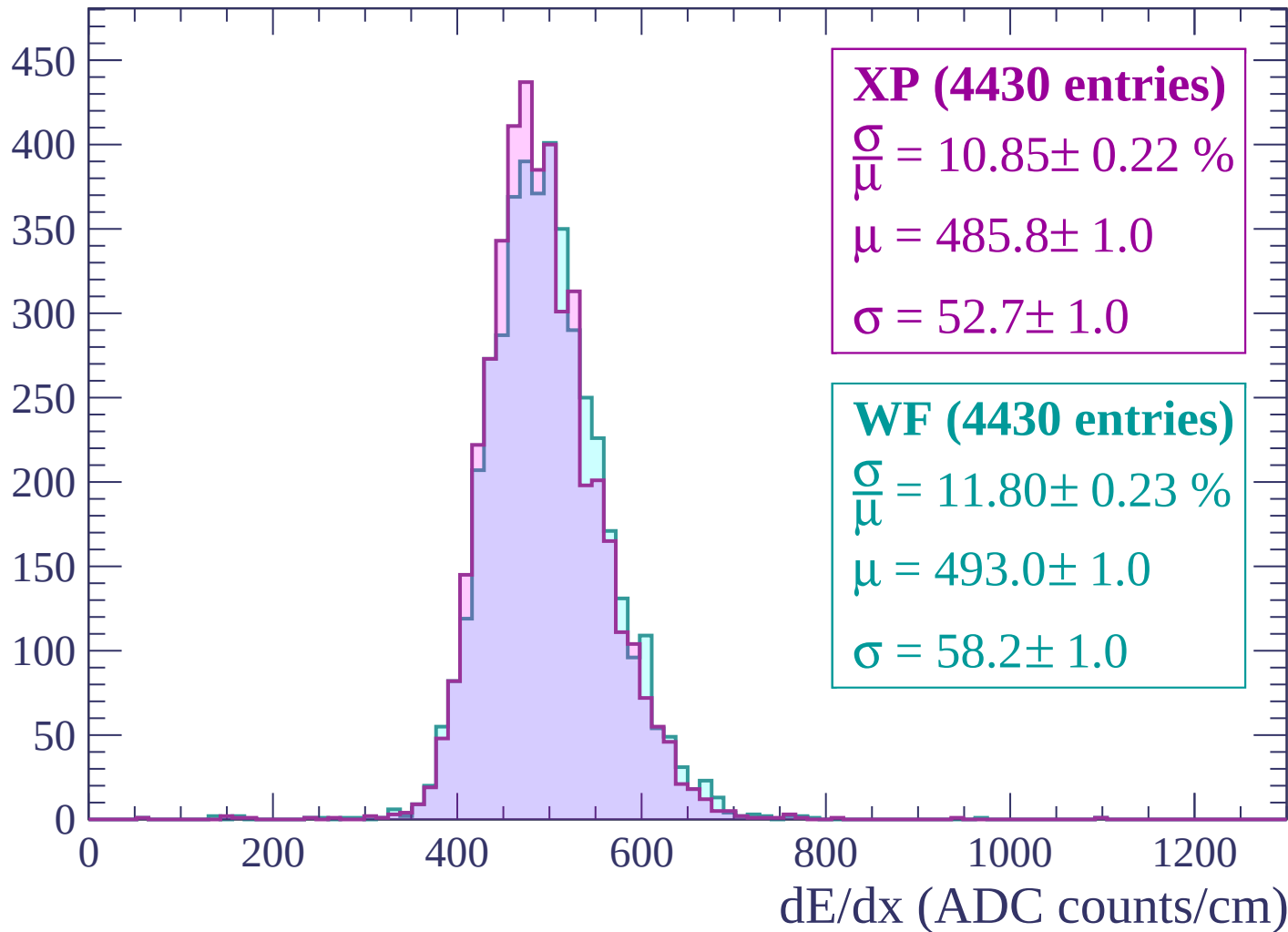
Energy loss | -1980 < p < -1920

Count



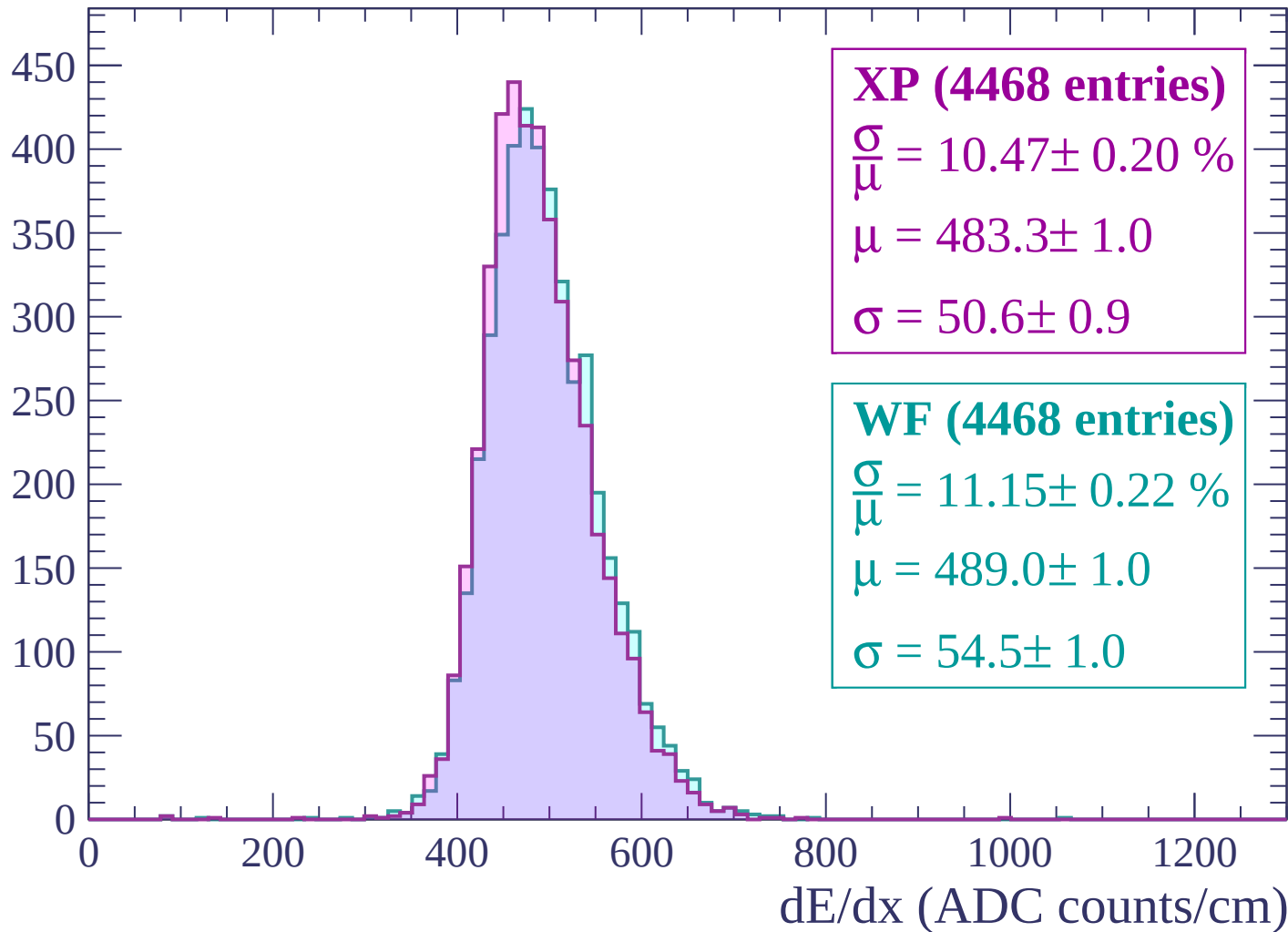
Energy loss | -1920 < p < -1860

Count



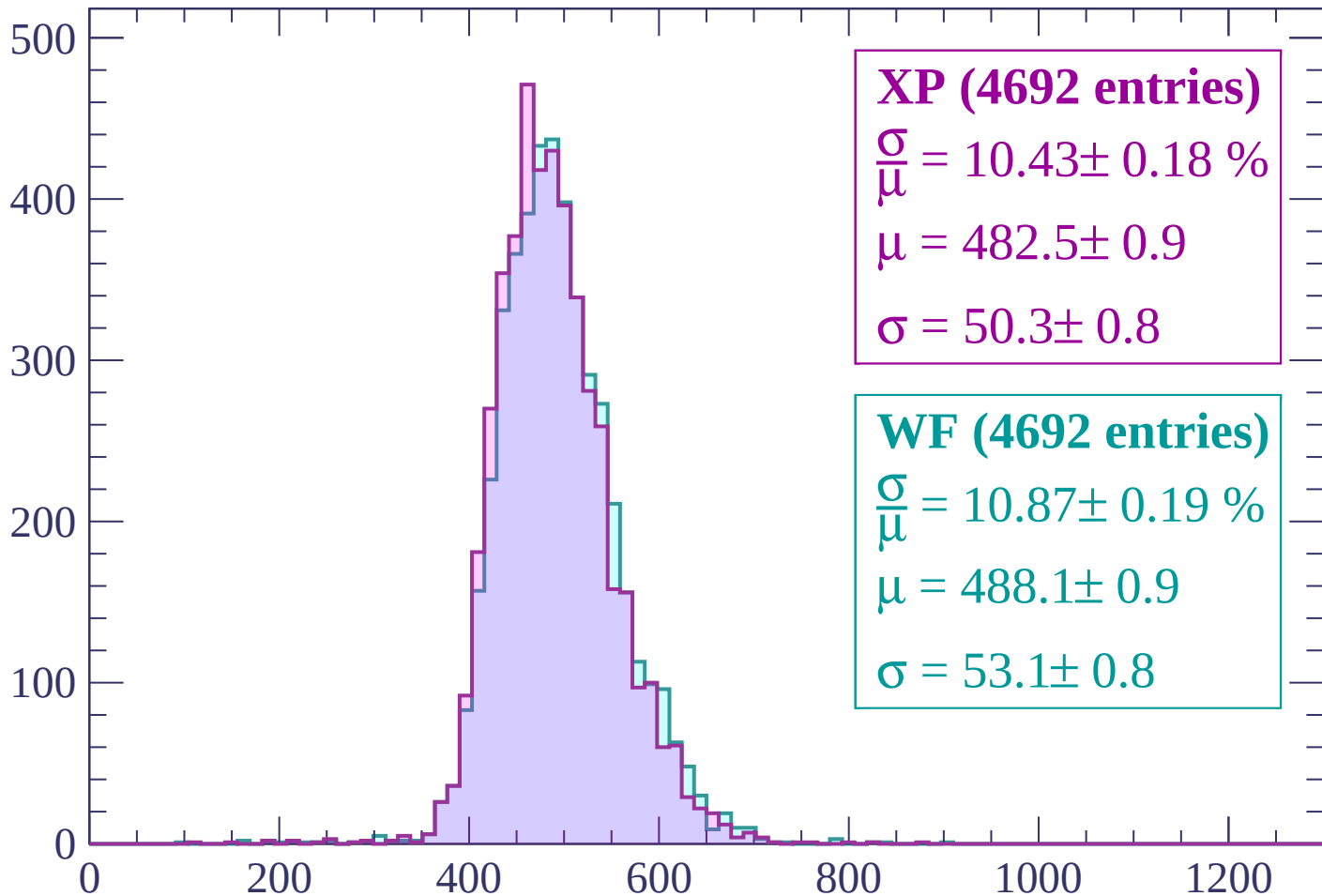
Energy loss | $-1860 < p < -1800$

Count



Energy loss | -1800 < p < -1740

Count



XP (4692 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.18 \%$

$\mu = 482.5 \pm 0.9$

$\sigma = 50.3 \pm 0.8$

WF (4692 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.19 \%$

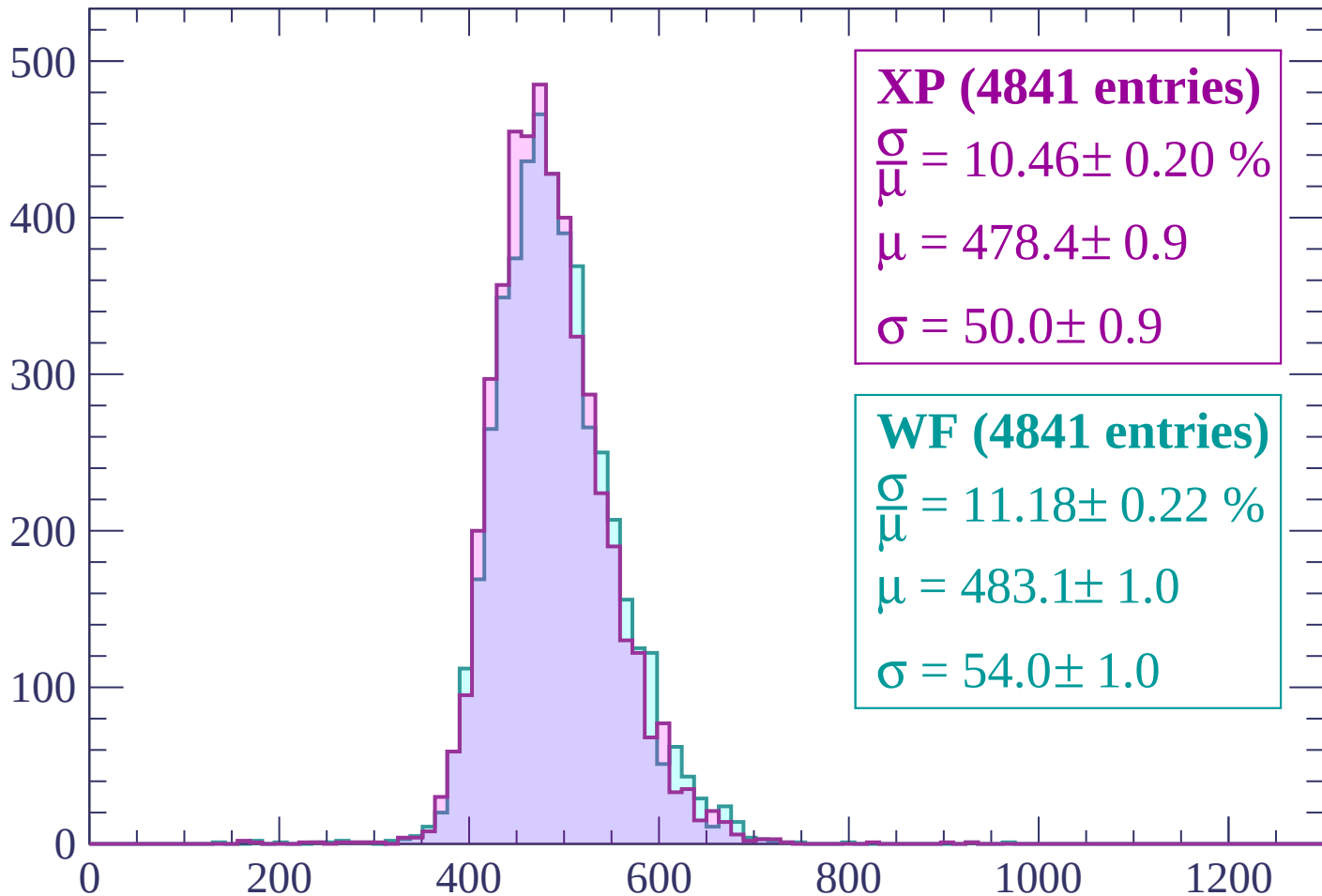
$\mu = 488.1 \pm 0.9$

$\sigma = 53.1 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (4841 entries)

$$\frac{\sigma}{\mu} = 10.46 \pm 0.20 \%$$

$$\mu = 478.4 \pm 0.9$$

$$\sigma = 50.0 \pm 0.9$$

WF (4841 entries)

$$\frac{\sigma}{\mu} = 11.18 \pm 0.22 \%$$

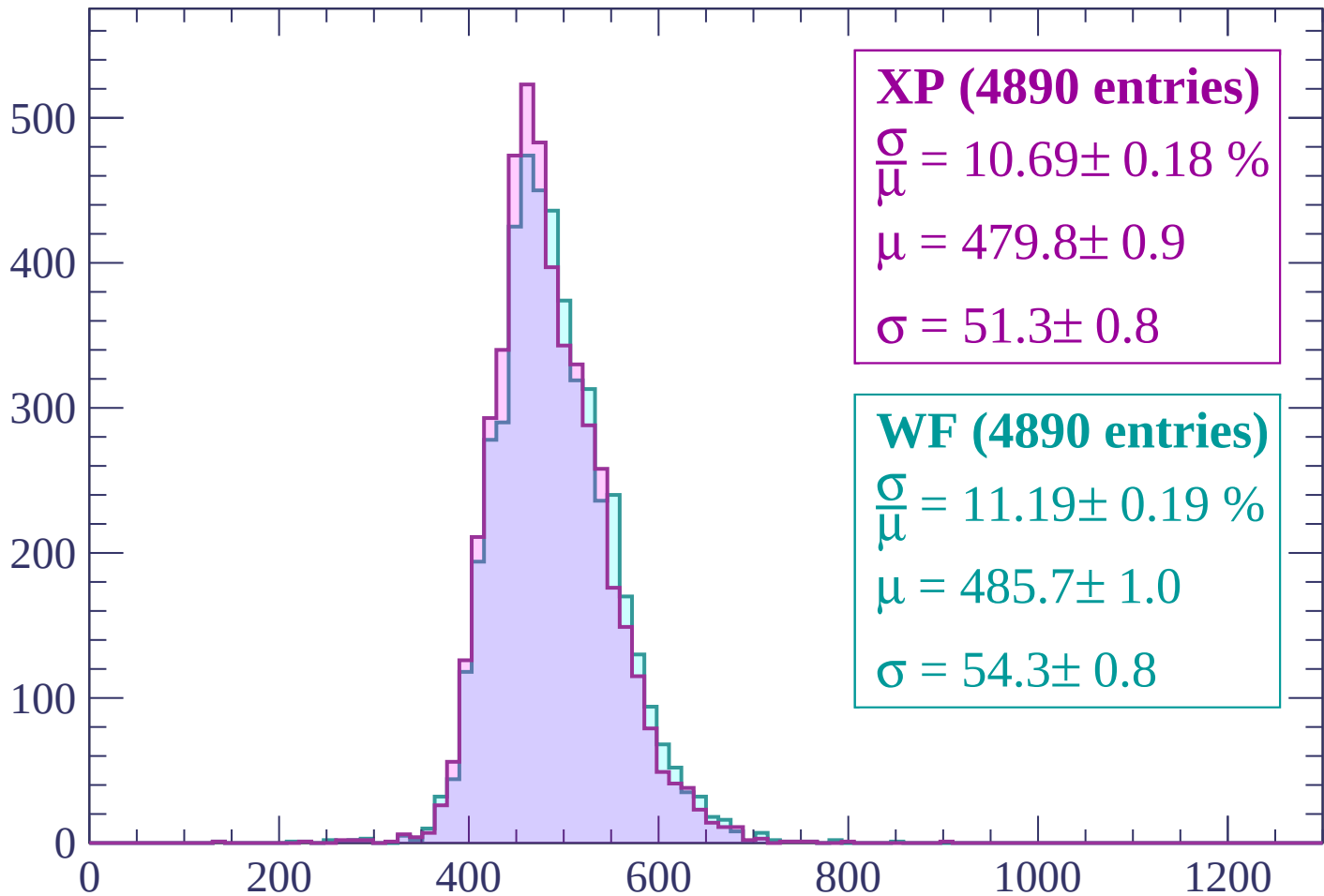
$$\mu = 483.1 \pm 1.0$$

$$\sigma = 54.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (4890 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.18 \%$$

$$\mu = 479.8 \pm 0.9$$

$$\sigma = 51.3 \pm 0.8$$

WF (4890 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.19 \%$$

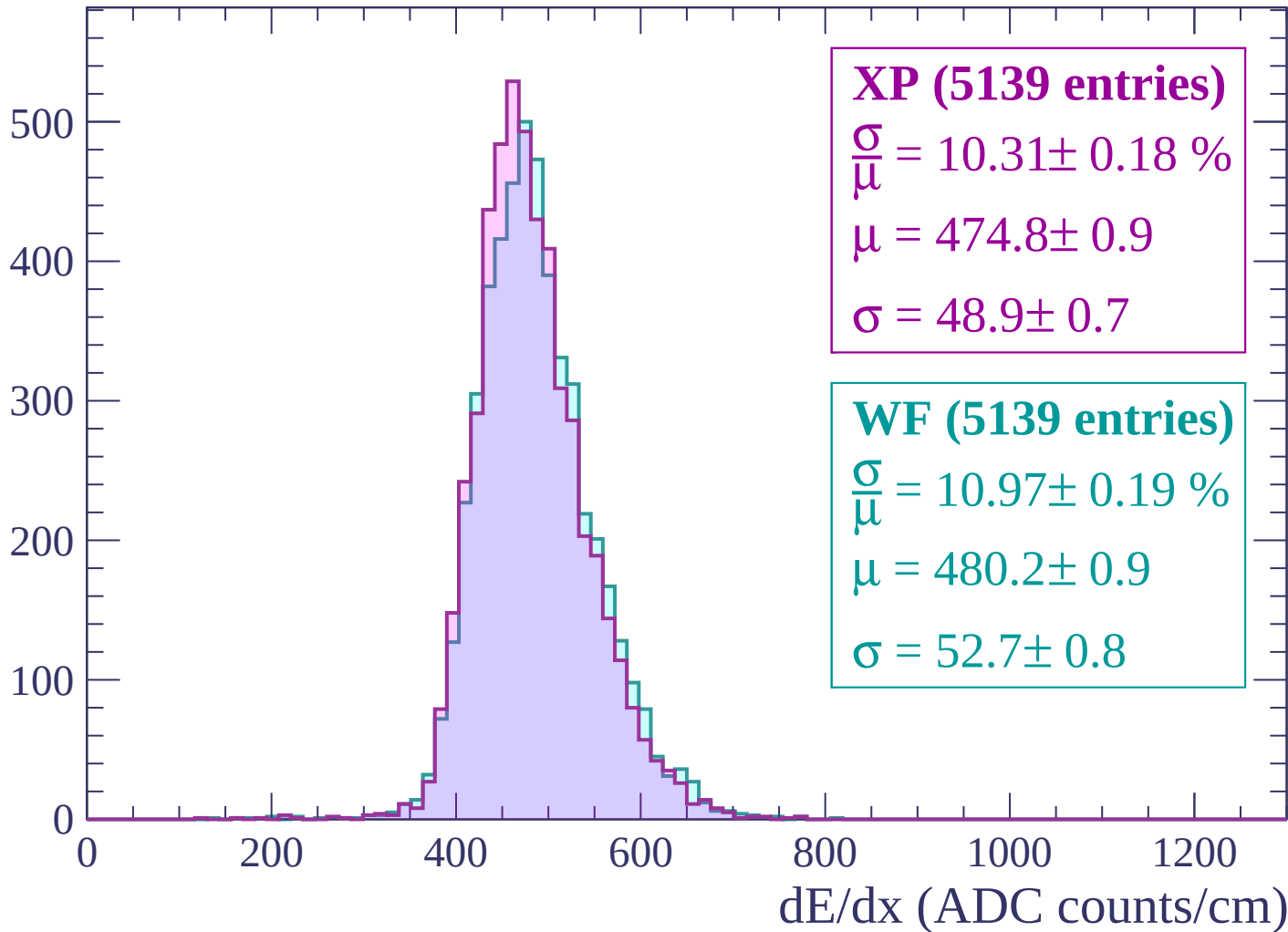
$$\mu = 485.7 \pm 1.0$$

$$\sigma = 54.3 \pm 0.8$$

dE/dx (ADC counts/cm)

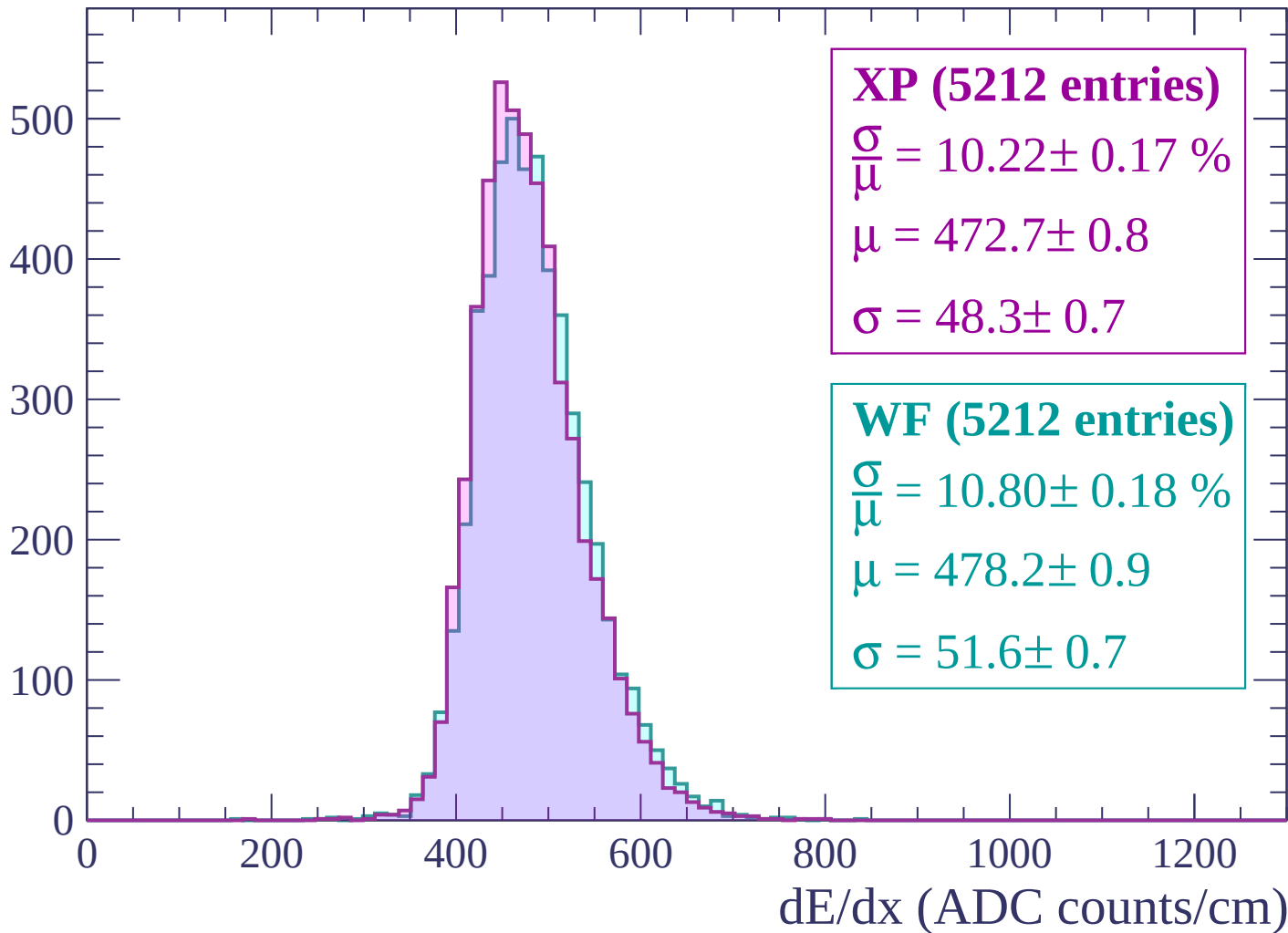
Energy loss | $-1620 < p < -1560$

Count



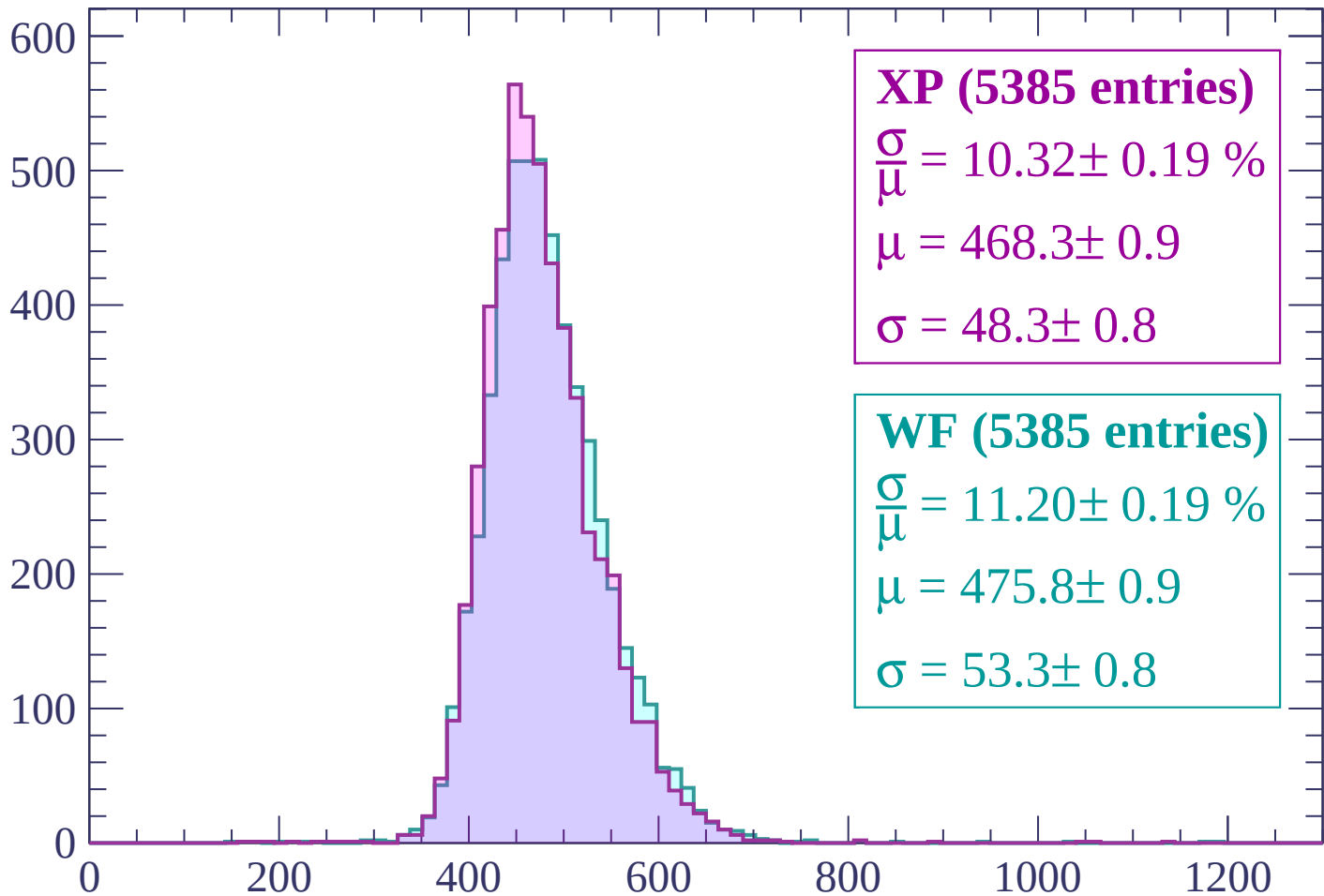
Energy loss | $-1560 < p < -1500$

Count



Energy loss | $-1500 < p < -1440$

Count



XP (5385 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.19 \%$$

$$\mu = 468.3 \pm 0.9$$

$$\sigma = 48.3 \pm 0.8$$

WF (5385 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.19 \%$$

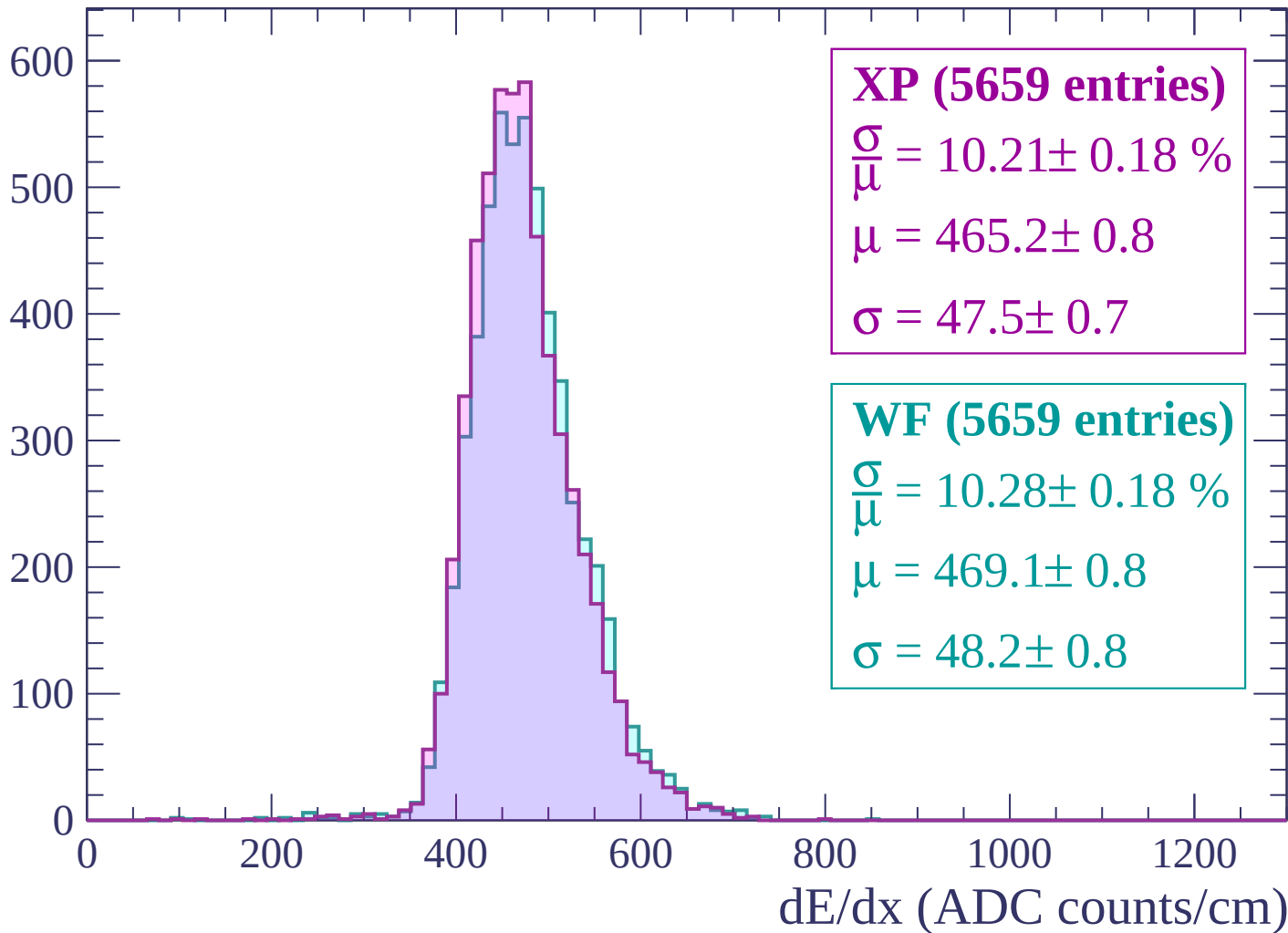
$$\mu = 475.8 \pm 0.9$$

$$\sigma = 53.3 \pm 0.8$$

dE/dx (ADC counts/cm)

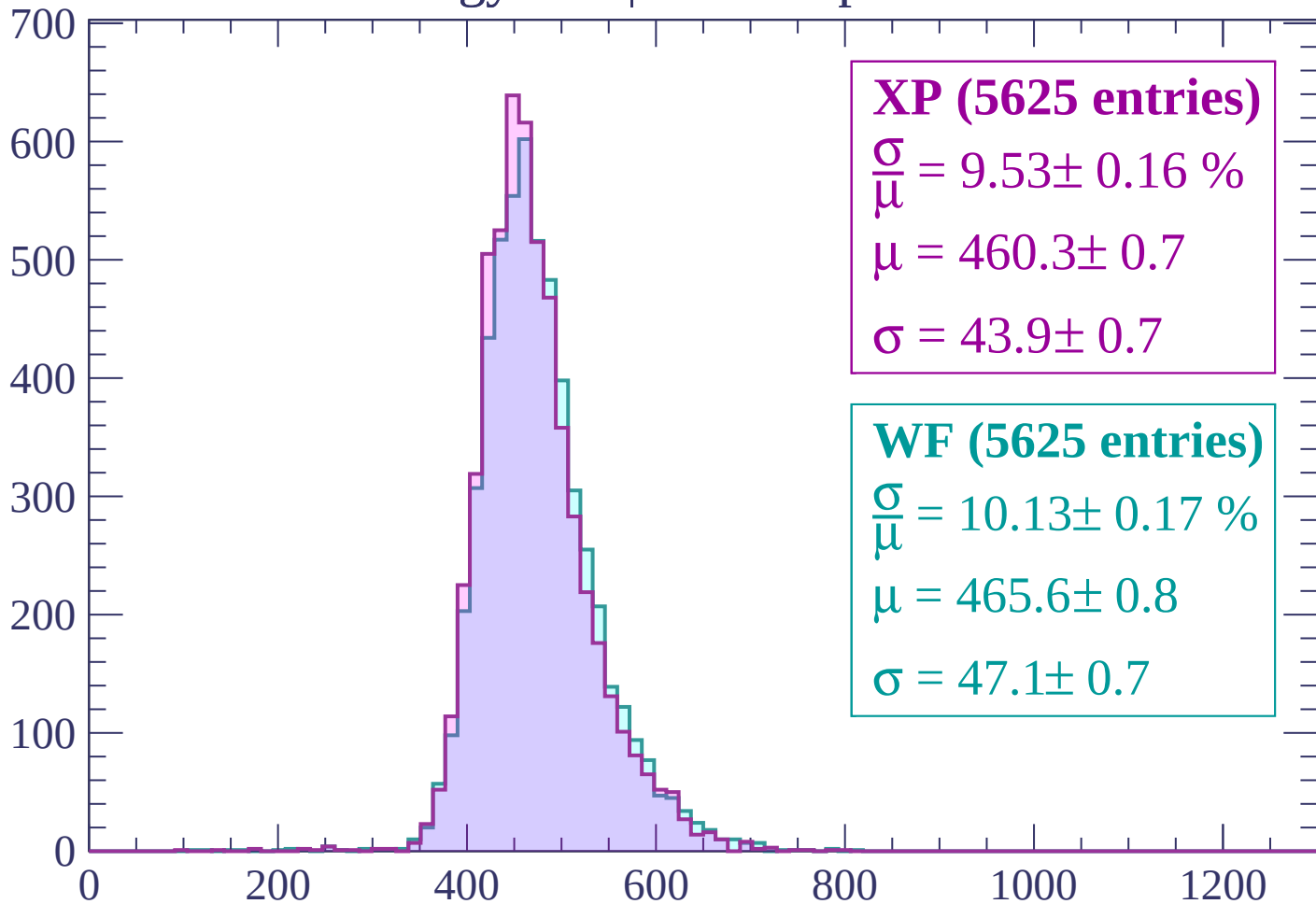
Energy loss | -1440 < p < -1380

Count



Energy loss | $-1380 < p < -1320$

Count



XP (5625 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$$

$$\mu = 460.3 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (5625 entries)

$$\frac{\sigma}{\mu} = 10.13 \pm 0.17 \%$$

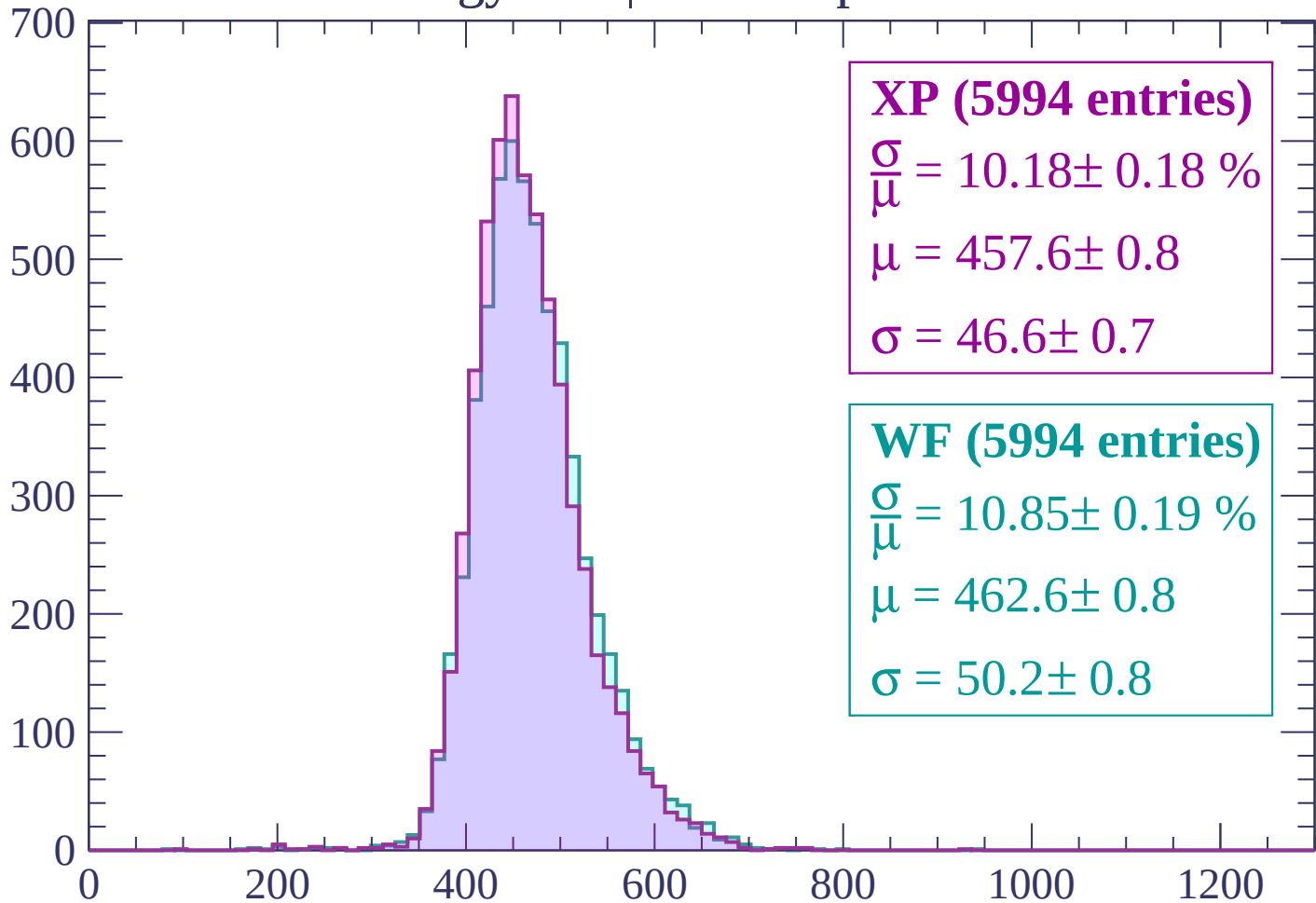
$$\mu = 465.6 \pm 0.8$$

$$\sigma = 47.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count



XP (5994 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.18 \%$$

$$\mu = 457.6 \pm 0.8$$

$$\sigma = 46.6 \pm 0.7$$

WF (5994 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.19 \%$$

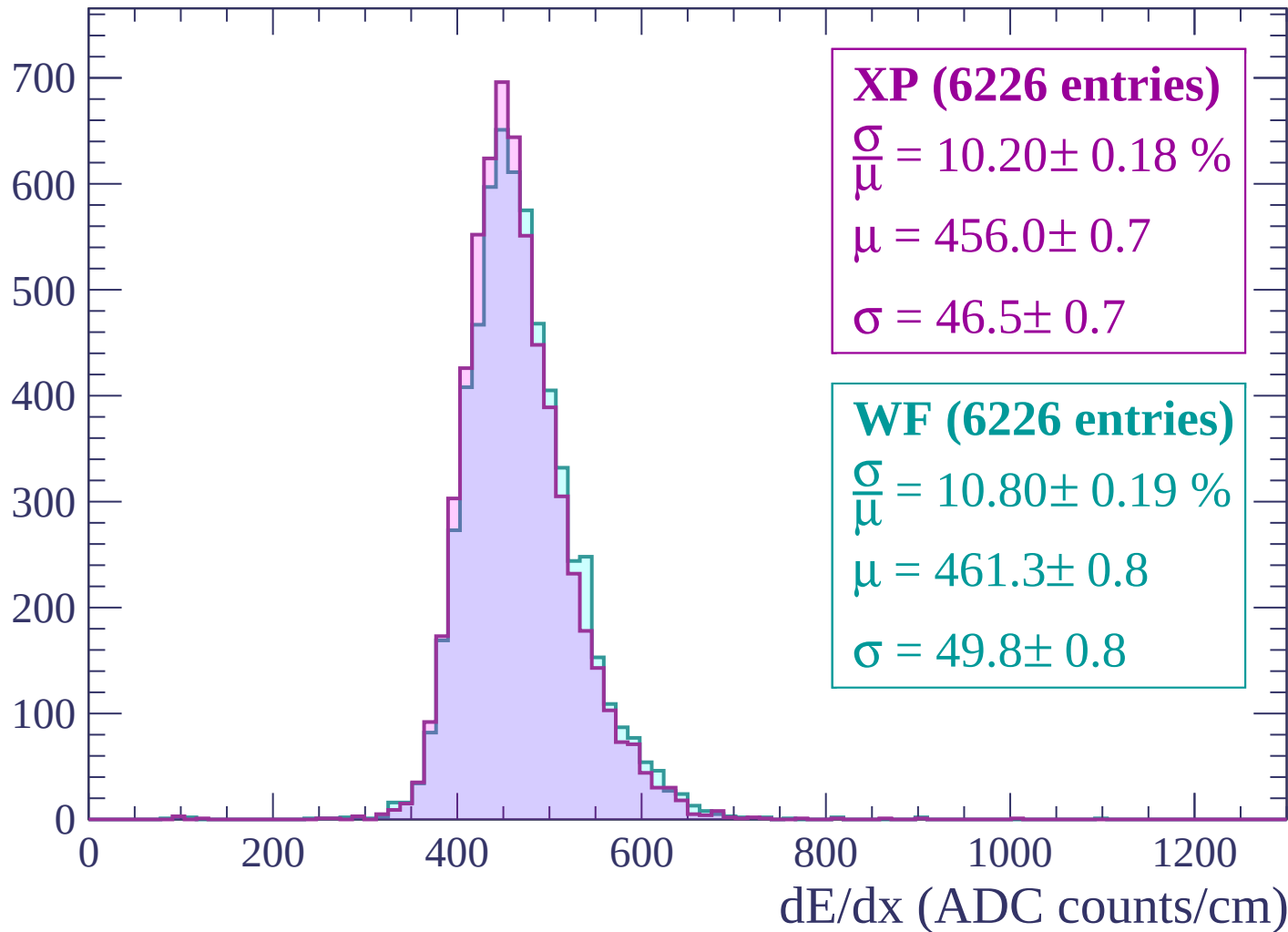
$$\mu = 462.6 \pm 0.8$$

$$\sigma = 50.2 \pm 0.8$$

dE/dx (ADC counts/cm)

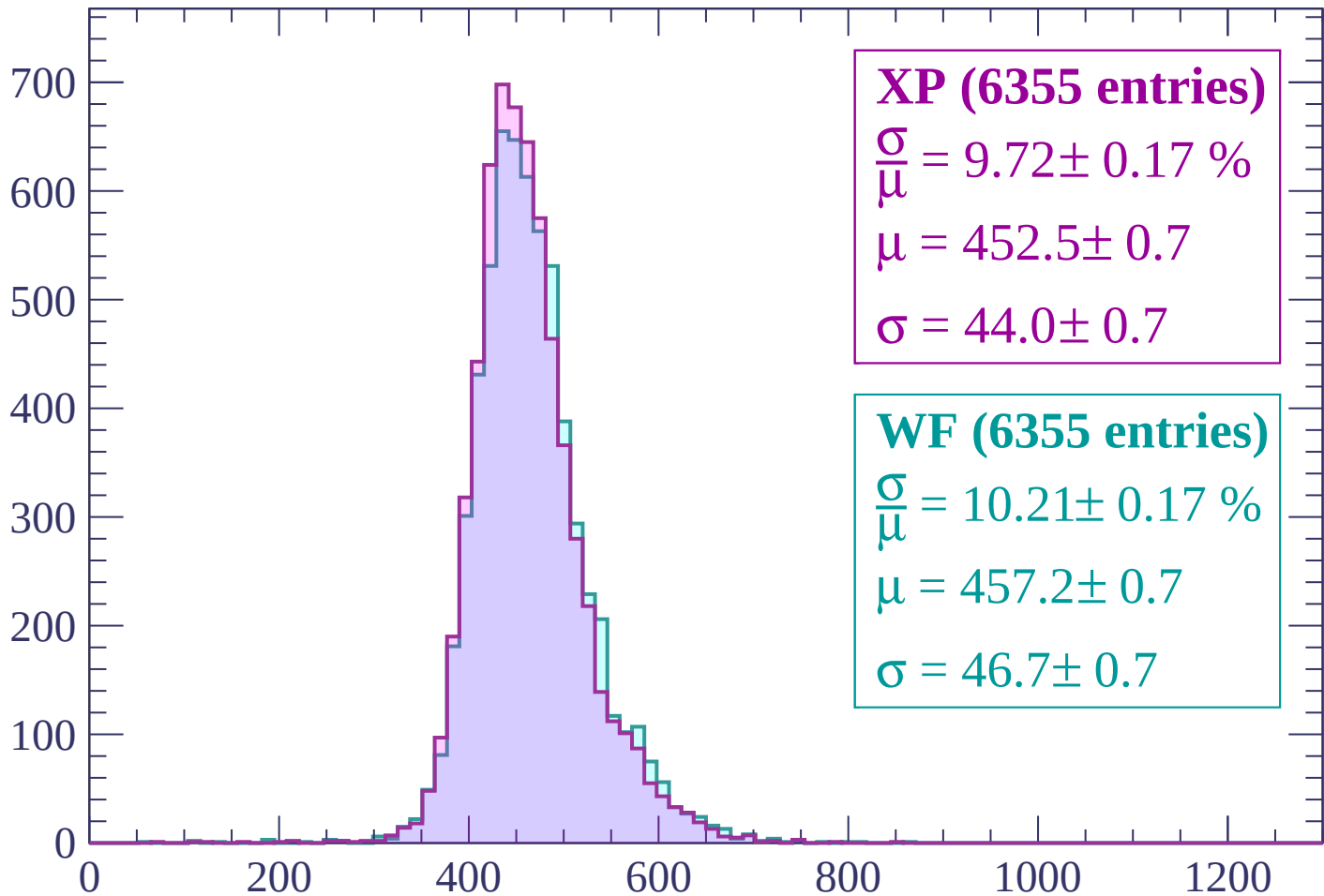
Energy loss | $-1260 < p < -1200$

Count



Energy loss | -1200 < p < -1140

Count



XP (6355 entries)

$$\frac{\sigma}{\mu} = 9.72 \pm 0.17 \%$$

$$\mu = 452.5 \pm 0.7$$

$$\sigma = 44.0 \pm 0.7$$

WF (6355 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.17 \%$$

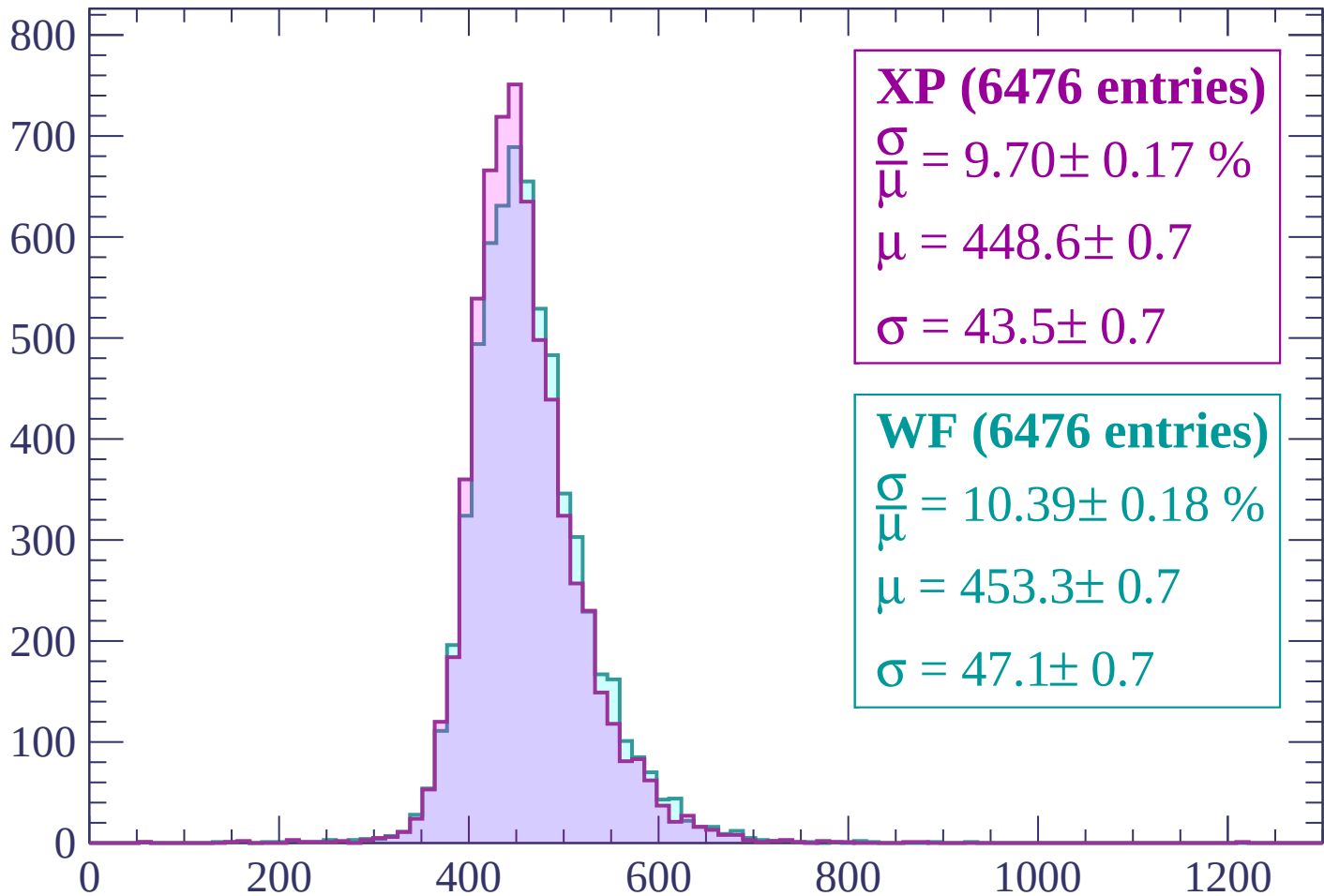
$$\mu = 457.2 \pm 0.7$$

$$\sigma = 46.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (6476 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.17 \%$$

$$\mu = 448.6 \pm 0.7$$

$$\sigma = 43.5 \pm 0.7$$

WF (6476 entries)

$$\frac{\sigma}{\mu} = 10.39 \pm 0.18 \%$$

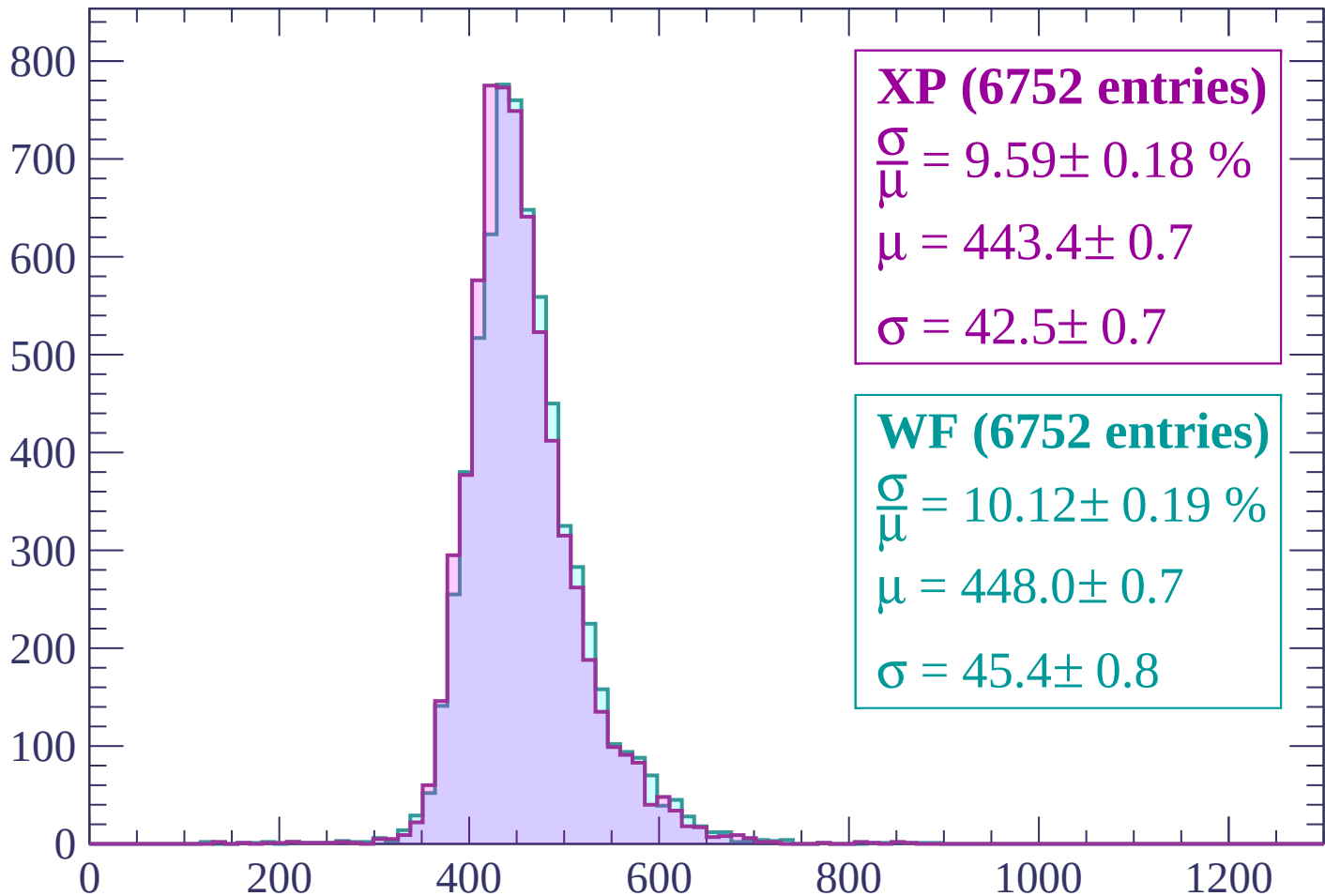
$$\mu = 453.3 \pm 0.7$$

$$\sigma = 47.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (6752 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.18 \%$$

$$\mu = 443.4 \pm 0.7$$

$$\sigma = 42.5 \pm 0.7$$

WF (6752 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.19 \%$$

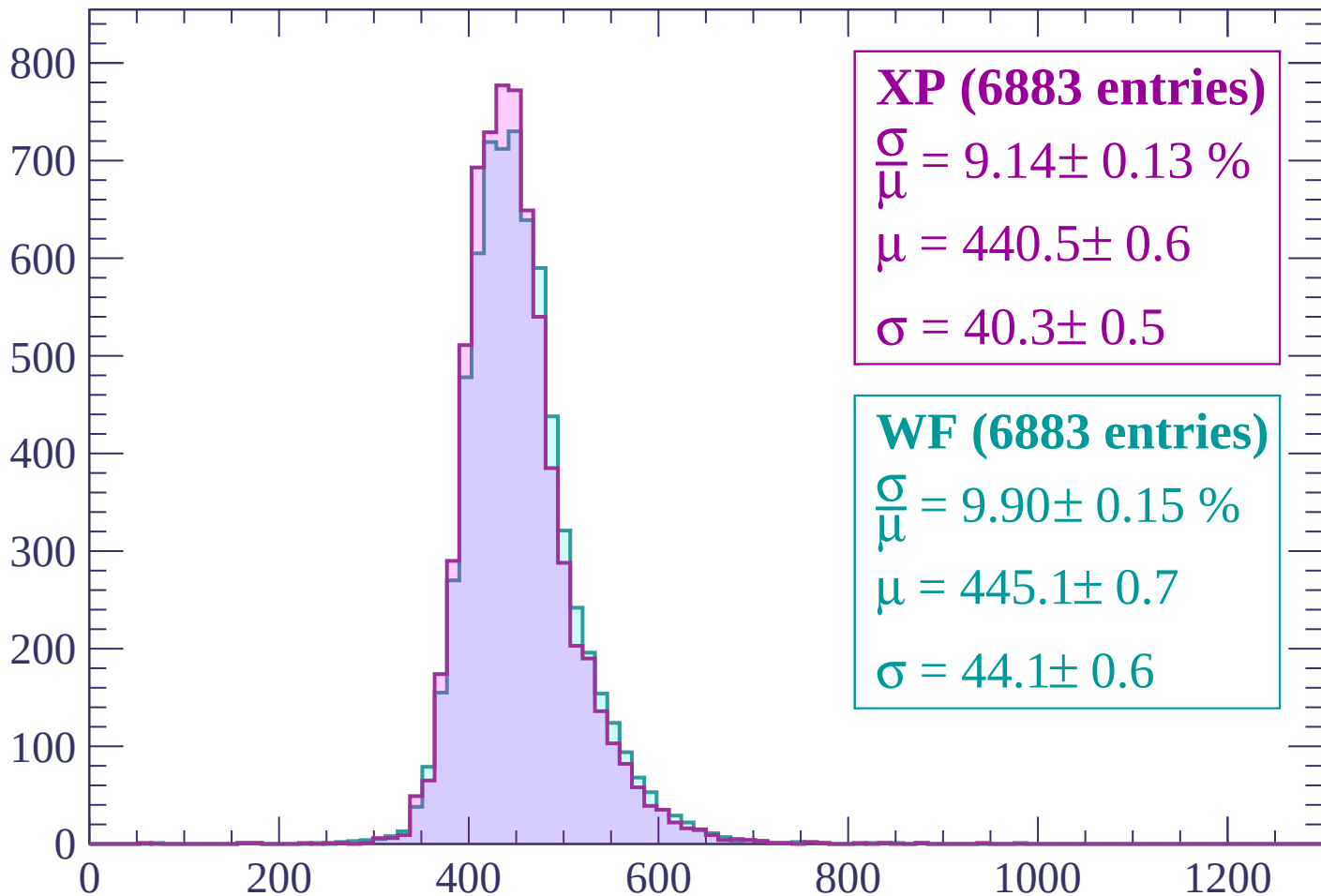
$$\mu = 448.0 \pm 0.7$$

$$\sigma = 45.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6883 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.13 \%$$

$$\mu = 440.5 \pm 0.6$$

$$\sigma = 40.3 \pm 0.5$$

WF (6883 entries)

$$\frac{\sigma}{\mu} = 9.90 \pm 0.15 \%$$

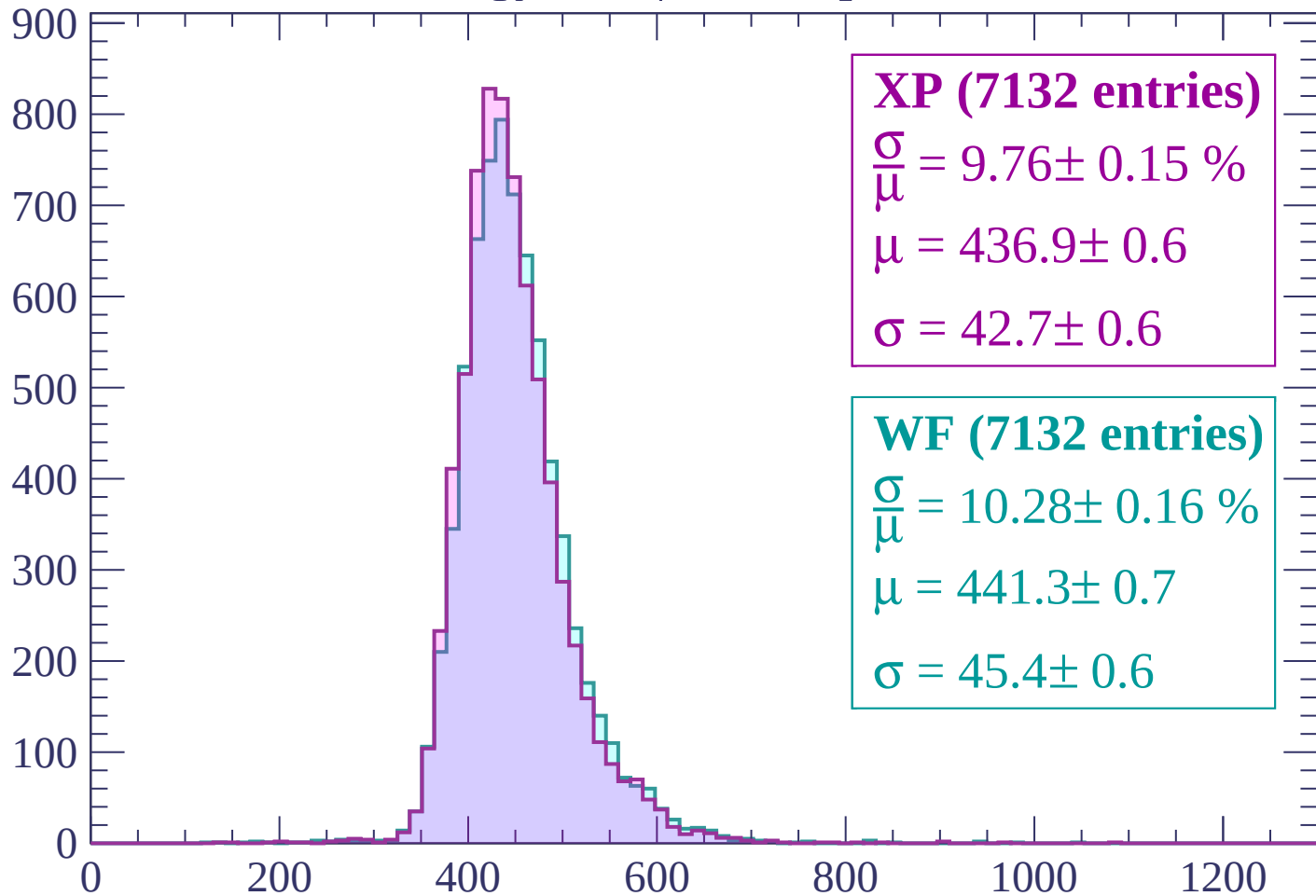
$$\mu = 445.1 \pm 0.7$$

$$\sigma = 44.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (7132 entries)

$$\frac{\sigma}{\mu} = 9.76 \pm 0.15 \%$$

$$\mu = 436.9 \pm 0.6$$

$$\sigma = 42.7 \pm 0.6$$

WF (7132 entries)

$$\frac{\sigma}{\mu} = 10.28 \pm 0.16 \%$$

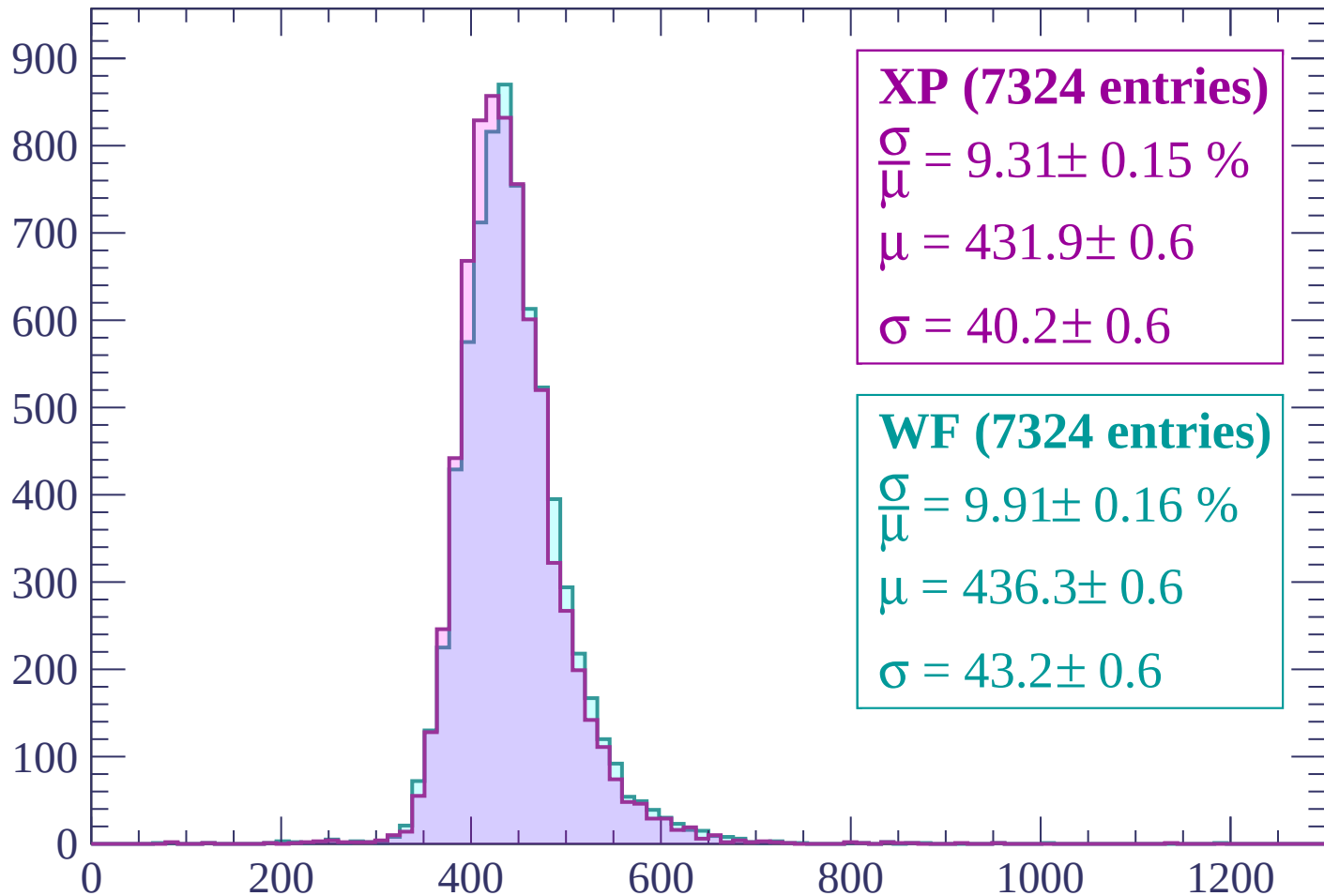
$$\mu = 441.3 \pm 0.7$$

$$\sigma = 45.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7324 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.15 \%$$

$$\mu = 431.9 \pm 0.6$$

$$\sigma = 40.2 \pm 0.6$$

WF (7324 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

$$\mu = 436.3 \pm 0.6$$

$$\sigma = 43.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1000

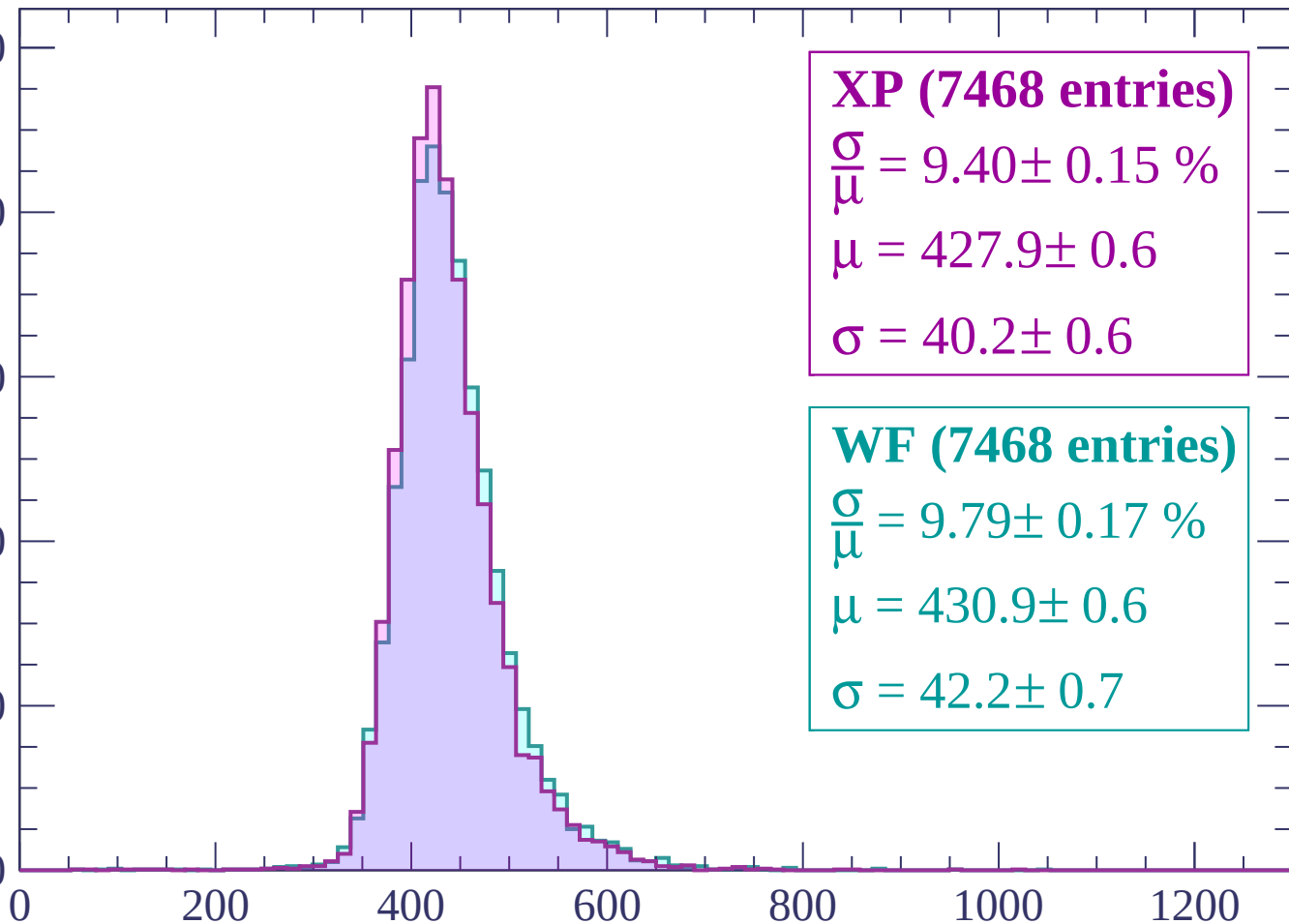
800

600

400

200

0



XP (7468 entries)

$\frac{\sigma}{\mu} = 9.40 \pm 0.15 \%$

$\mu = 427.9 \pm 0.6$

$\sigma = 40.2 \pm 0.6$

WF (7468 entries)

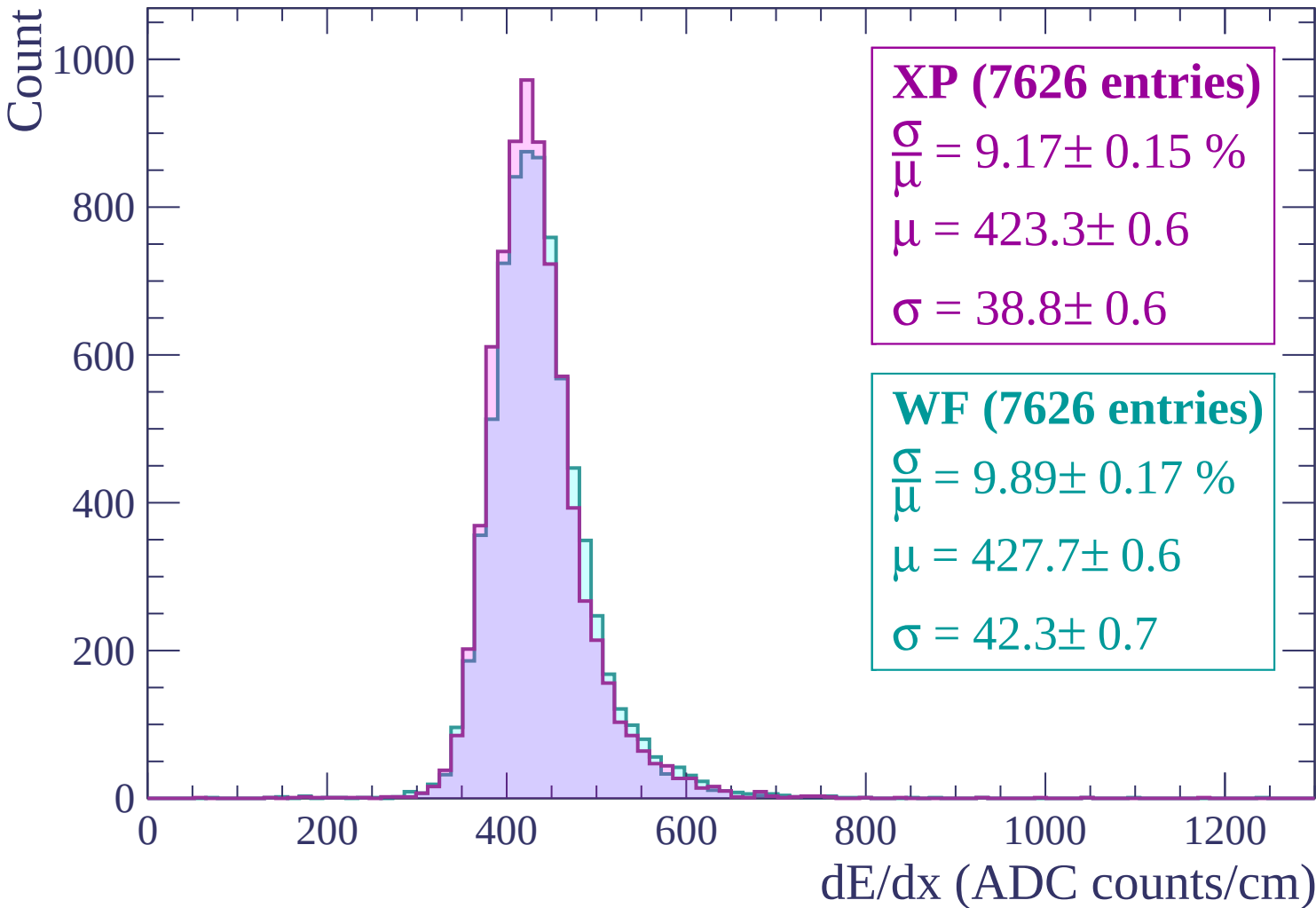
$\frac{\sigma}{\mu} = 9.79 \pm 0.17 \%$

$\mu = 430.9 \pm 0.6$

$\sigma = 42.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$



Energy loss | $-720 < p < -660$

Count

1000

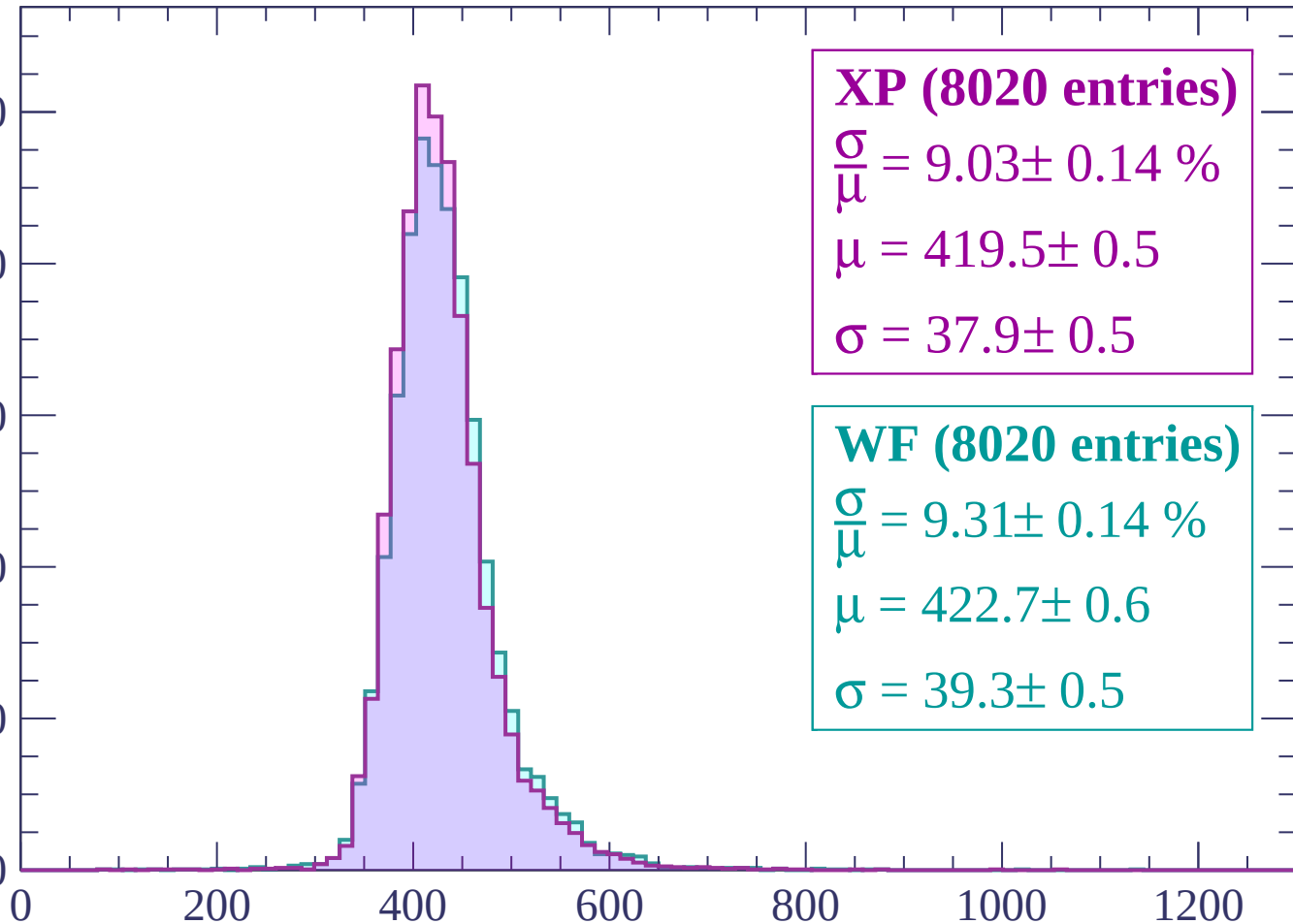
800

600

400

200

0



XP (8020 entries)

$\frac{\sigma}{\mu} = 9.03 \pm 0.14 \%$

$\mu = 419.5 \pm 0.5$

$\sigma = 37.9 \pm 0.5$

WF (8020 entries)

$\frac{\sigma}{\mu} = 9.31 \pm 0.14 \%$

$\mu = 422.7 \pm 0.6$

$\sigma = 39.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8080 entries)

$\frac{\sigma}{\mu} = 9.03 \pm 0.12 \%$

$\mu = 415.6 \pm 0.5$

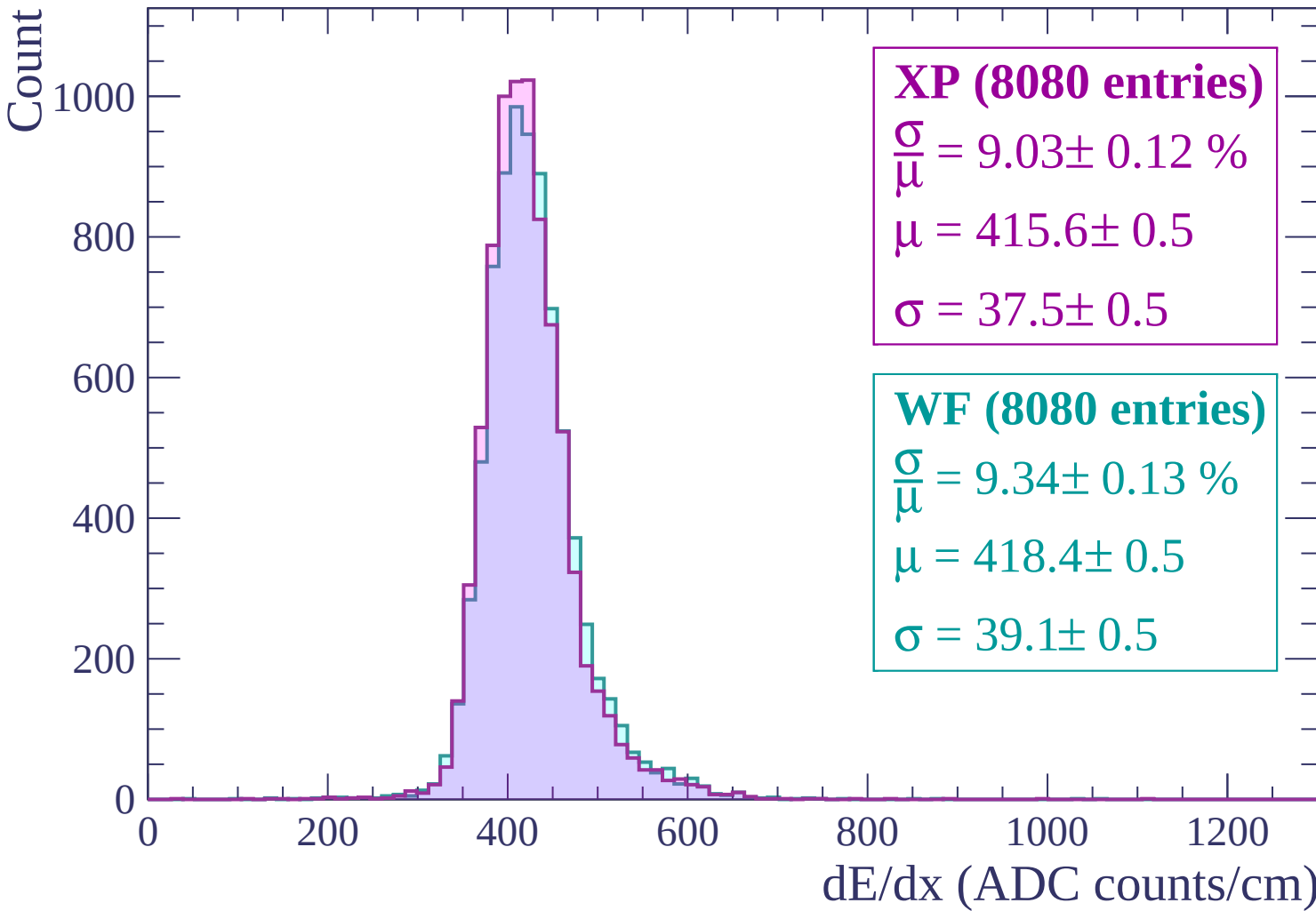
$\sigma = 37.5 \pm 0.5$

WF (8080 entries)

$\frac{\sigma}{\mu} = 9.34 \pm 0.13 \%$

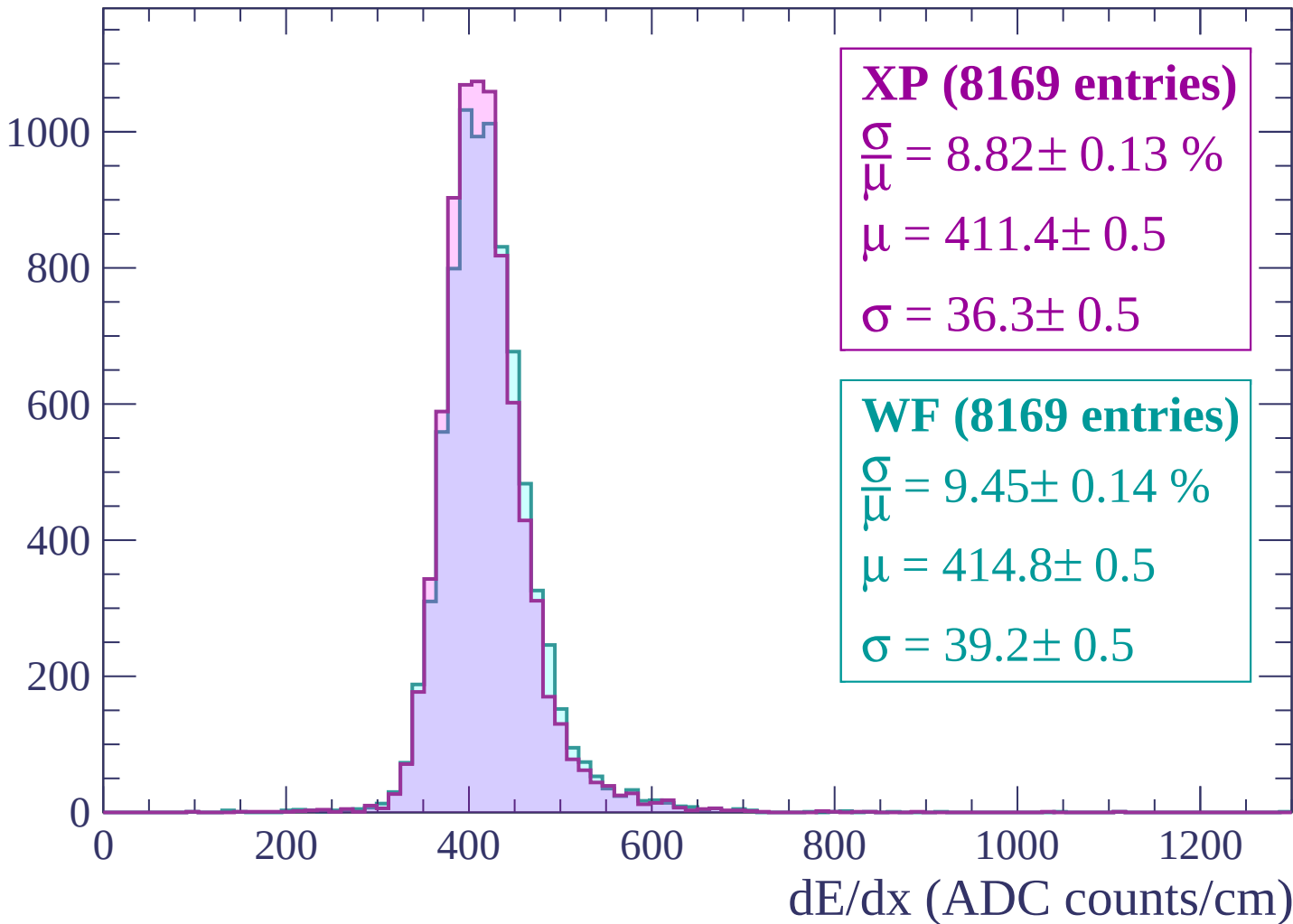
$\mu = 418.4 \pm 0.5$

$\sigma = 39.1 \pm 0.5$



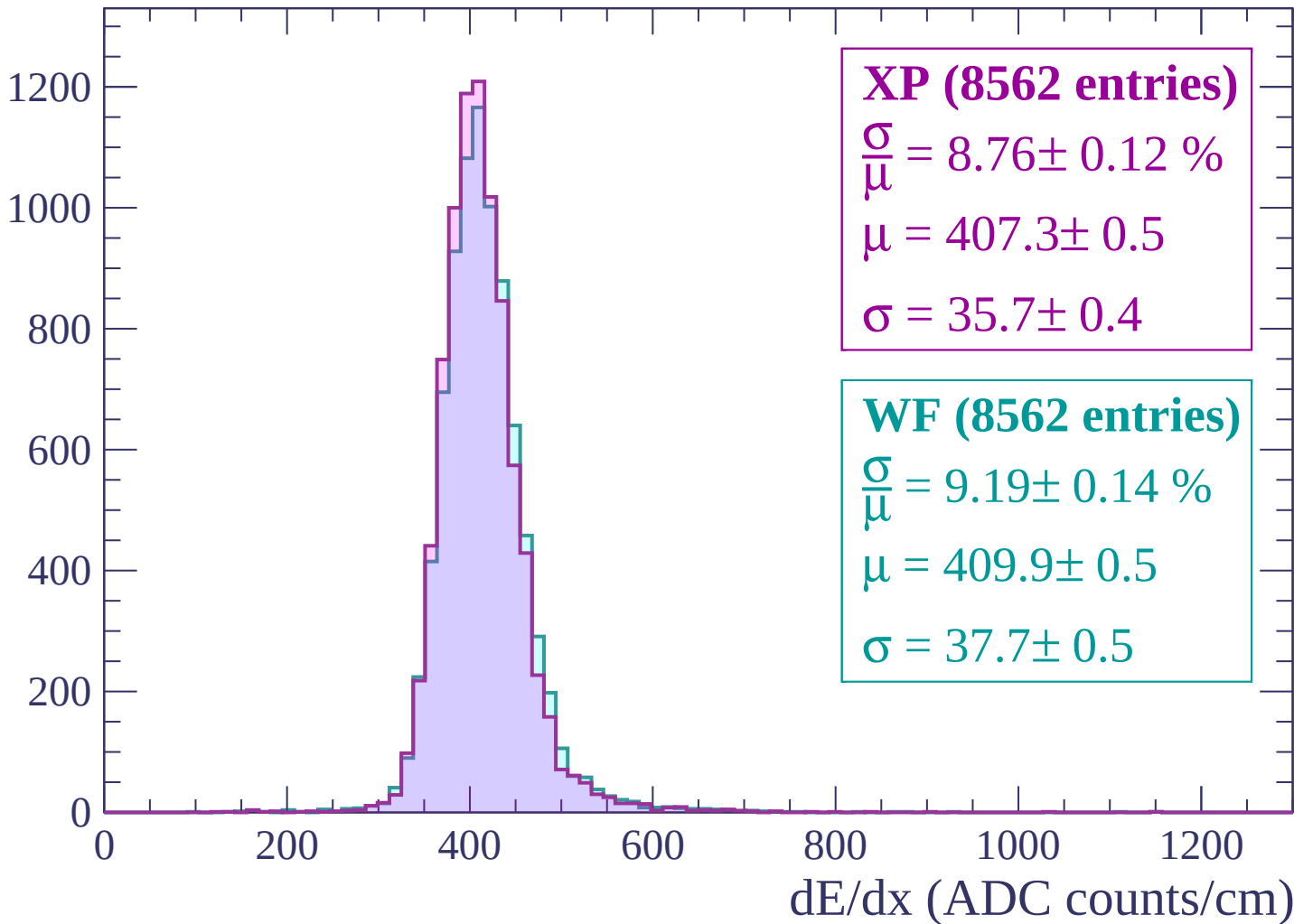
Energy loss | $-600 < p < -540$

Count

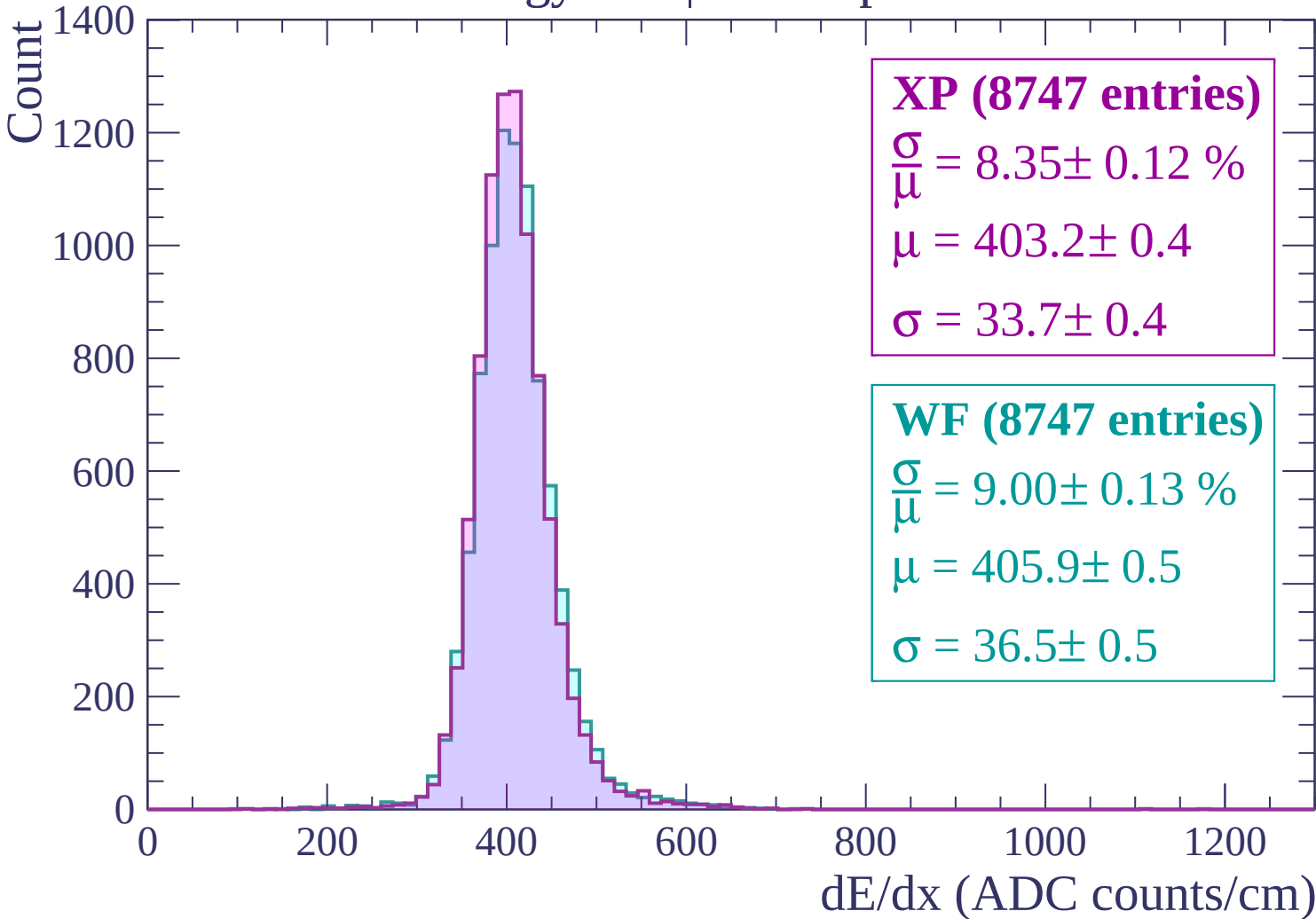


Energy loss | $-540 < p < -480$

Count

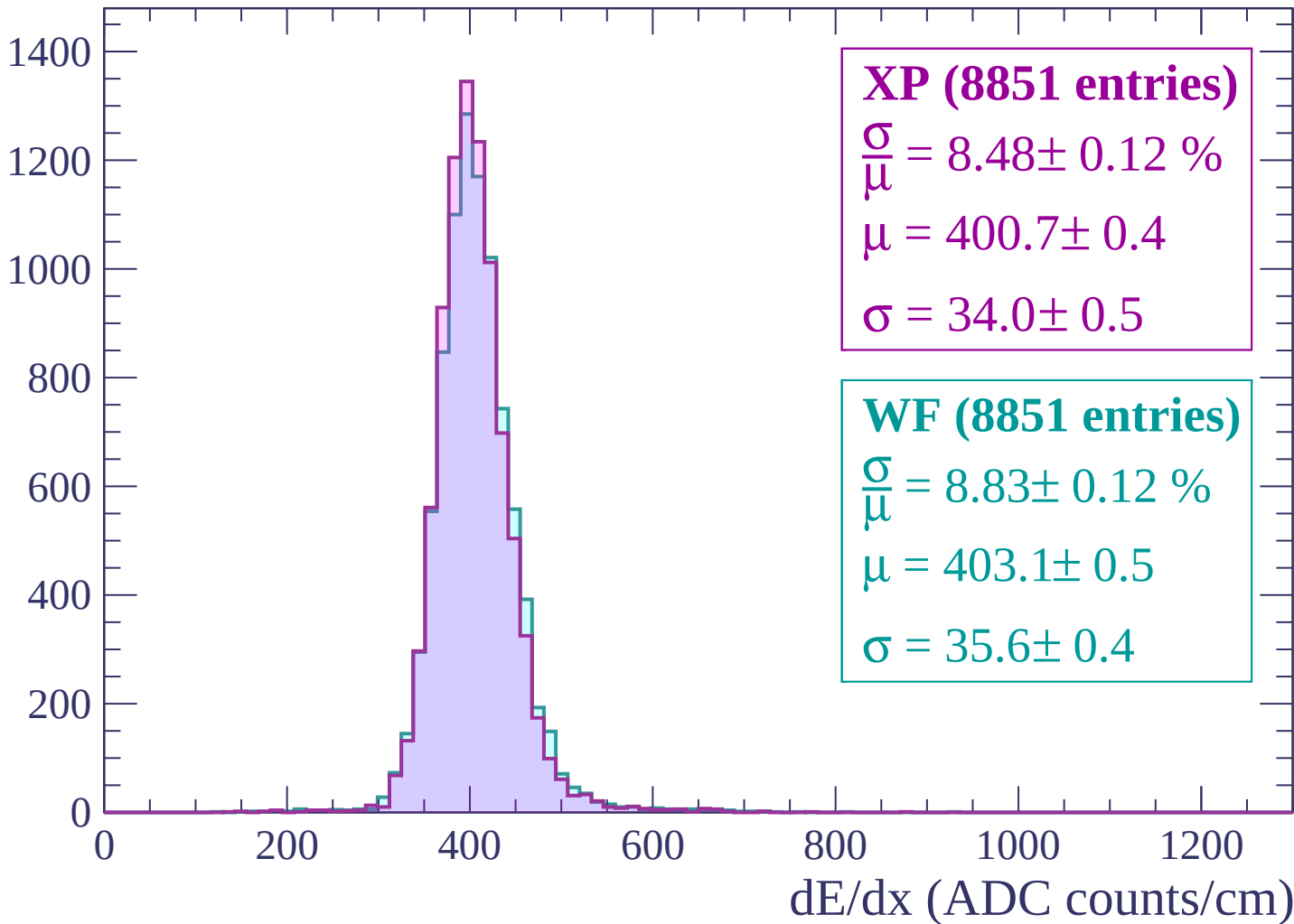


Energy loss | $-480 < p < -420$



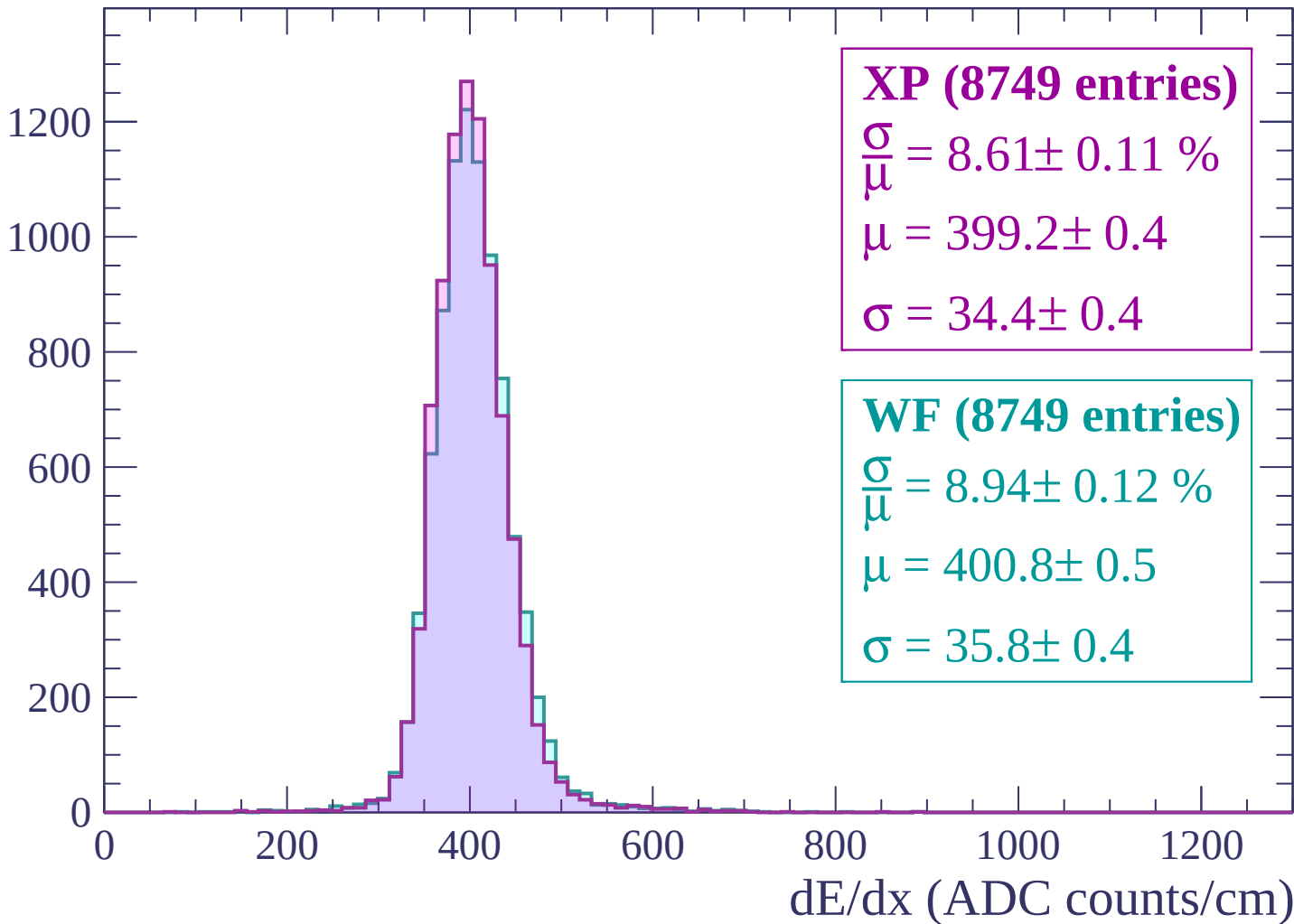
Energy loss | $-420 < p < -360$

Count



Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count

1200

1000

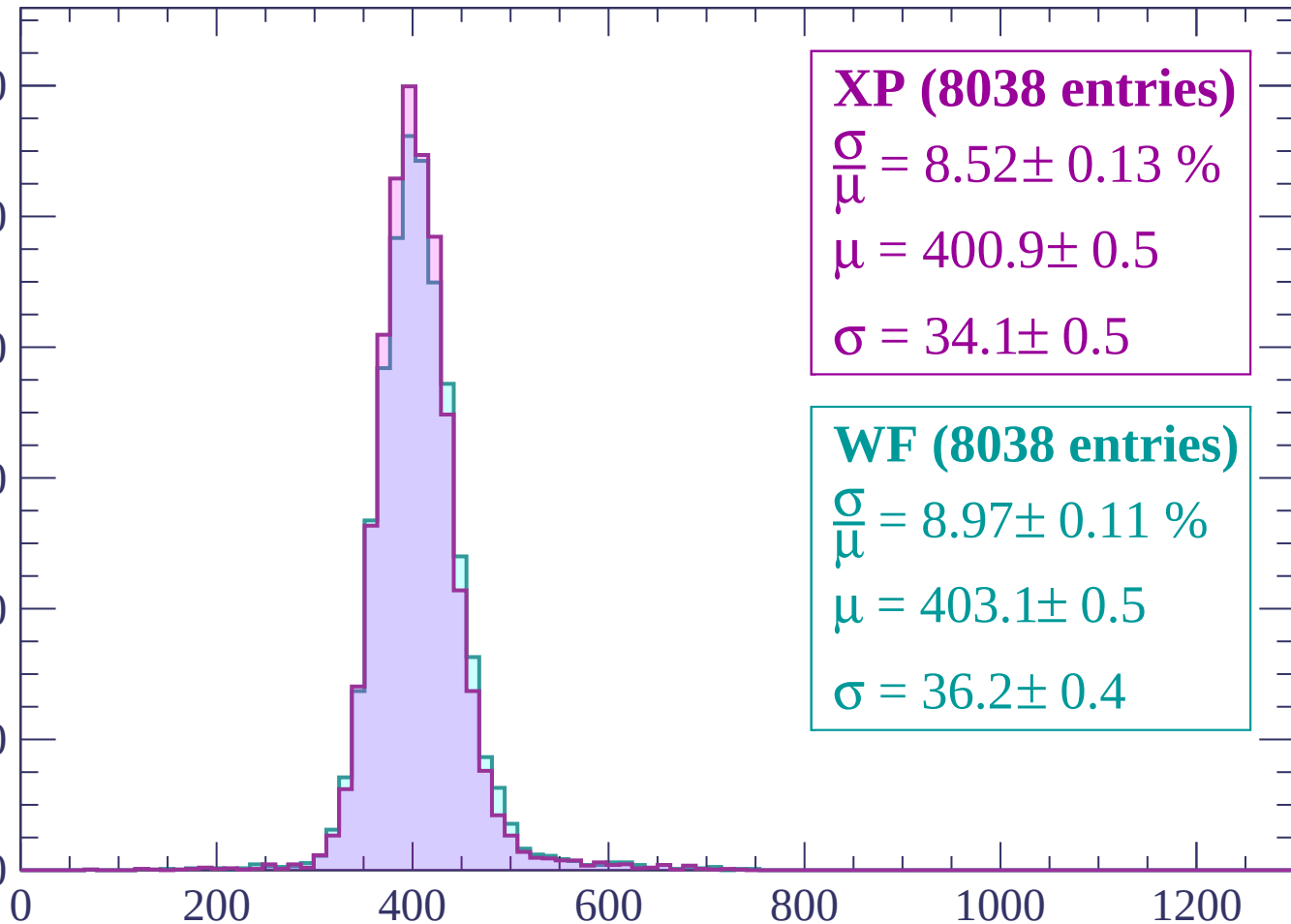
800

600

400

200

0



XP (8038 entries)

$\frac{\sigma}{\mu} = 8.52 \pm 0.13 \%$

$\mu = 400.9 \pm 0.5$

$\sigma = 34.1 \pm 0.5$

WF (8038 entries)

$\frac{\sigma}{\mu} = 8.97 \pm 0.11 \%$

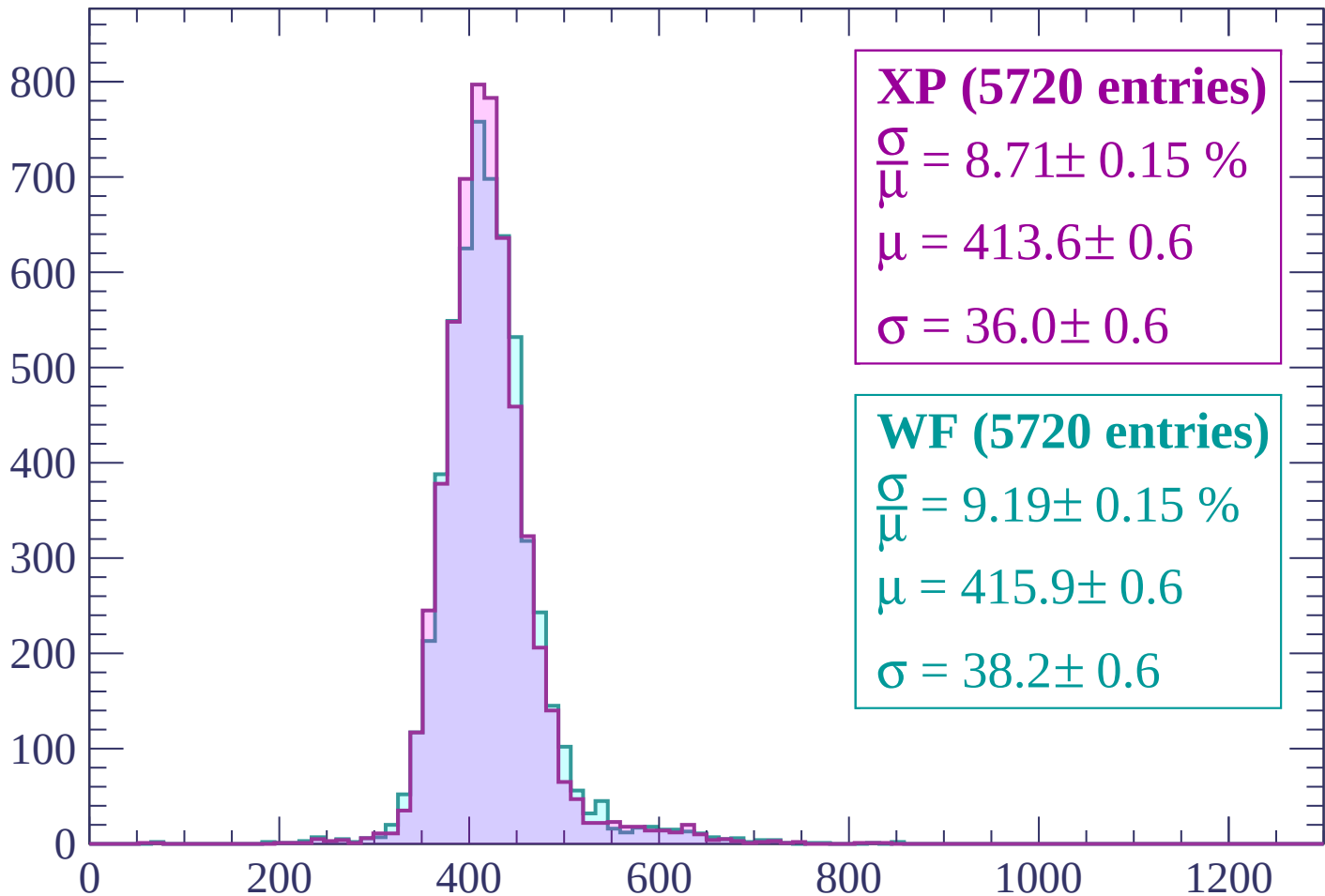
$\mu = 403.1 \pm 0.5$

$\sigma = 36.2 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (5720 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.15 \%$$

$$\mu = 413.6 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF (5720 entries)

$$\frac{\sigma}{\mu} = 9.19 \pm 0.15 \%$$

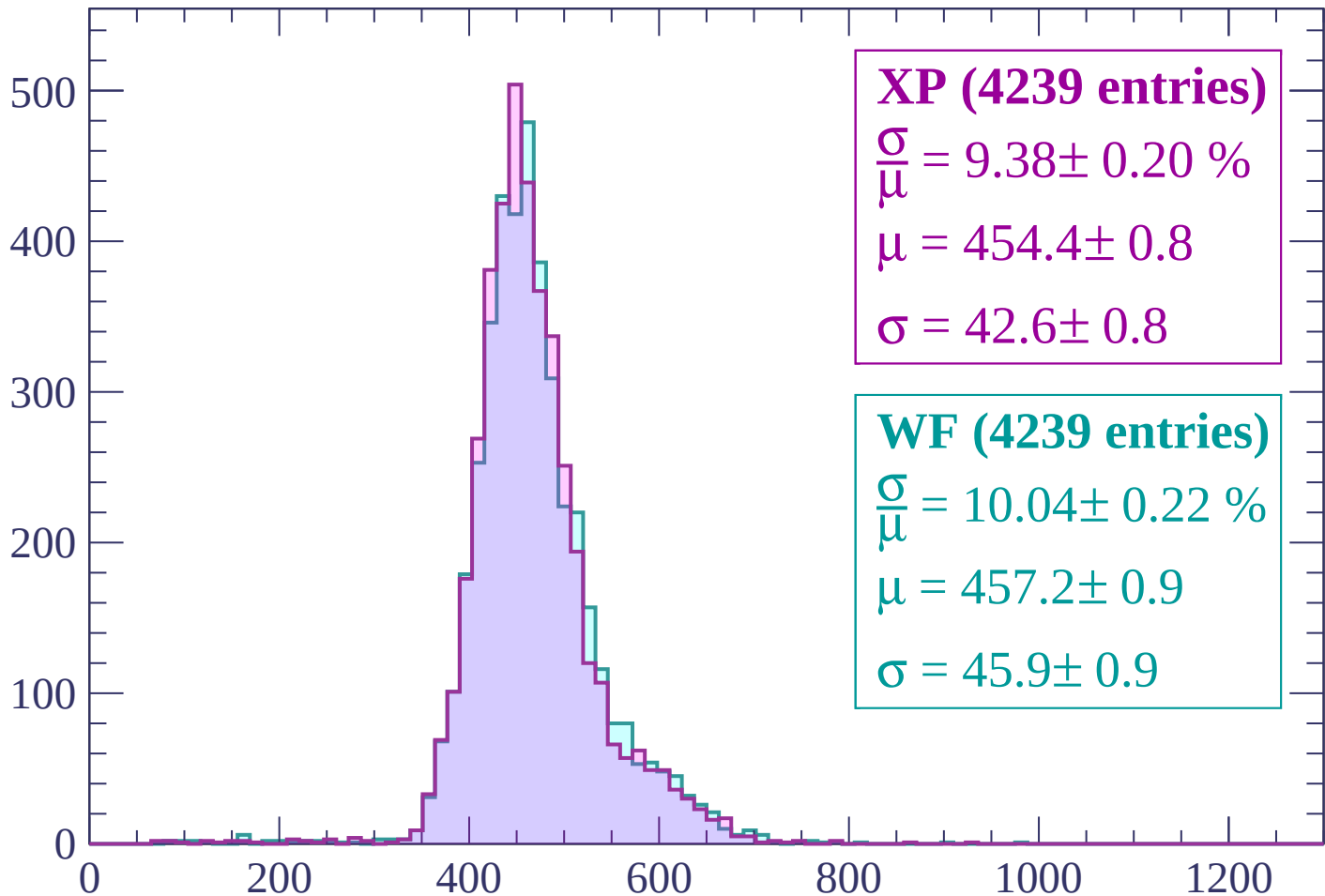
$$\mu = 415.9 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (4239 entries)

$$\frac{\sigma}{\mu} = 9.38 \pm 0.20 \%$$

$$\mu = 454.4 \pm 0.8$$

$$\sigma = 42.6 \pm 0.8$$

WF (4239 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.22 \%$$

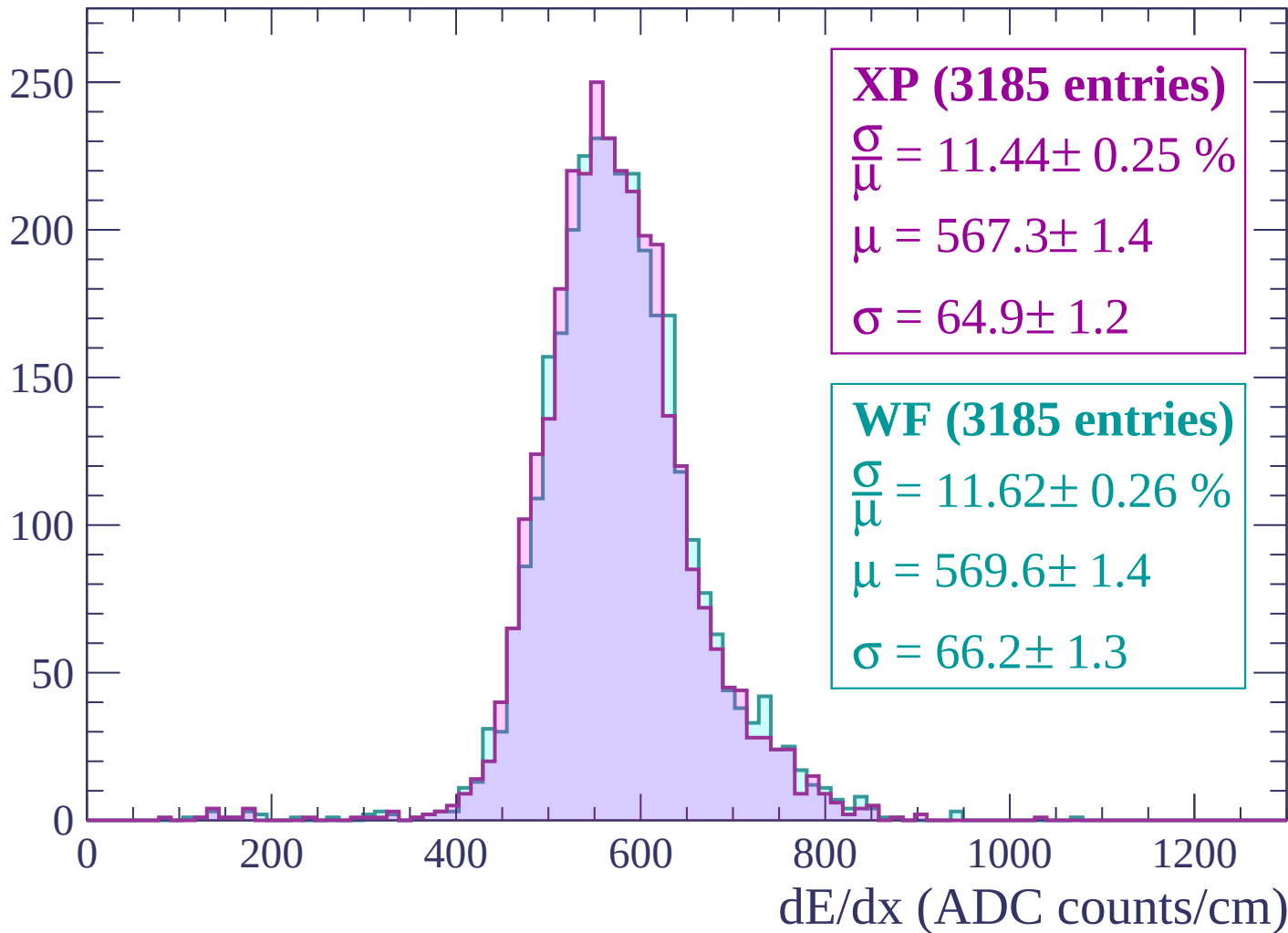
$$\mu = 457.2 \pm 0.9$$

$$\sigma = 45.9 \pm 0.9$$

dE/dx (ADC counts/cm)

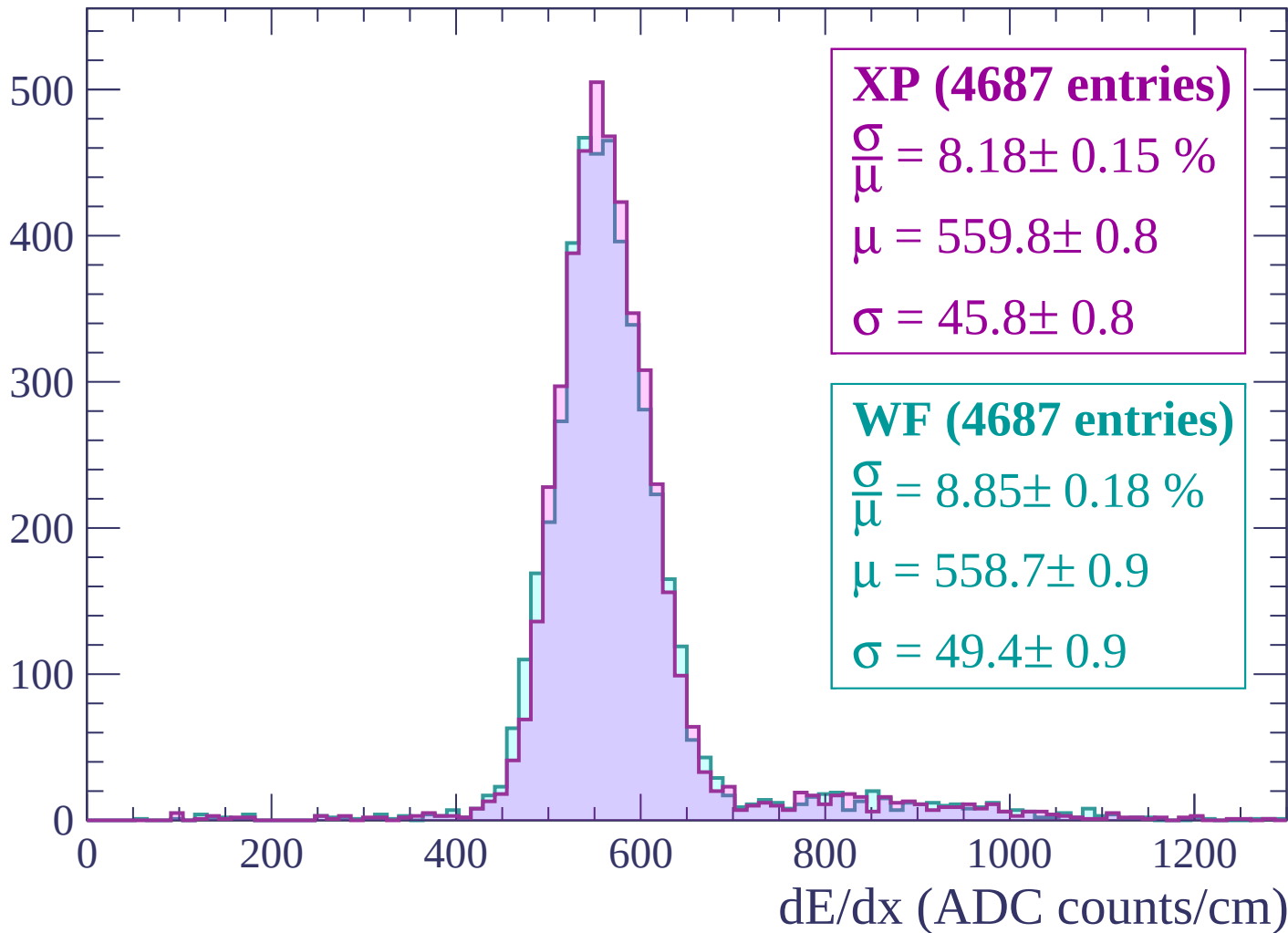
Energy loss | $-120 < p < -60$

Count



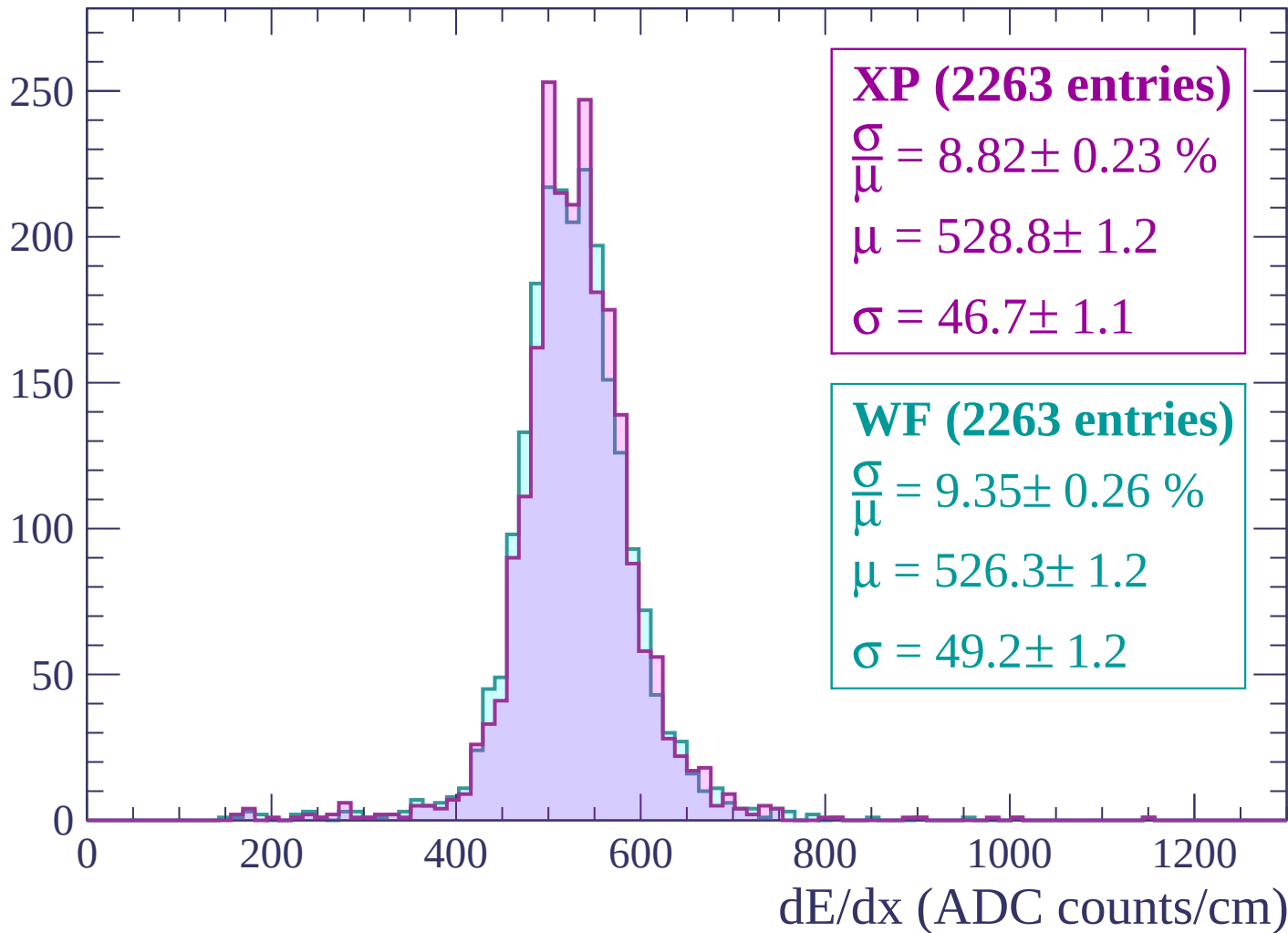
Energy loss | $-60 < p < 0$

Count



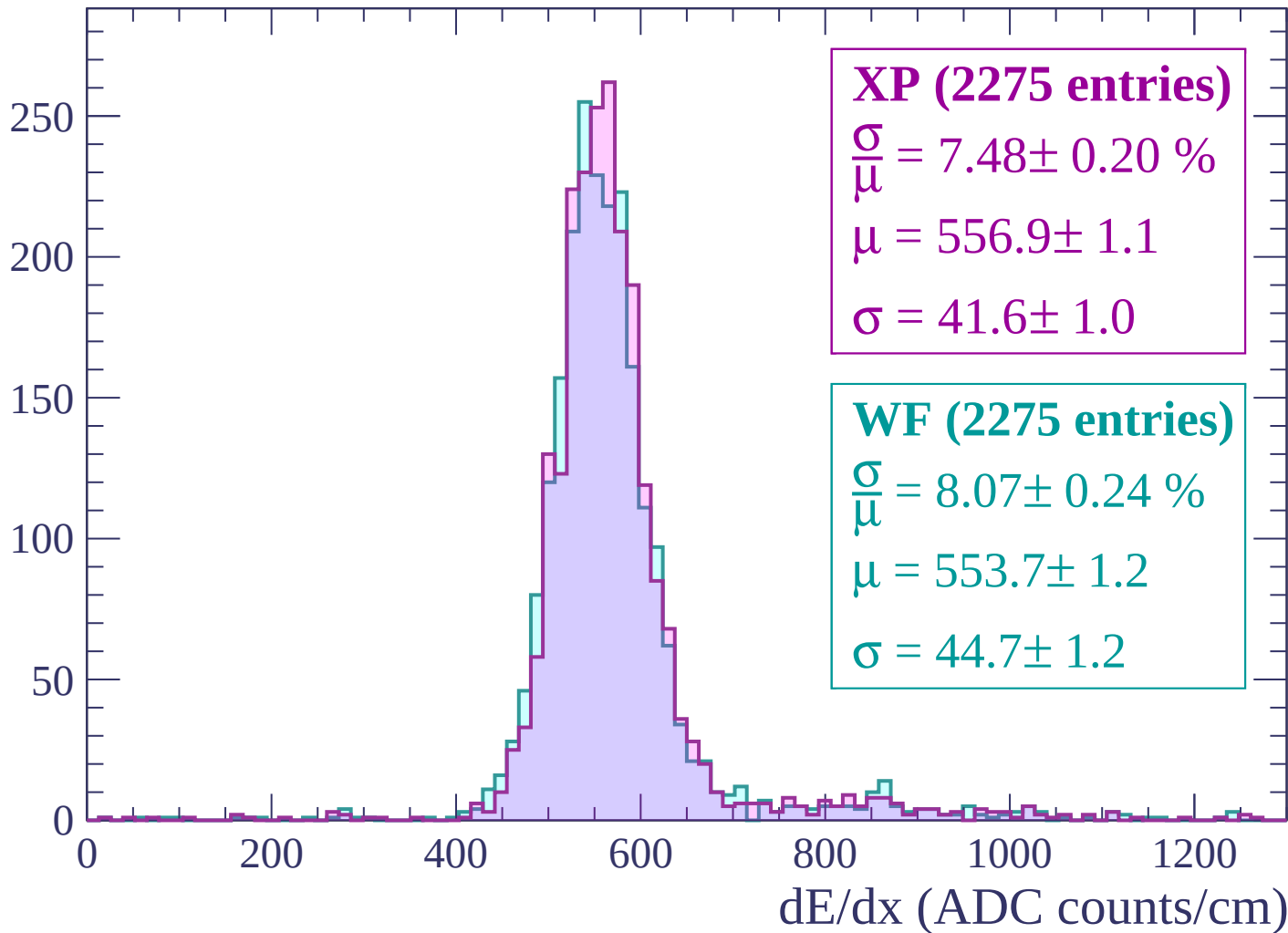
Energy loss | $0 < p < 60$

Count



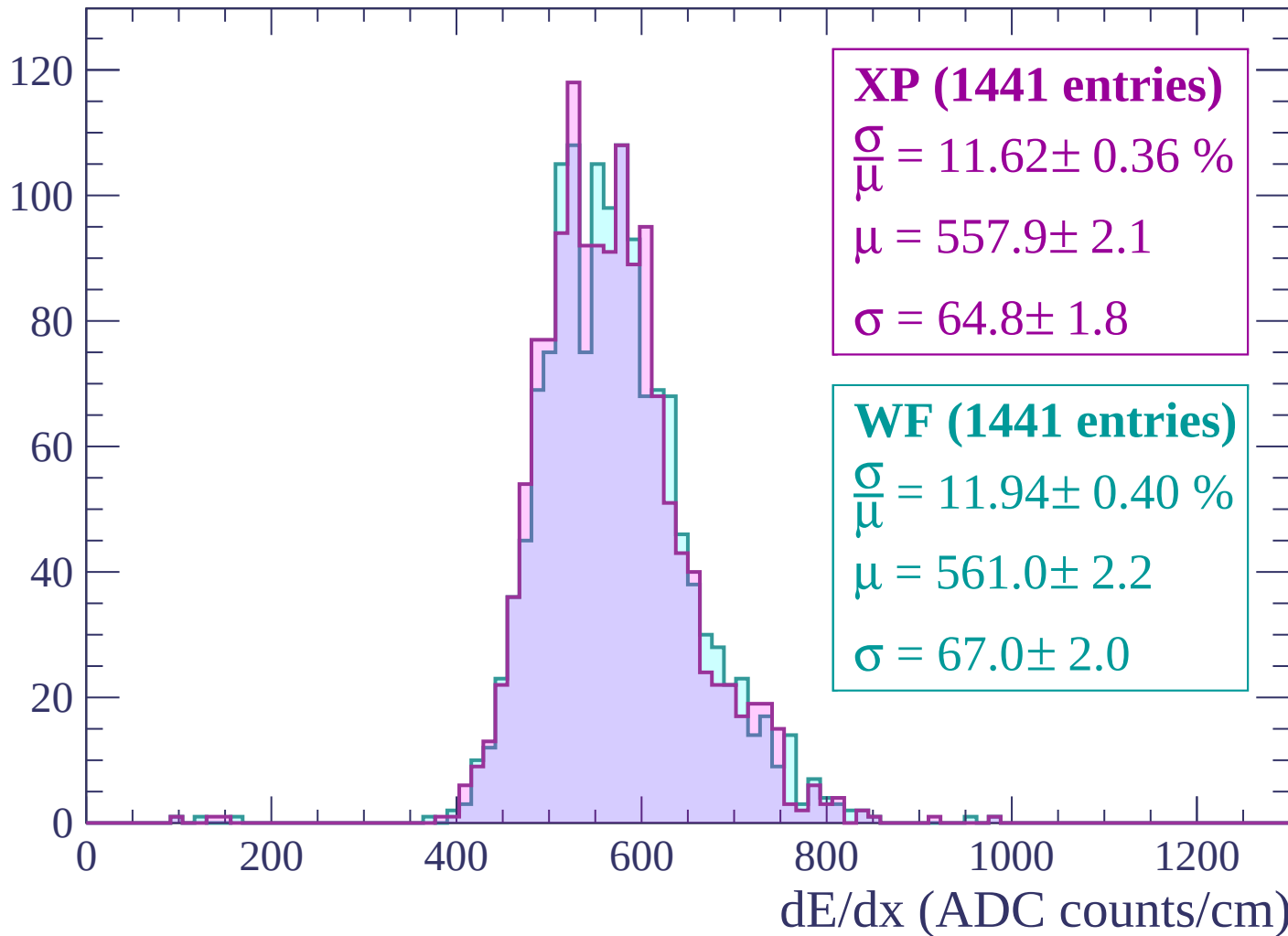
Energy loss | $60 < p < 120$

Count



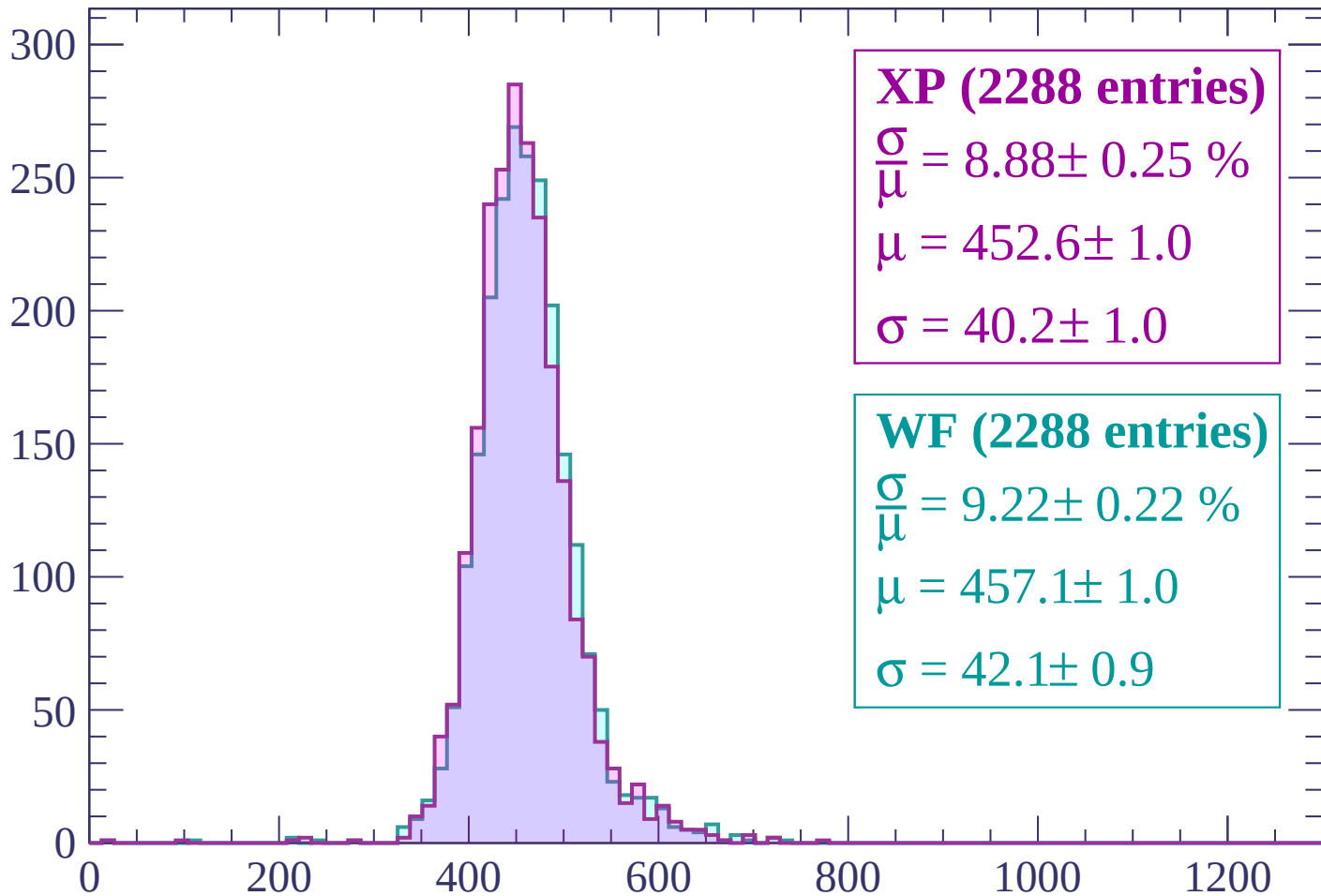
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (2288 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.25 \%$

$\mu = 452.6 \pm 1.0$

$\sigma = 40.2 \pm 1.0$

WF (2288 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.22 \%$

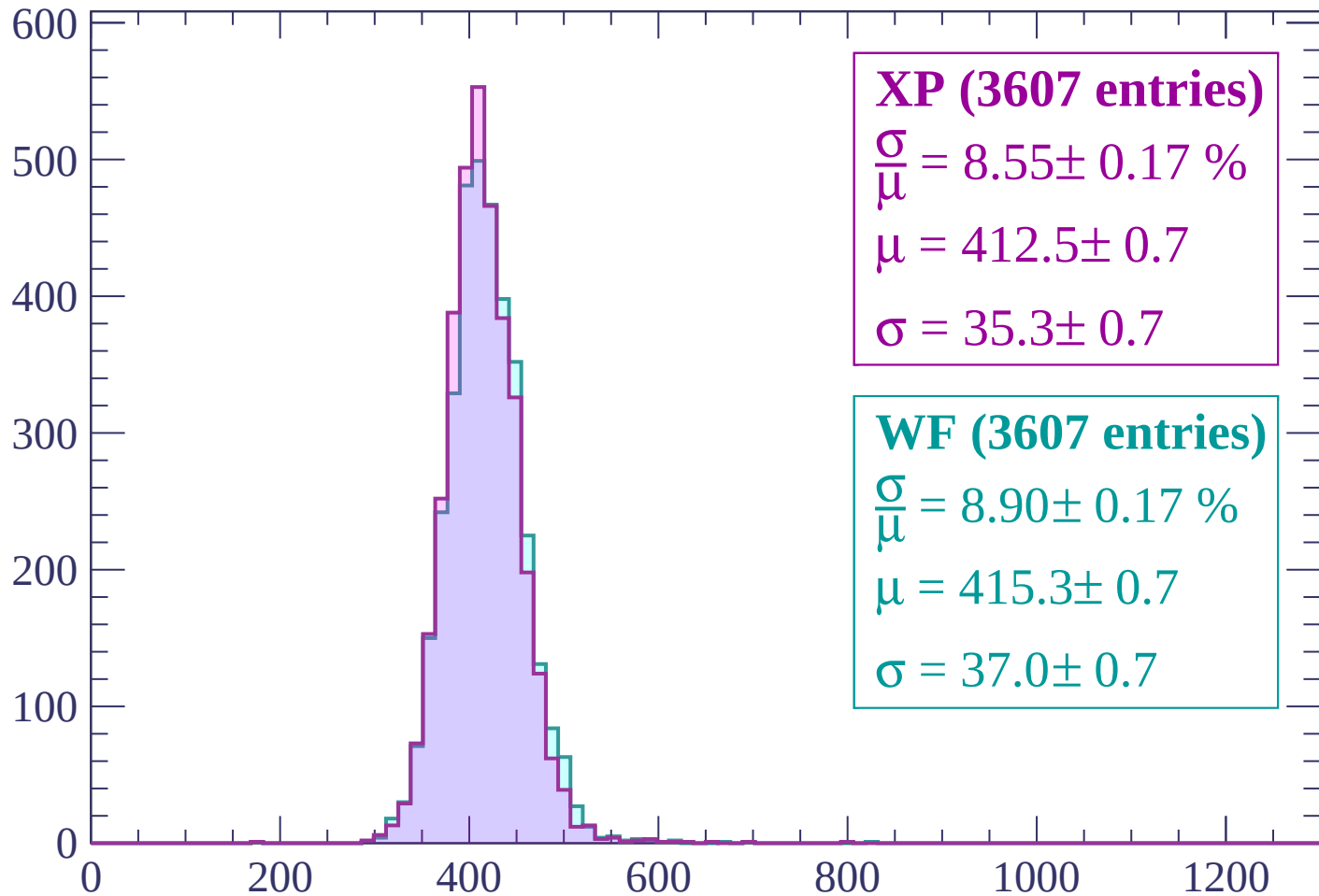
$\mu = 457.1 \pm 1.0$

$\sigma = 42.1 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (3607 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.17 \%$$

$$\mu = 412.5 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF (3607 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.17 \%$$

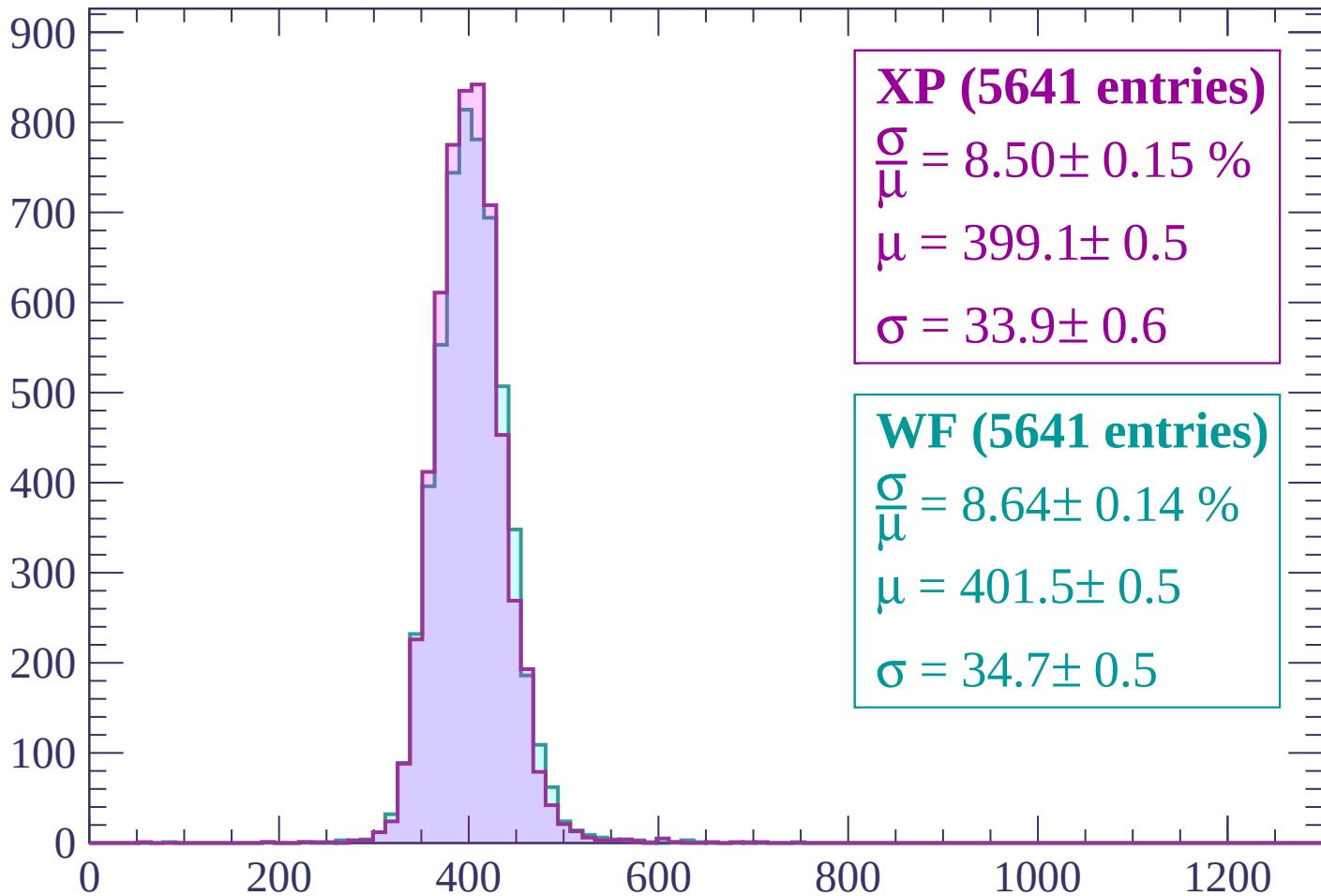
$$\mu = 415.3 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5641 entries)

$$\frac{\sigma}{\mu} = 8.50 \pm 0.15 \%$$

$$\mu = 399.1 \pm 0.5$$

$$\sigma = 33.9 \pm 0.6$$

WF (5641 entries)

$$\frac{\sigma}{\mu} = 8.64 \pm 0.14 \%$$

$$\mu = 401.5 \pm 0.5$$

$$\sigma = 34.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count

1000

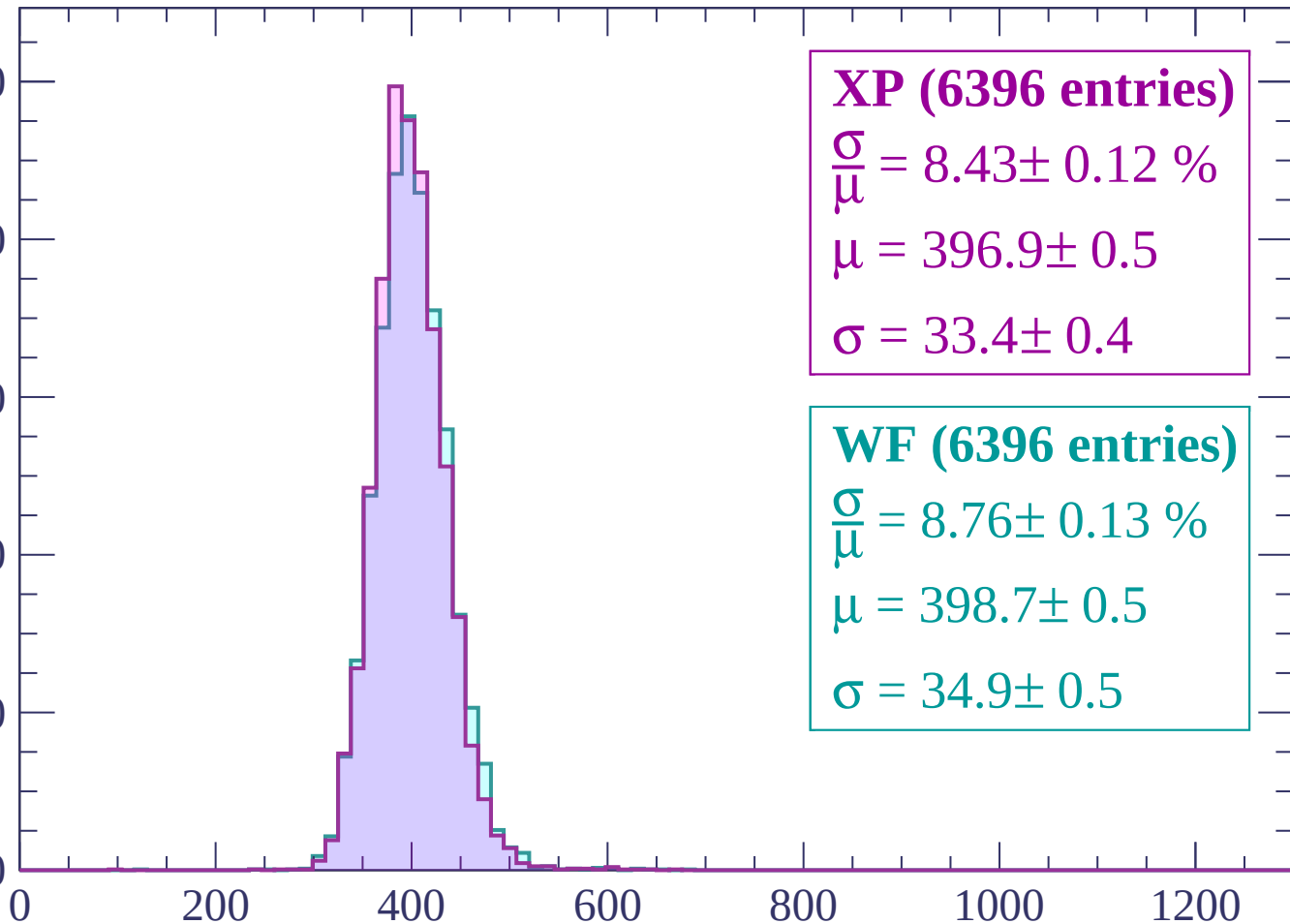
800

600

400

200

0



XP (6396 entries)

$\frac{\sigma}{\mu} = 8.43 \pm 0.12 \%$

$\mu = 396.9 \pm 0.5$

$\sigma = 33.4 \pm 0.4$

WF (6396 entries)

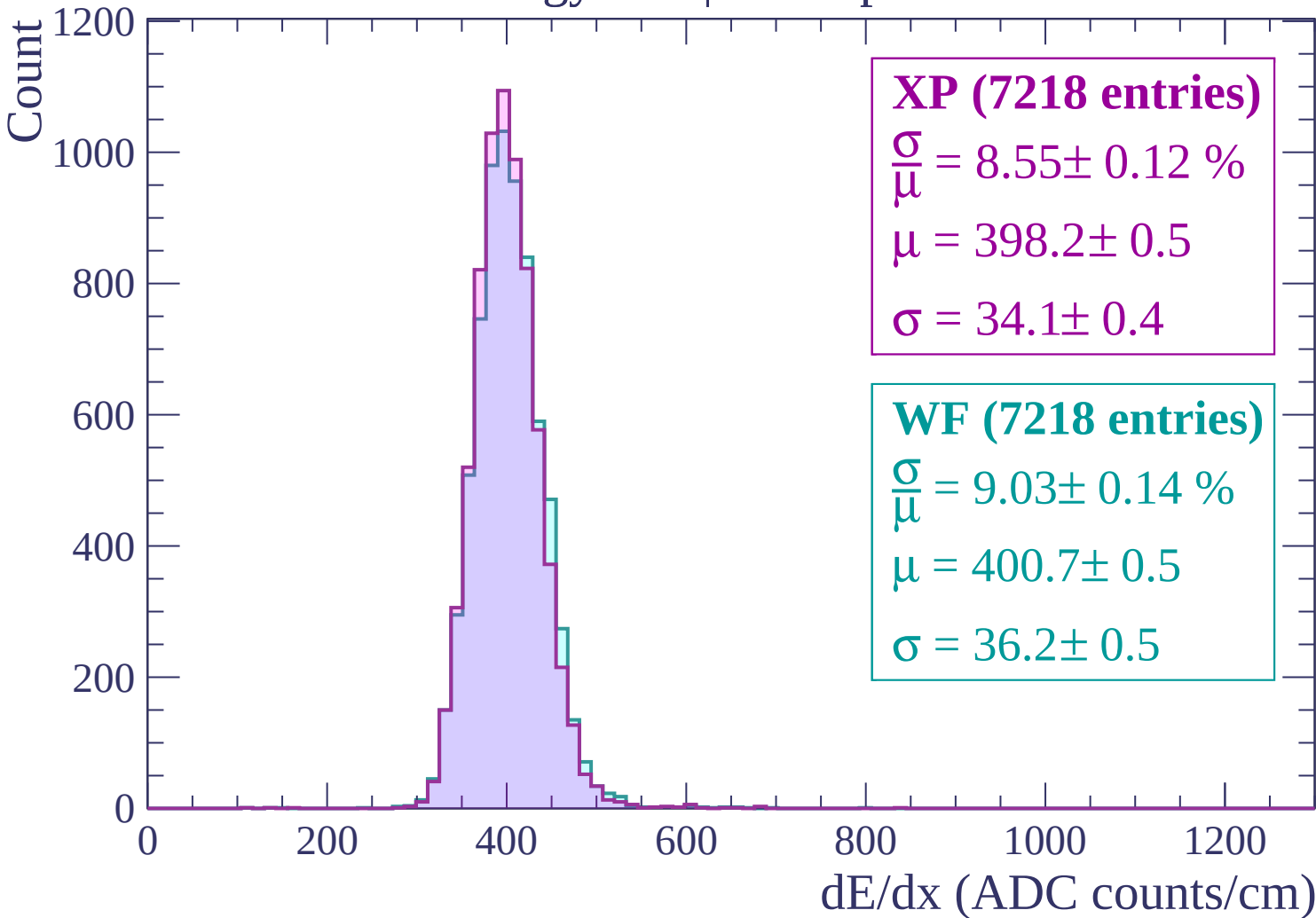
$\frac{\sigma}{\mu} = 8.76 \pm 0.13 \%$

$\mu = 398.7 \pm 0.5$

$\sigma = 34.9 \pm 0.5$

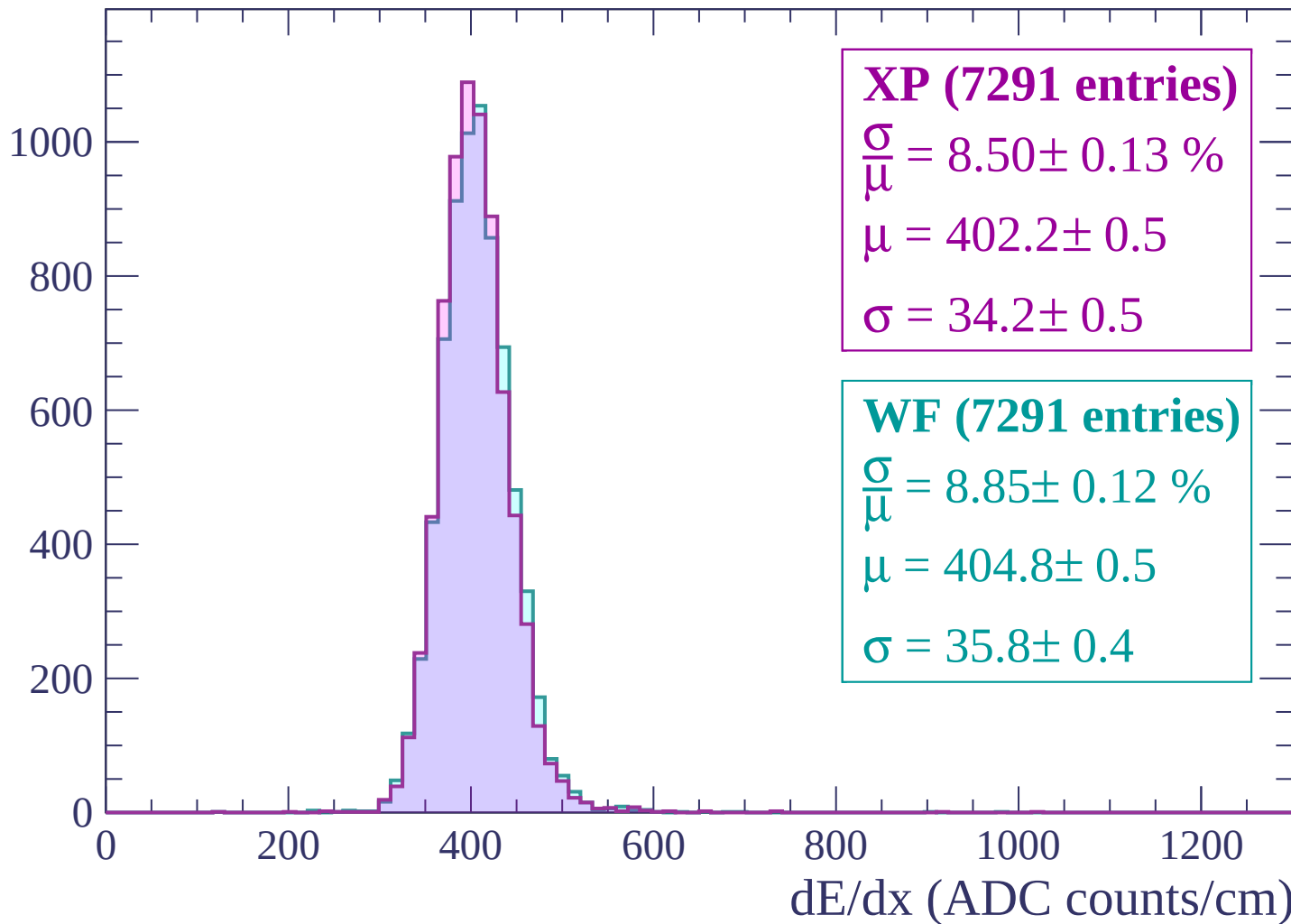
dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$



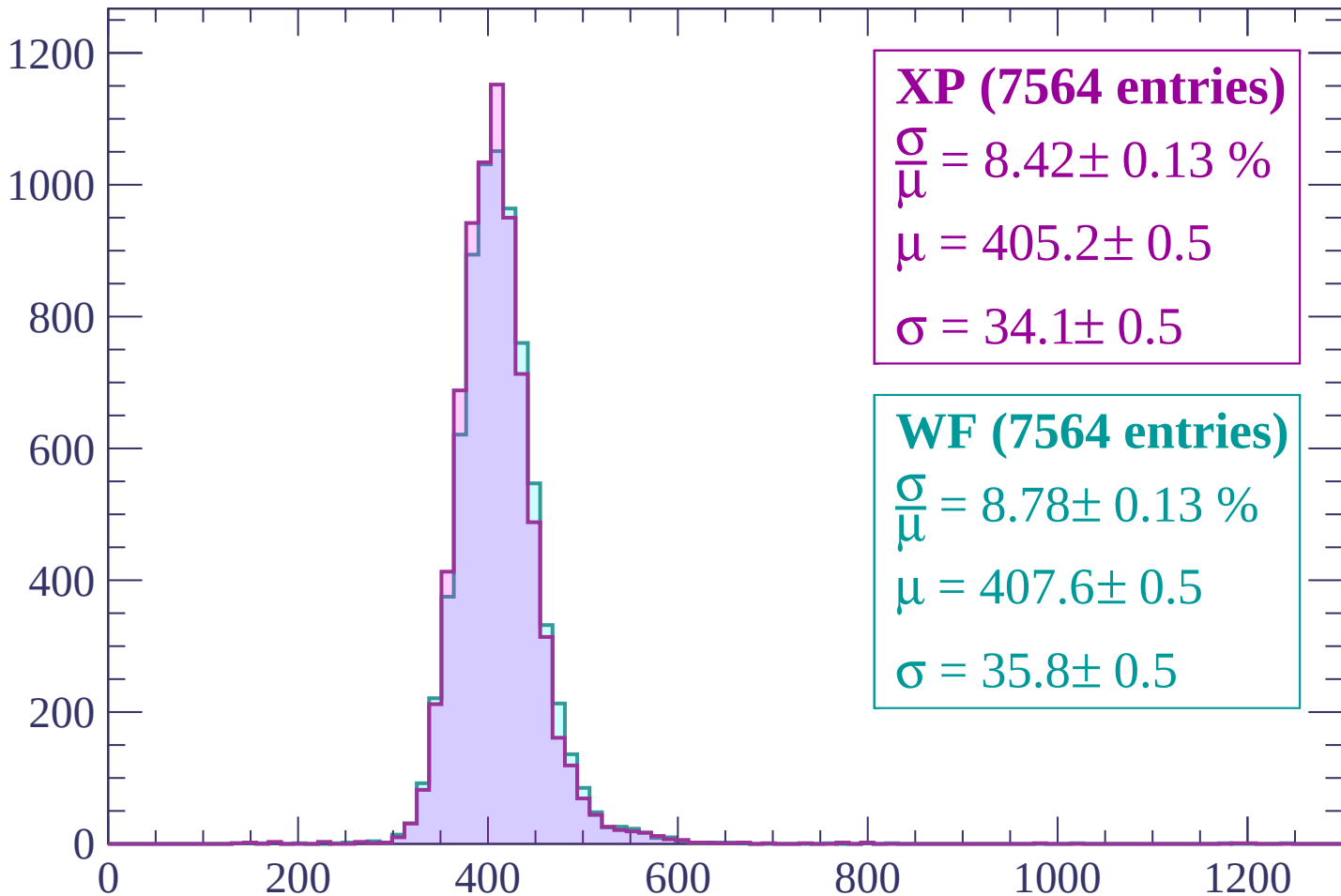
Energy loss | $480 < p < 540$

Count



Energy loss | $540 < p < 600$

Count



XP (7564 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.13 \%$$

$$\mu = 405.2 \pm 0.5$$

$$\sigma = 34.1 \pm 0.5$$

WF (7564 entries)

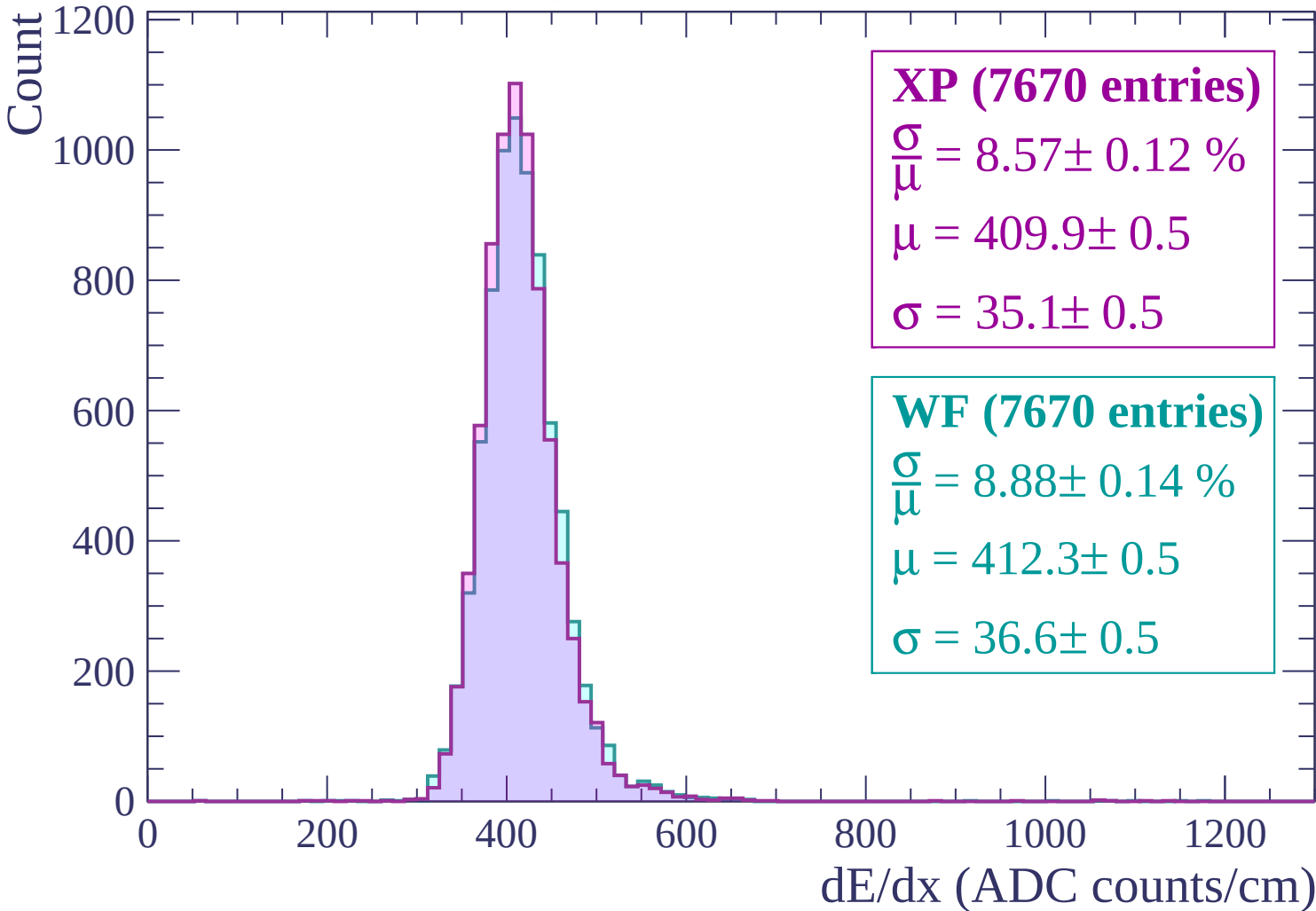
$$\frac{\sigma}{\mu} = 8.78 \pm 0.13 \%$$

$$\mu = 407.6 \pm 0.5$$

$$\sigma = 35.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$



Energy loss | $660 < p < 720$

Count

1000

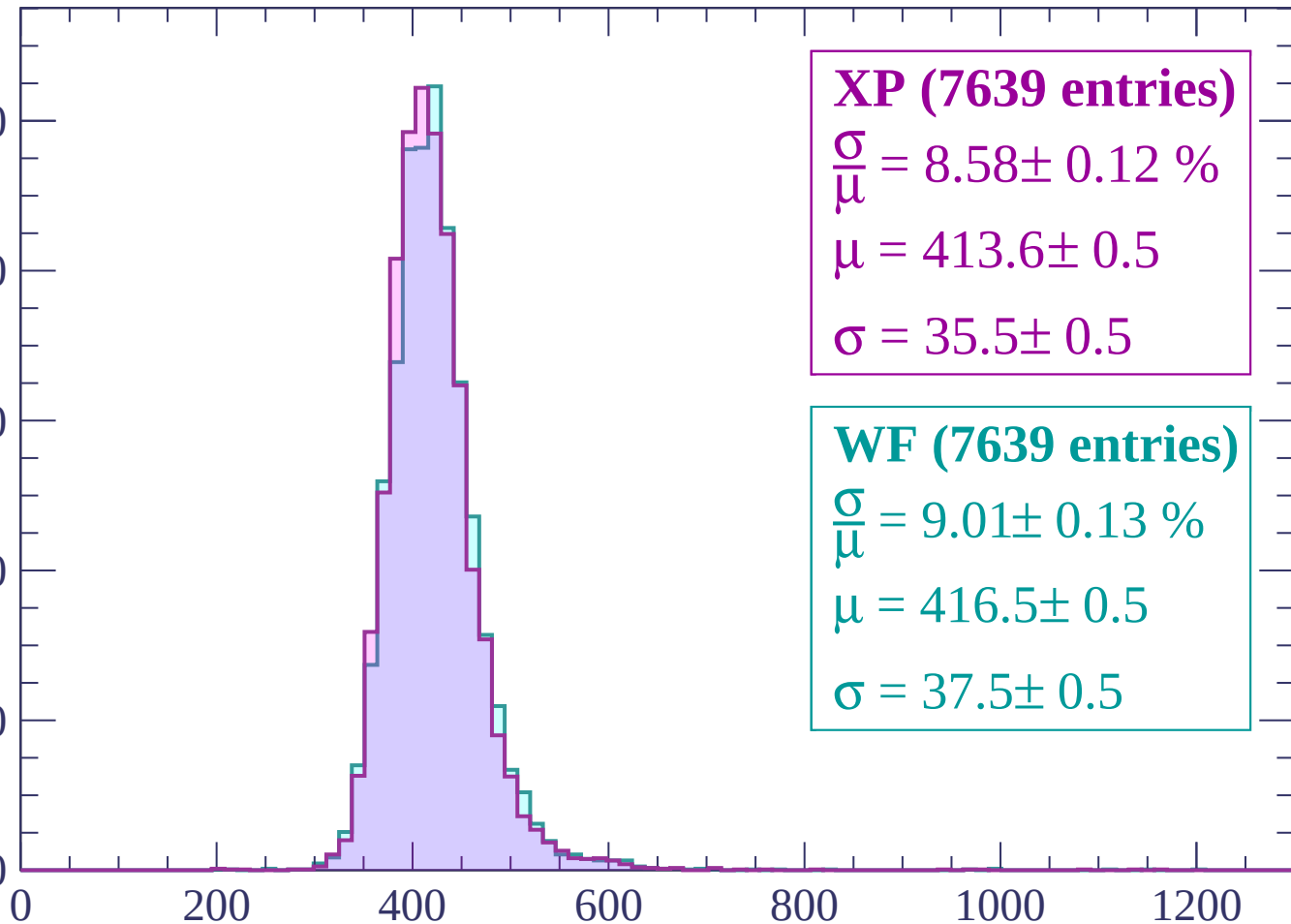
800

600

400

200

0



XP (7639 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.12 \%$

$\mu = 413.6 \pm 0.5$

$\sigma = 35.5 \pm 0.5$

WF (7639 entries)

$\frac{\sigma}{\mu} = 9.01 \pm 0.13 \%$

$\mu = 416.5 \pm 0.5$

$\sigma = 37.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1000

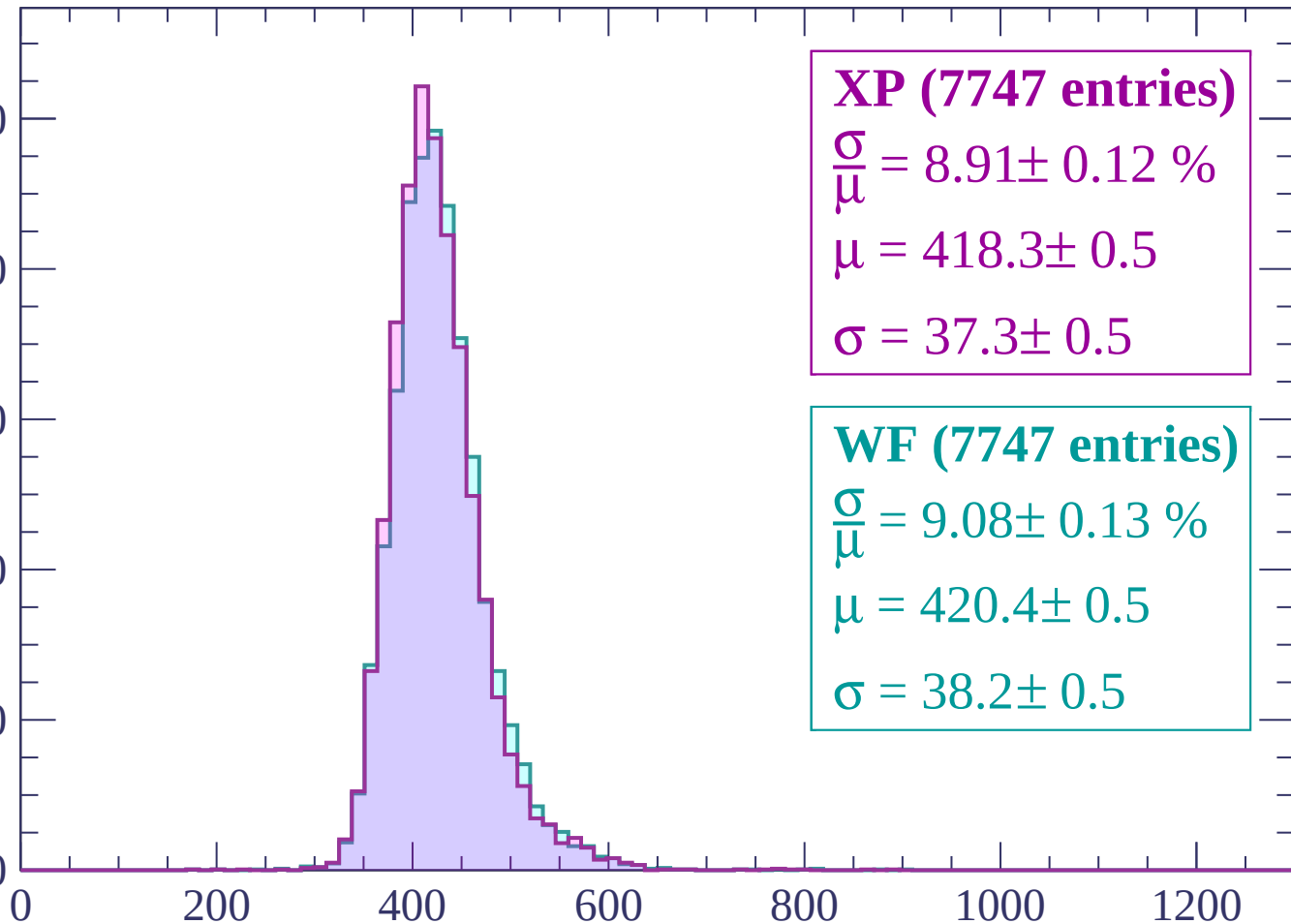
800

600

400

200

0



XP (7747 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.12 \%$

$\mu = 418.3 \pm 0.5$

$\sigma = 37.3 \pm 0.5$

WF (7747 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.13 \%$

$\mu = 420.4 \pm 0.5$

$\sigma = 38.2 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1000

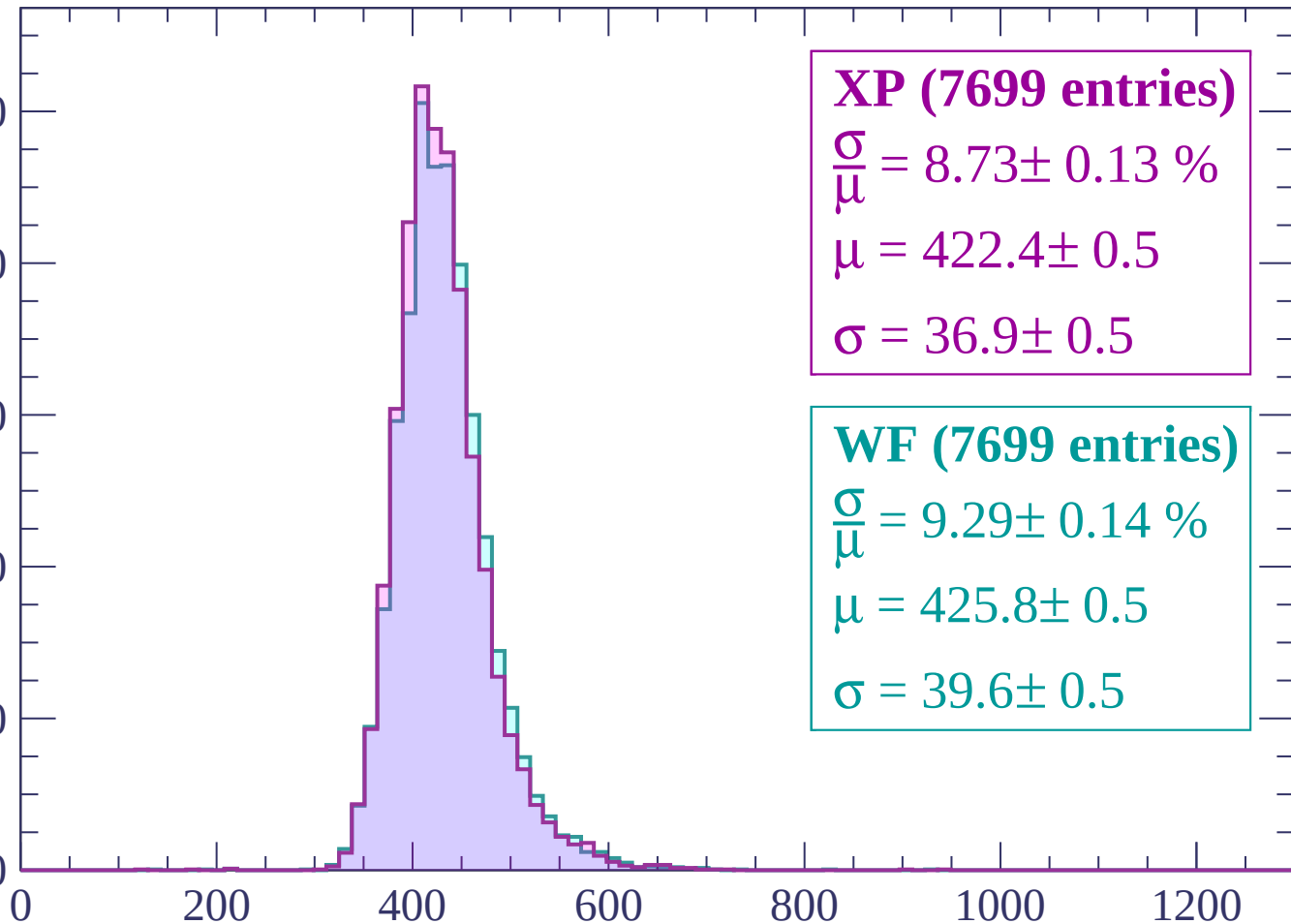
800

600

400

200

0



XP (7699 entries)

$\frac{\sigma}{\mu} = 8.73 \pm 0.13 \%$

$\mu = 422.4 \pm 0.5$

$\sigma = 36.9 \pm 0.5$

WF (7699 entries)

$\frac{\sigma}{\mu} = 9.29 \pm 0.14 \%$

$\mu = 425.8 \pm 0.5$

$\sigma = 39.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count

1000

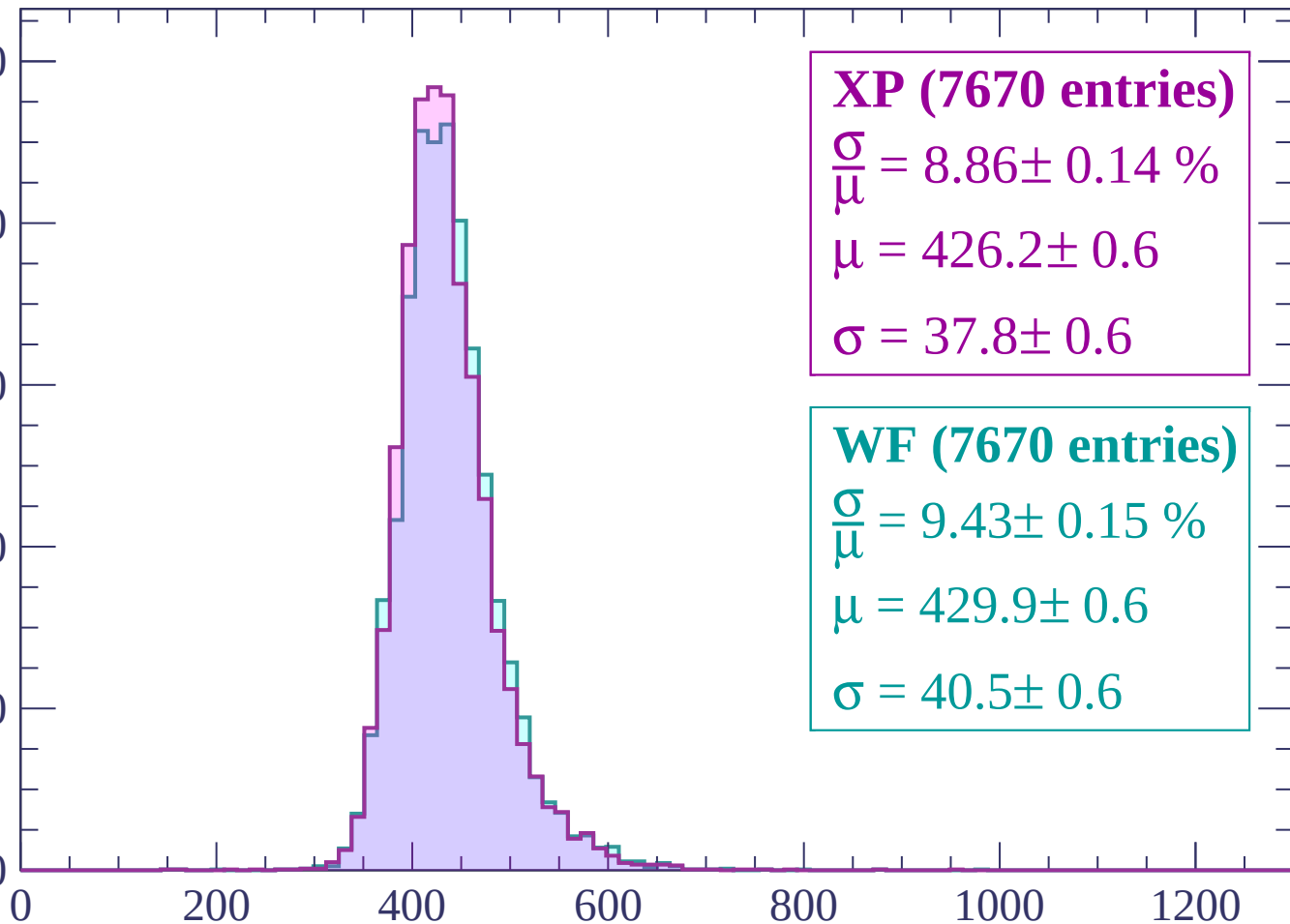
800

600

400

200

0



XP (7670 entries)

$\frac{\sigma}{\mu} = 8.86 \pm 0.14 \%$

$\mu = 426.2 \pm 0.6$

$\sigma = 37.8 \pm 0.6$

WF (7670 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.15 \%$

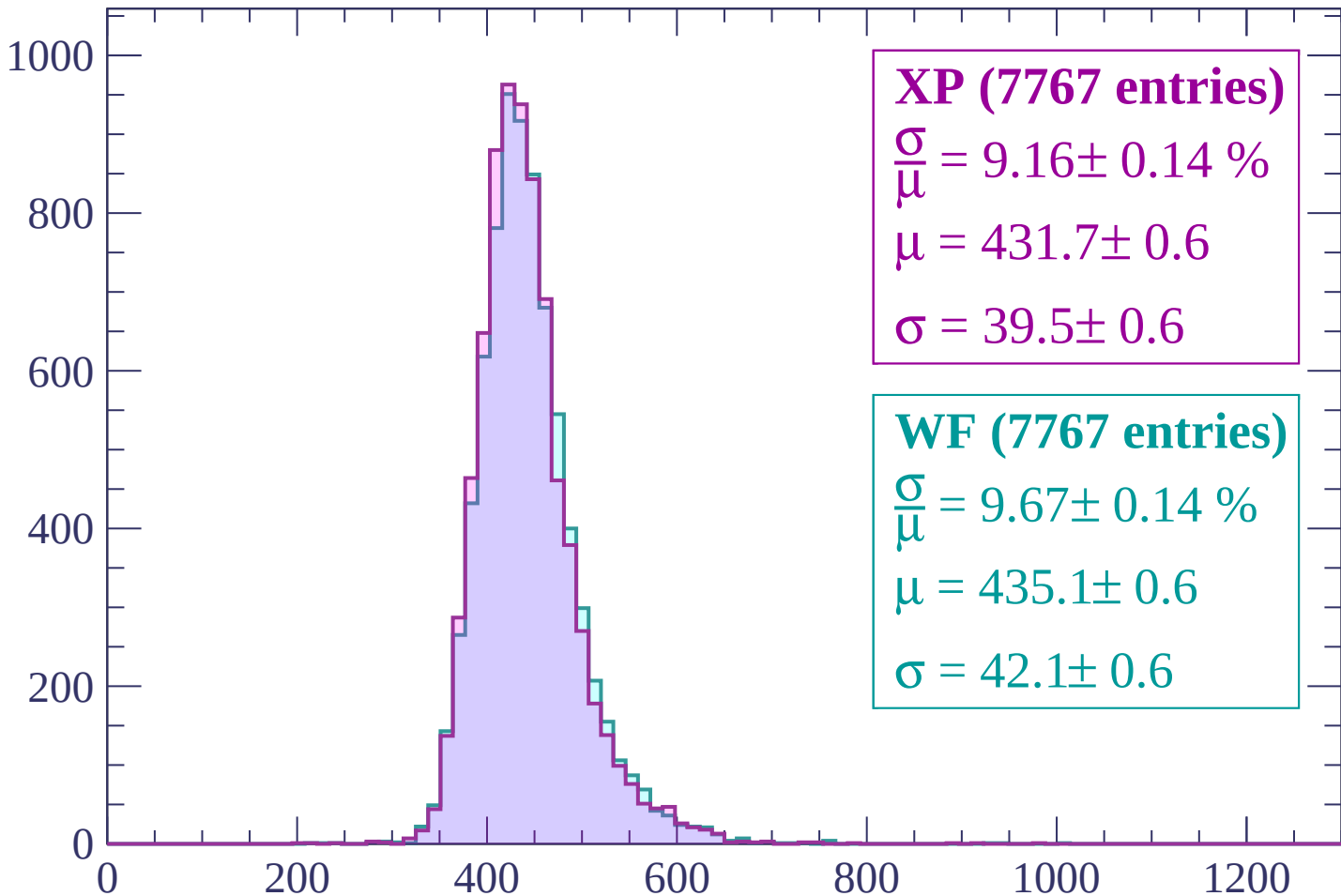
$\mu = 429.9 \pm 0.6$

$\sigma = 40.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (7767 entries)

$\frac{\sigma}{\mu} = 9.16 \pm 0.14 \%$

$\mu = 431.7 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

WF (7767 entries)

$\frac{\sigma}{\mu} = 9.67 \pm 0.14 \%$

$\mu = 435.1 \pm 0.6$

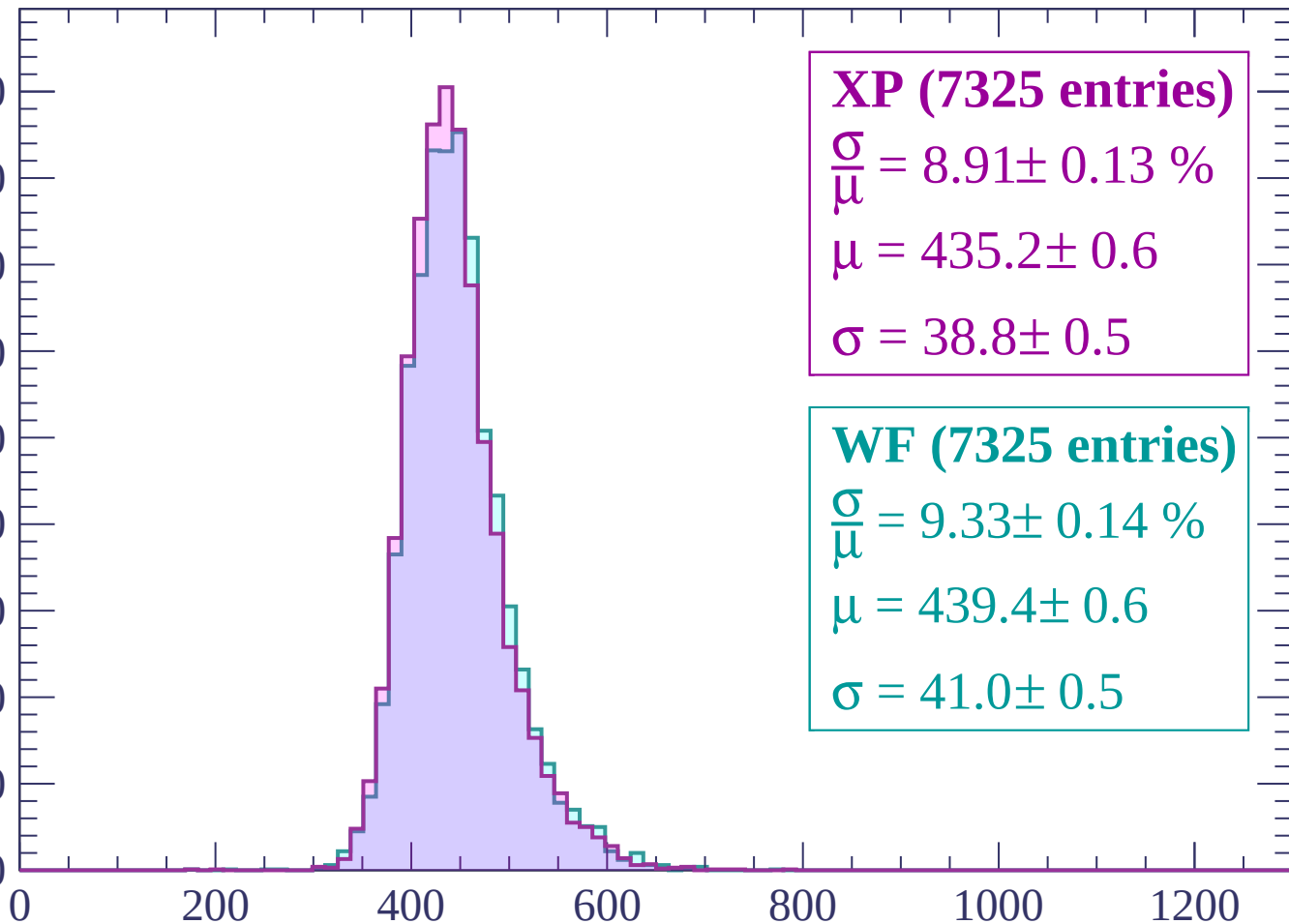
$\sigma = 42.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

900
800
700
600
500
400
300
200
100
0



XP (7325 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$$

$$\mu = 435.2 \pm 0.6$$

$$\sigma = 38.8 \pm 0.5$$

WF (7325 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.14 \%$$

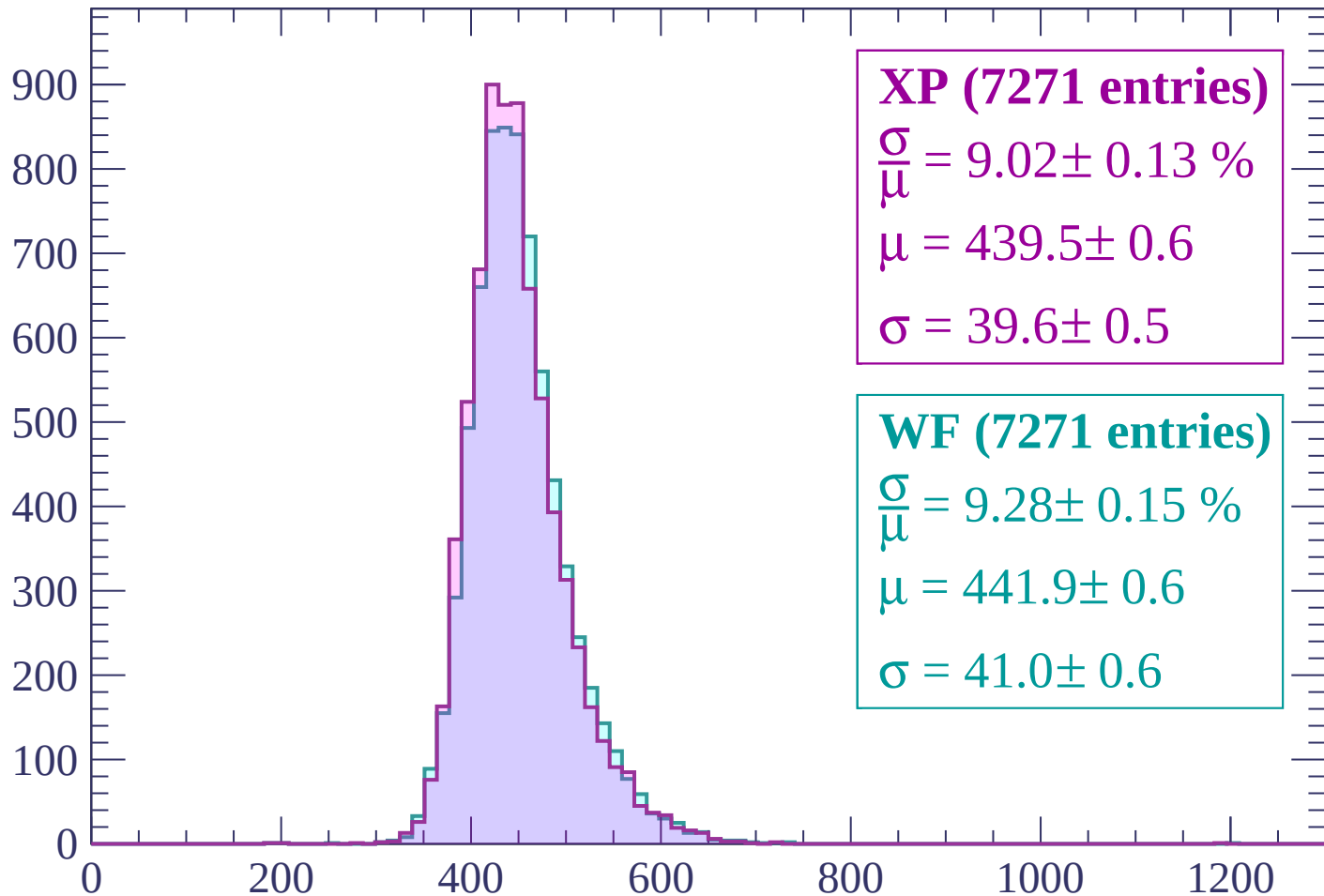
$$\mu = 439.4 \pm 0.6$$

$$\sigma = 41.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (7271 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.13 \%$$

$$\mu = 439.5 \pm 0.6$$

$$\sigma = 39.6 \pm 0.5$$

WF (7271 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.15 \%$$

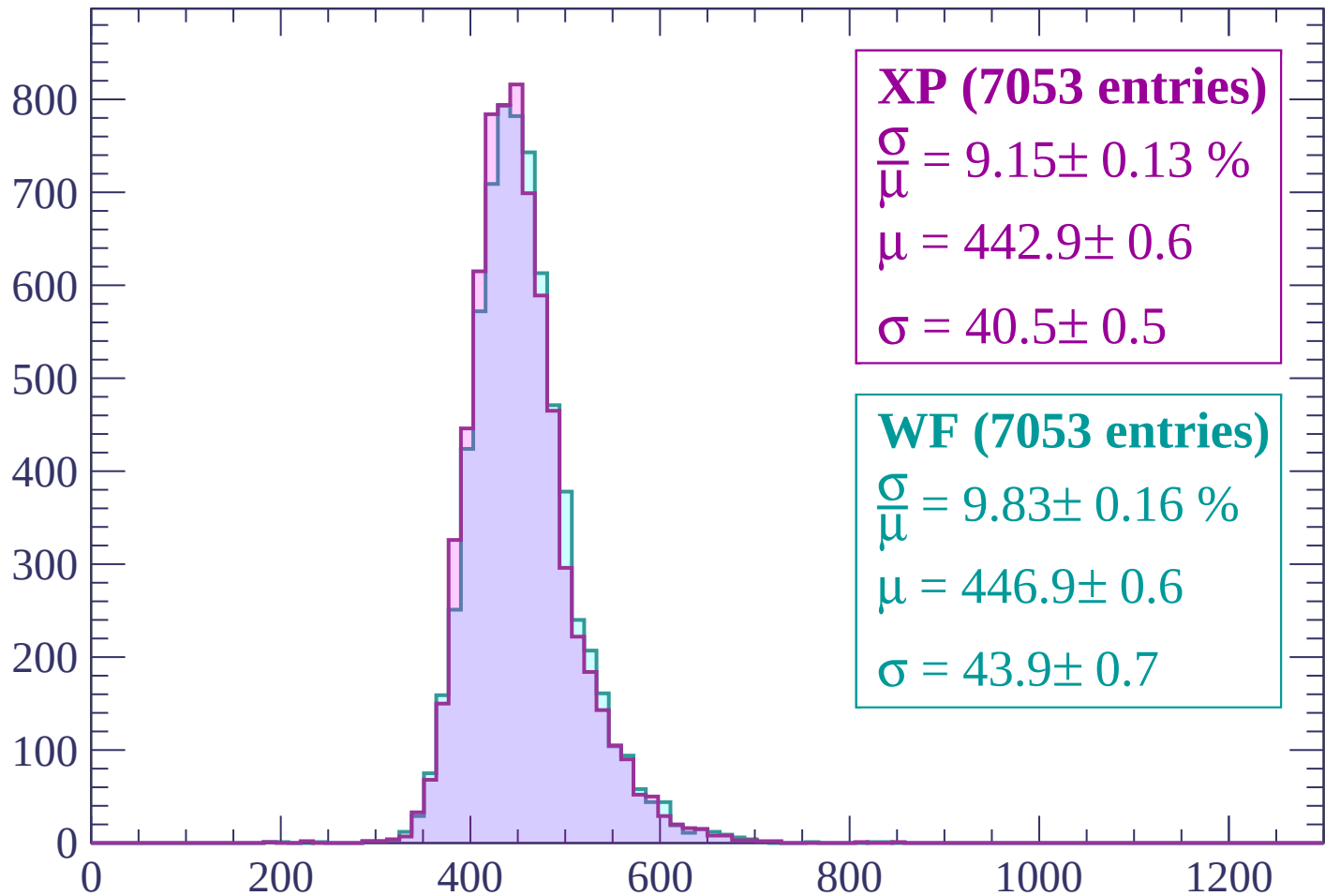
$$\mu = 441.9 \pm 0.6$$

$$\sigma = 41.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (7053 entries)

$$\frac{\sigma}{\mu} = 9.15 \pm 0.13 \%$$

$$\mu = 442.9 \pm 0.6$$

$$\sigma = 40.5 \pm 0.5$$

WF (7053 entries)

$$\frac{\sigma}{\mu} = 9.83 \pm 0.16 \%$$

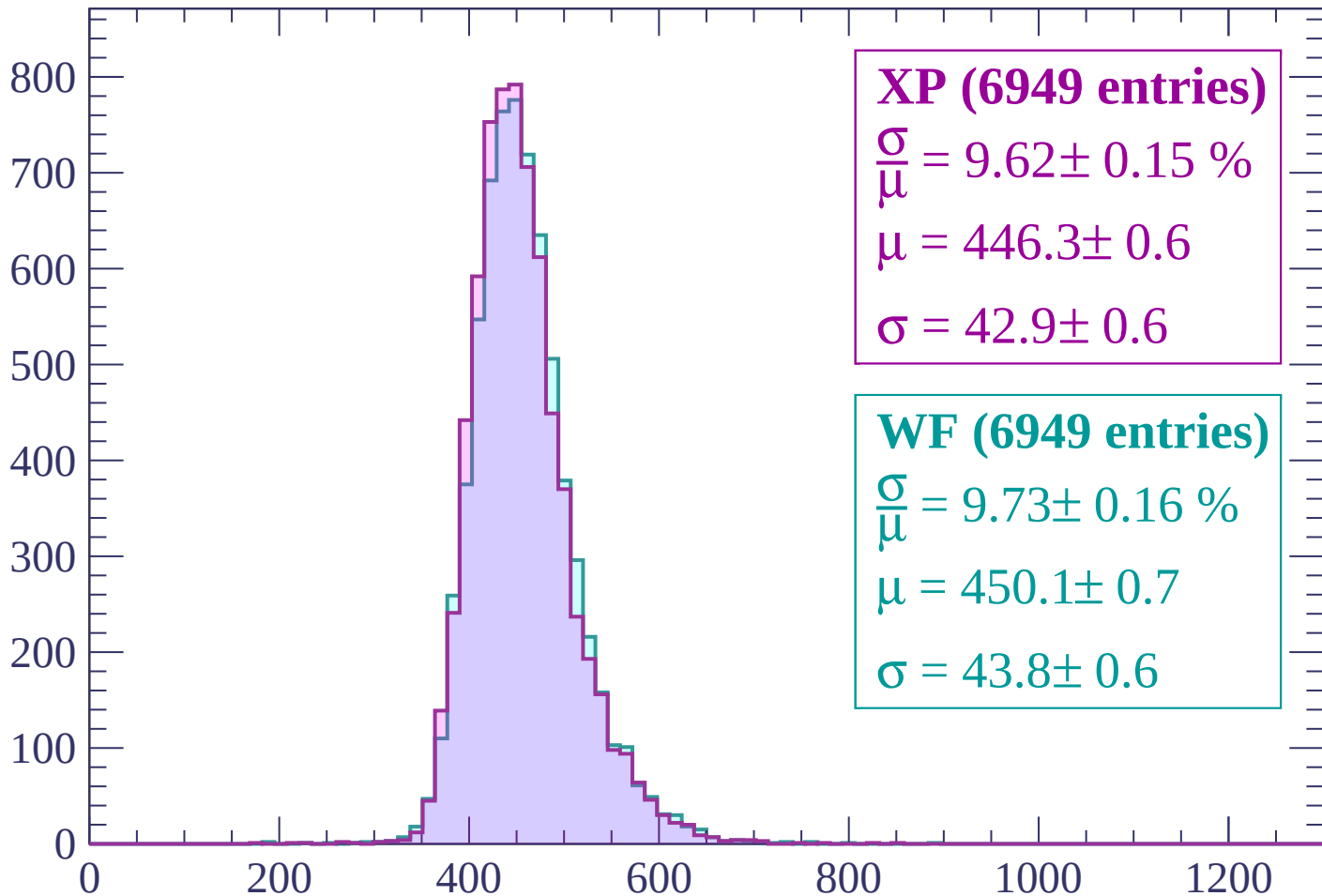
$$\mu = 446.9 \pm 0.6$$

$$\sigma = 43.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (6949 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.15 \%$$

$$\mu = 446.3 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

WF (6949 entries)

$$\frac{\sigma}{\mu} = 9.73 \pm 0.16 \%$$

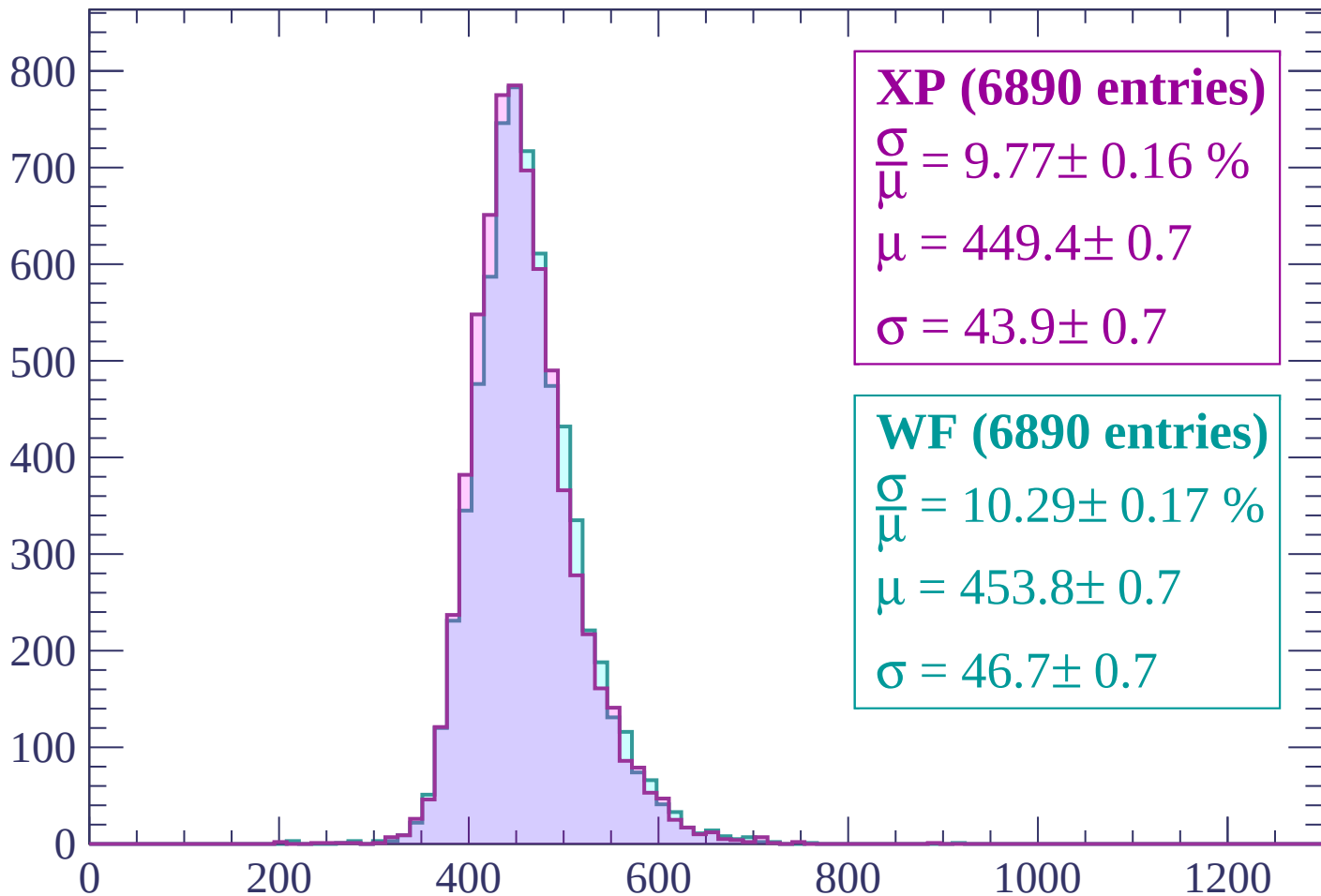
$$\mu = 450.1 \pm 0.7$$

$$\sigma = 43.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (6890 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.16 \%$$

$$\mu = 449.4 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (6890 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$$

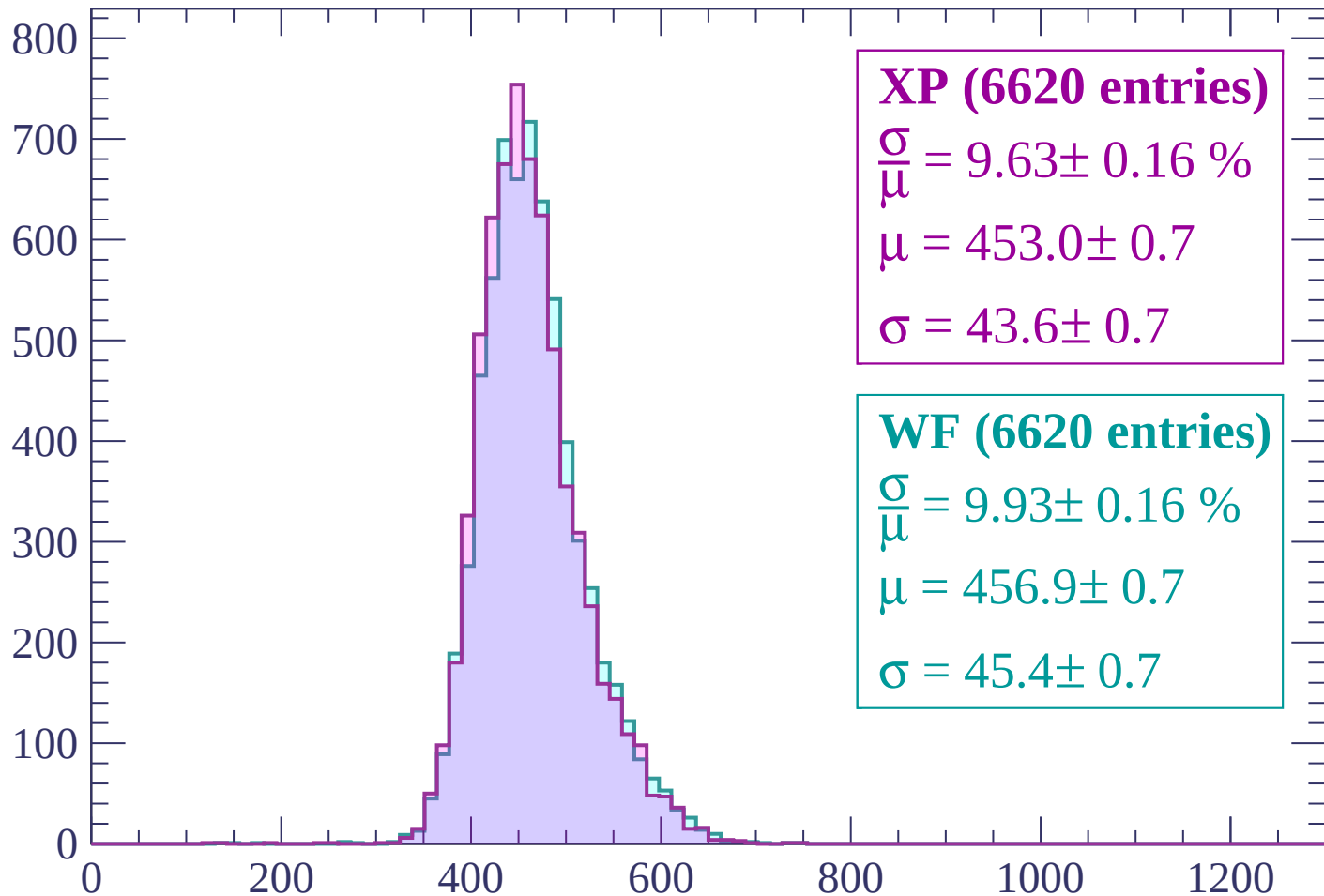
$$\mu = 453.8 \pm 0.7$$

$$\sigma = 46.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (6620 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.16 \%$$

$$\mu = 453.0 \pm 0.7$$

$$\sigma = 43.6 \pm 0.7$$

WF (6620 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.16 \%$$

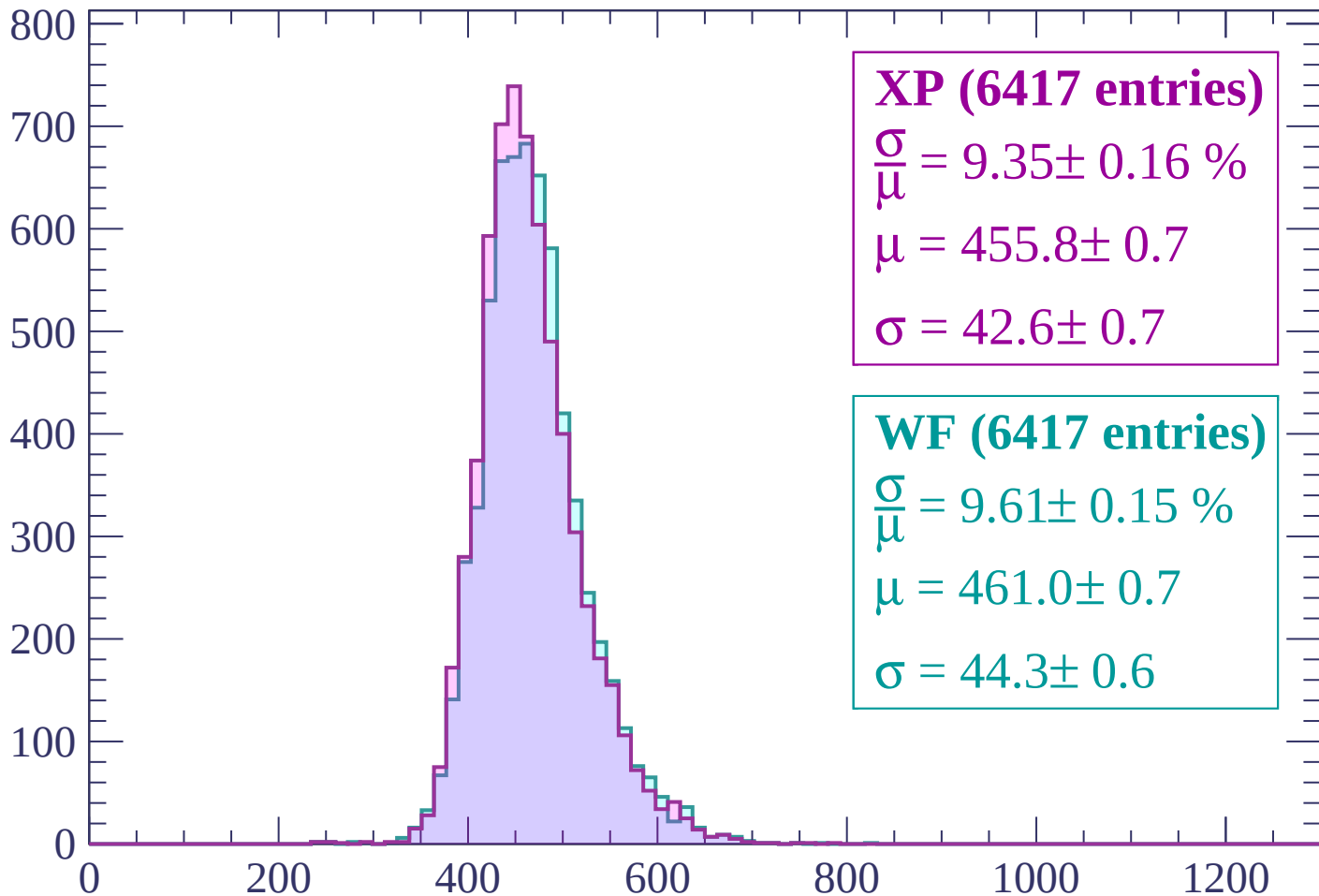
$$\mu = 456.9 \pm 0.7$$

$$\sigma = 45.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (6417 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.16 \%$$

$$\mu = 455.8 \pm 0.7$$

$$\sigma = 42.6 \pm 0.7$$

WF (6417 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.15 \%$$

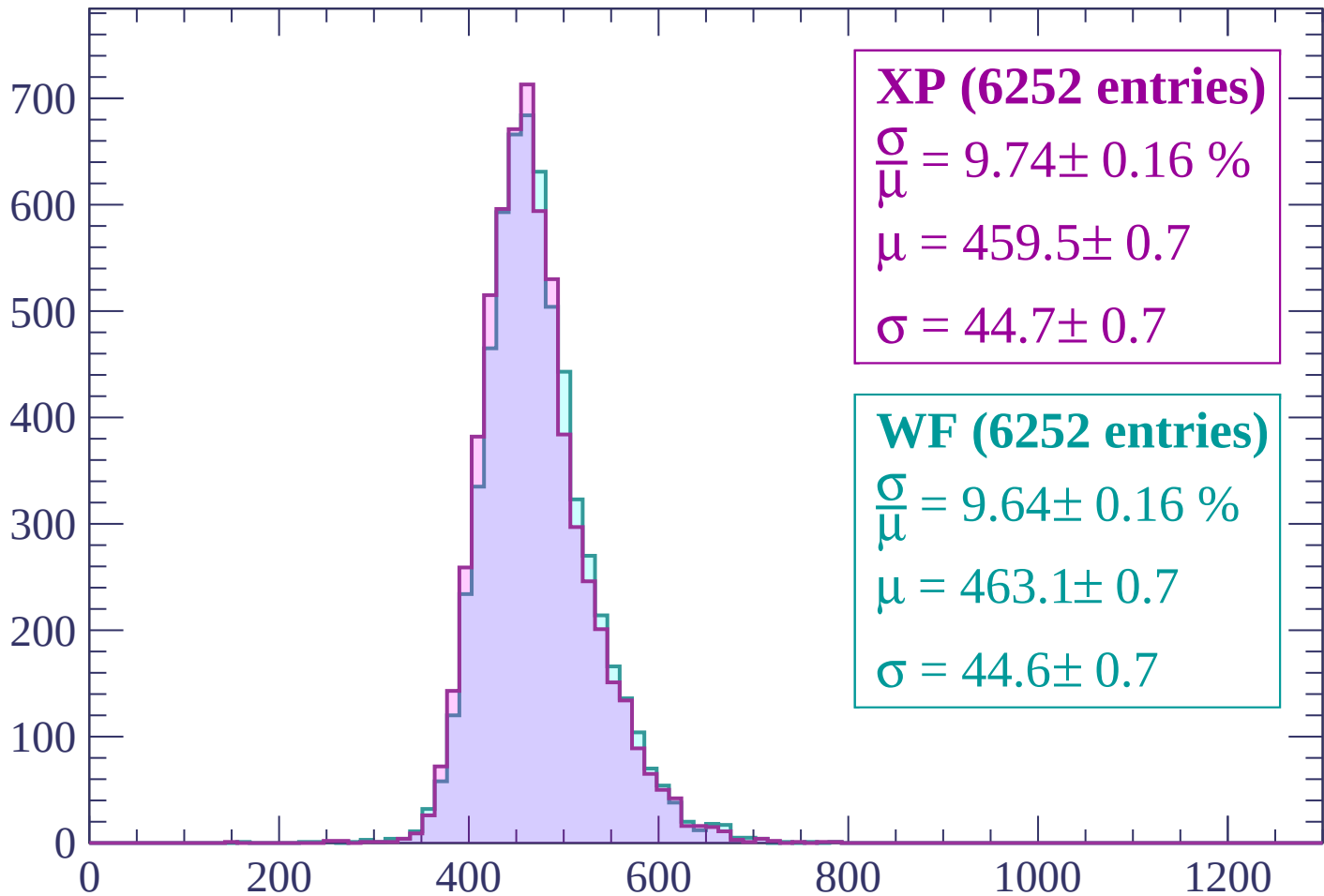
$$\mu = 461.0 \pm 0.7$$

$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (6252 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.16 \%$$

$$\mu = 459.5 \pm 0.7$$

$$\sigma = 44.7 \pm 0.7$$

WF (6252 entries)

$$\frac{\sigma}{\mu} = 9.64 \pm 0.16 \%$$

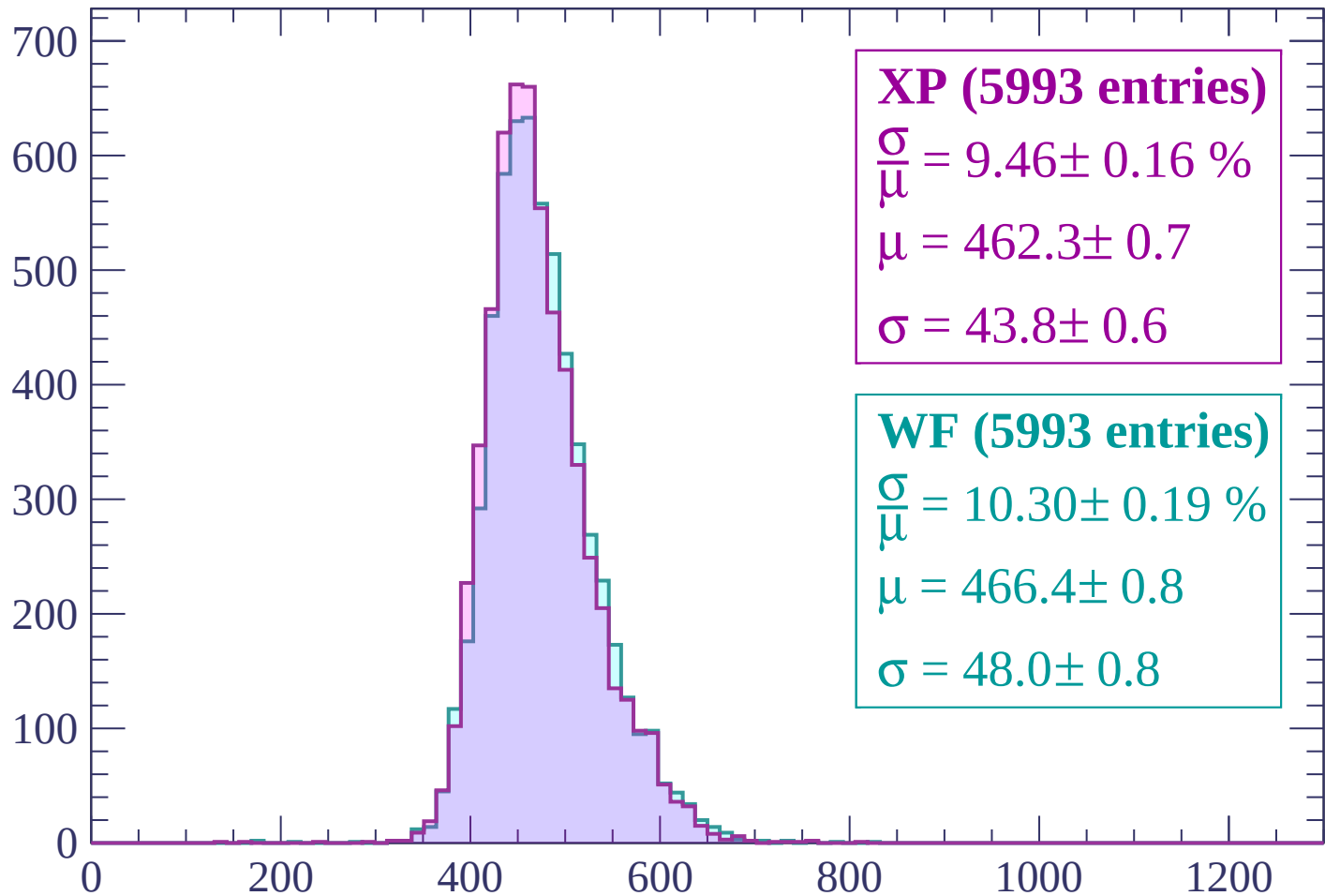
$$\mu = 463.1 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (5993 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.16 \%$$

$$\mu = 462.3 \pm 0.7$$

$$\sigma = 43.8 \pm 0.6$$

WF (5993 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.19 \%$$

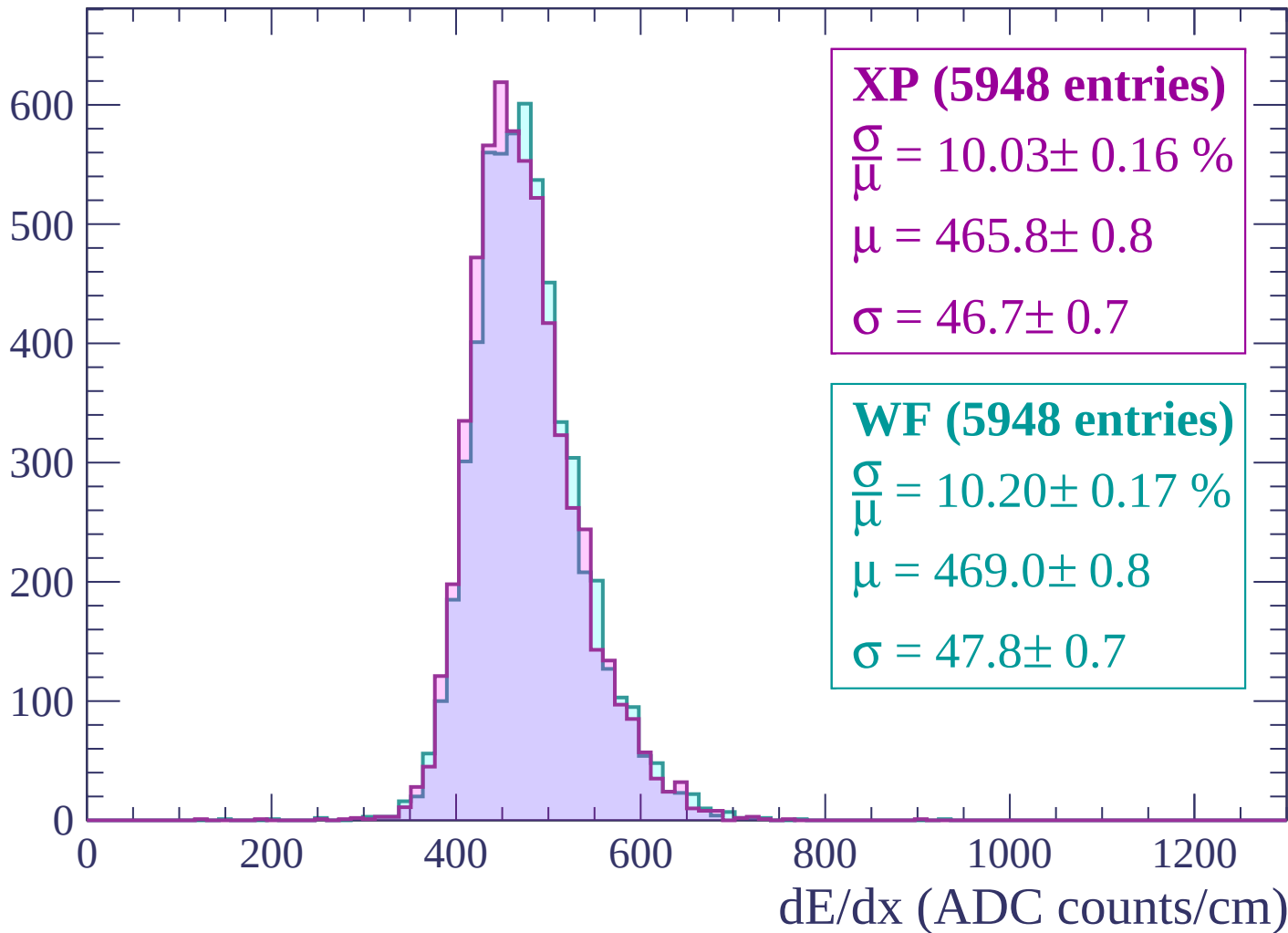
$$\mu = 466.4 \pm 0.8$$

$$\sigma = 48.0 \pm 0.8$$

dE/dx (ADC counts/cm)

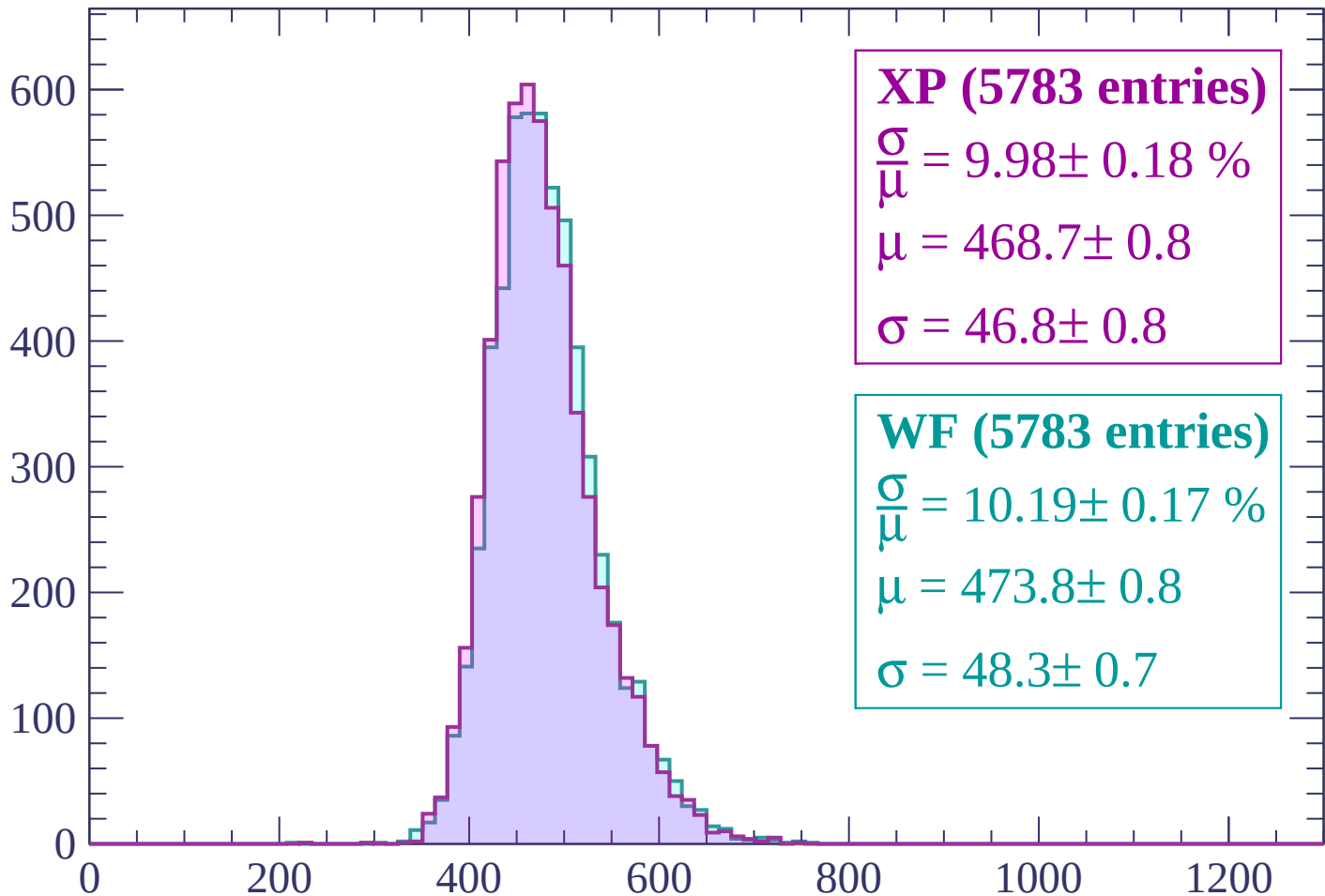
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (5783 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.18 \%$$

$$\mu = 468.7 \pm 0.8$$

$$\sigma = 46.8 \pm 0.8$$

WF (5783 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.17 \%$$

$$\mu = 473.8 \pm 0.8$$

$$\sigma = 48.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count

600
500
400
300
200
100
0

XP (5633 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.19 \%$

$\mu = 470.3 \pm 0.8$

$\sigma = 47.2 \pm 0.8$

WF (5633 entries)

$\frac{\sigma}{\mu} = 10.45 \pm 0.19 \%$

$\mu = 475.6 \pm 0.8$

$\sigma = 49.7 \pm 0.8$

0

200

400

600

800

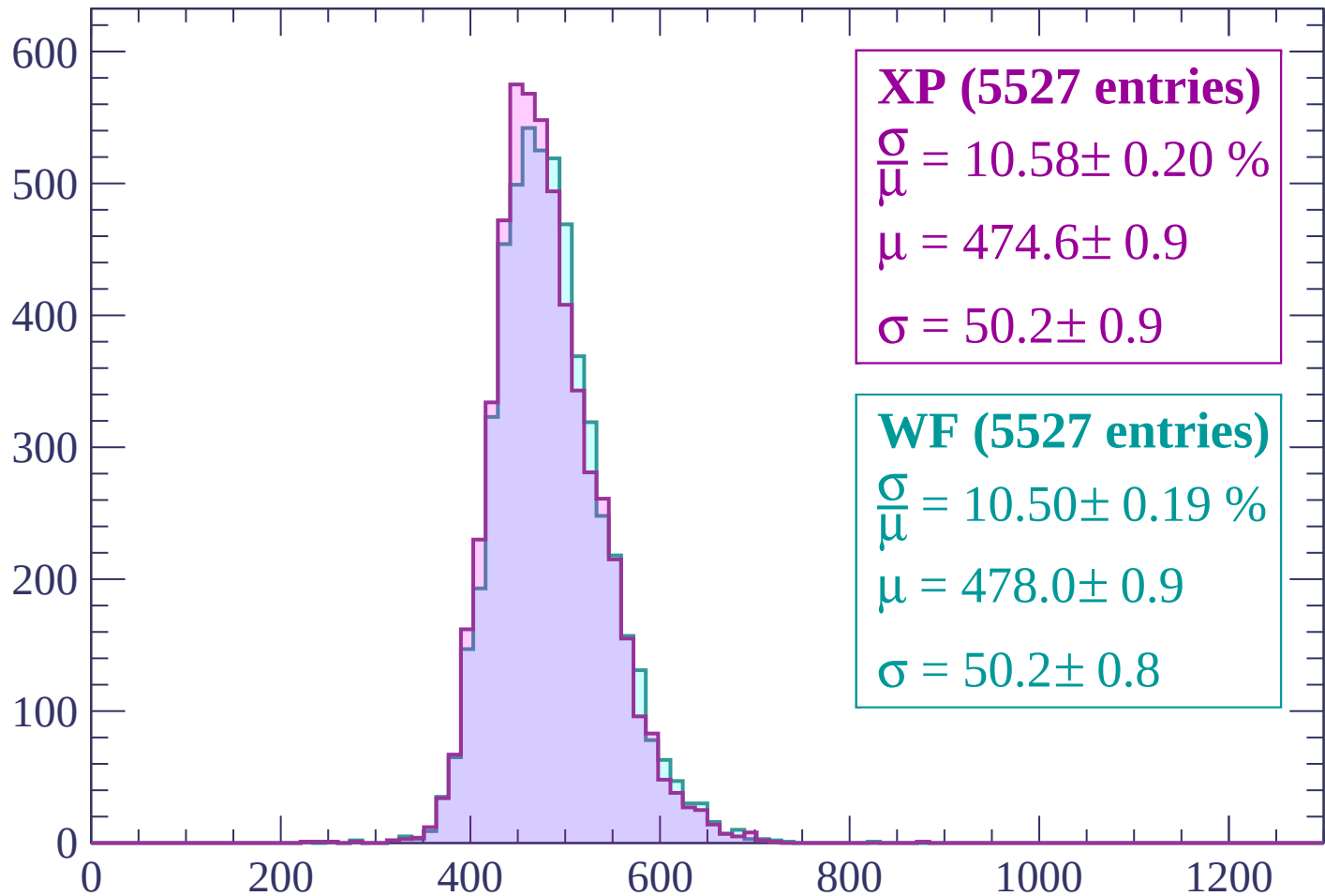
1000

1200

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (5527 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.20 \%$

$\mu = 474.6 \pm 0.9$

$\sigma = 50.2 \pm 0.9$

WF (5527 entries)

$\frac{\sigma}{\mu} = 10.50 \pm 0.19 \%$

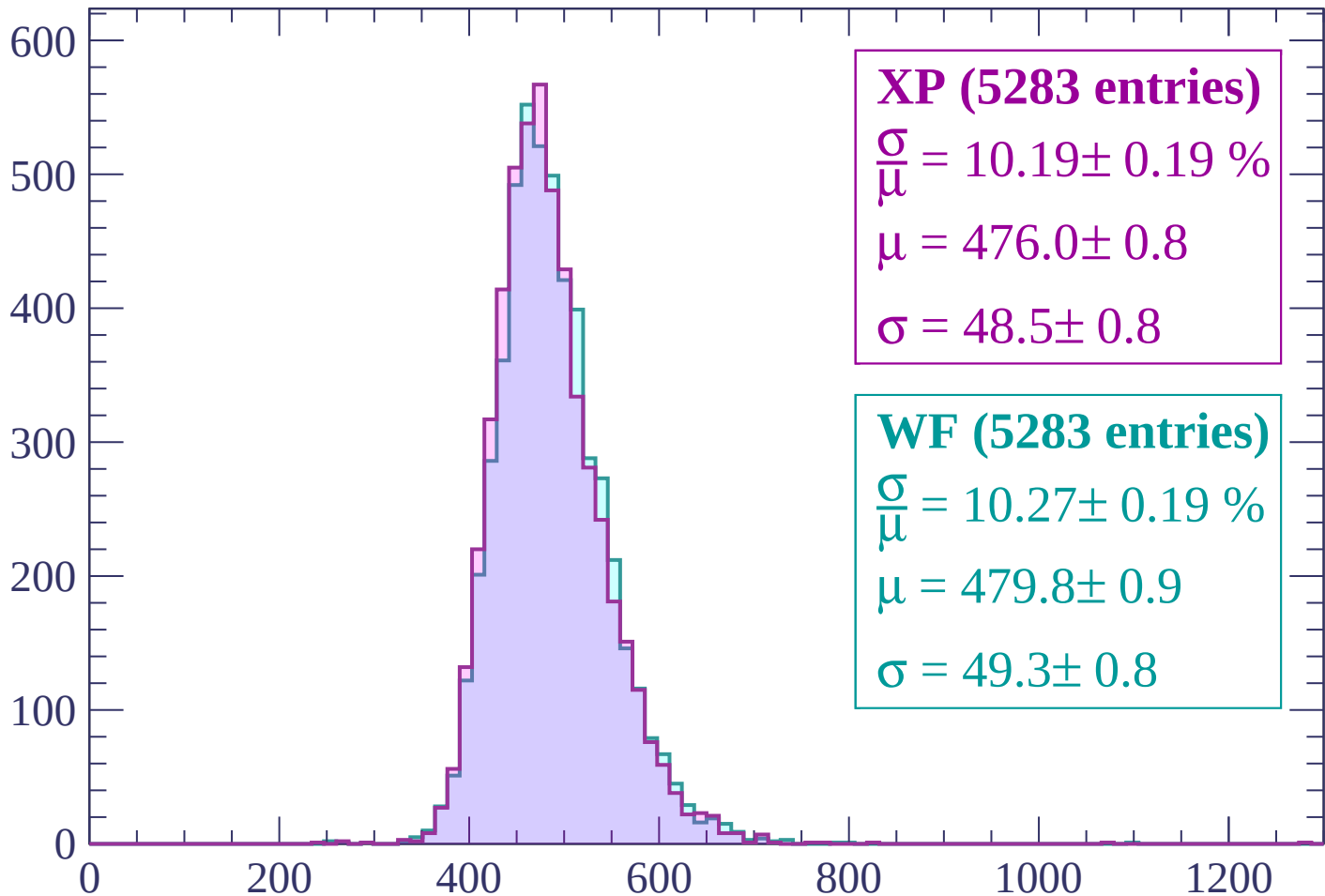
$\mu = 478.0 \pm 0.9$

$\sigma = 50.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (5283 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.19 \%$

$\mu = 476.0 \pm 0.8$

$\sigma = 48.5 \pm 0.8$

WF (5283 entries)

$\frac{\sigma}{\mu} = 10.27 \pm 0.19 \%$

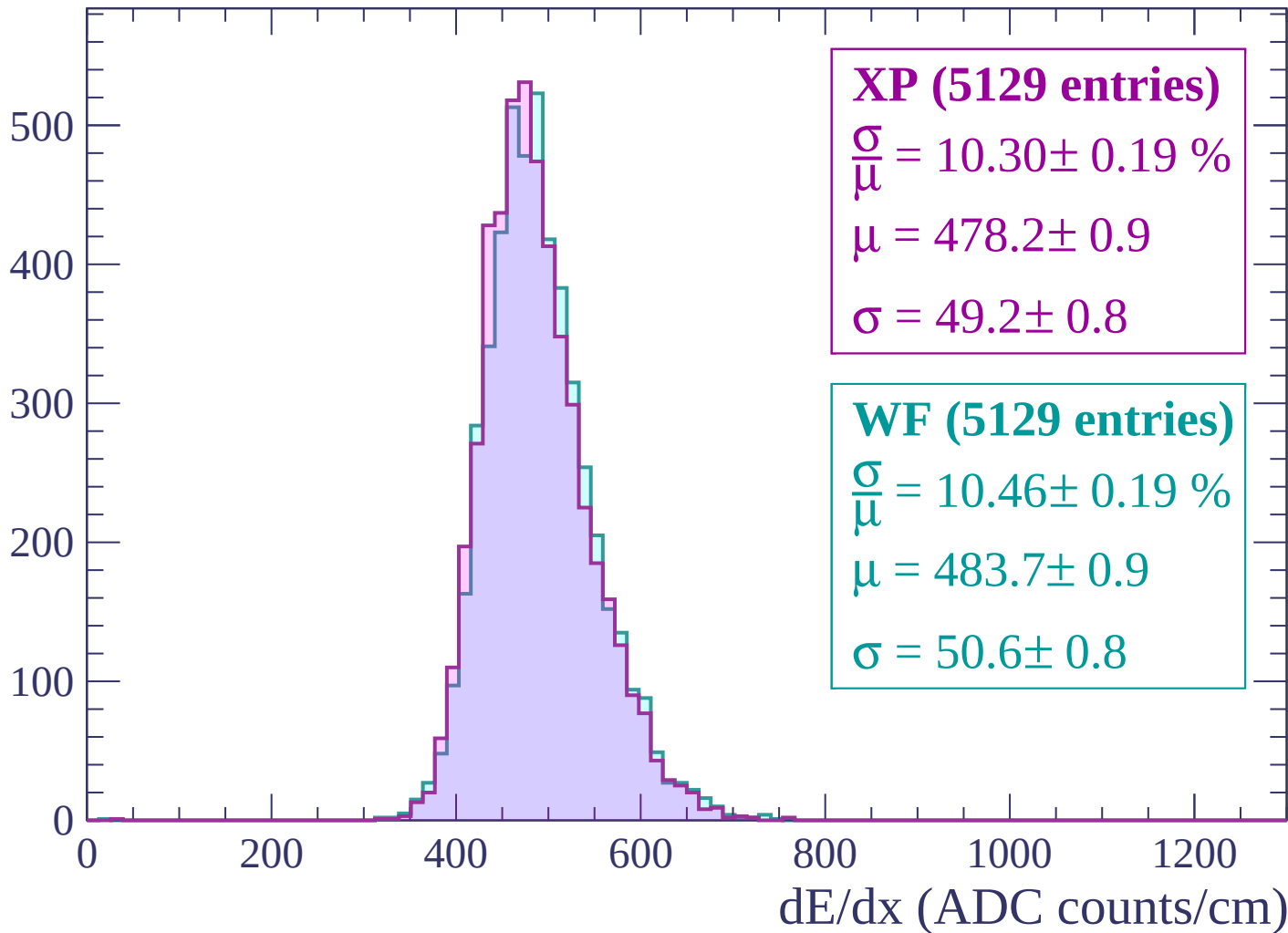
$\mu = 479.8 \pm 0.9$

$\sigma = 49.3 \pm 0.8$

dE/dx (ADC counts/cm)

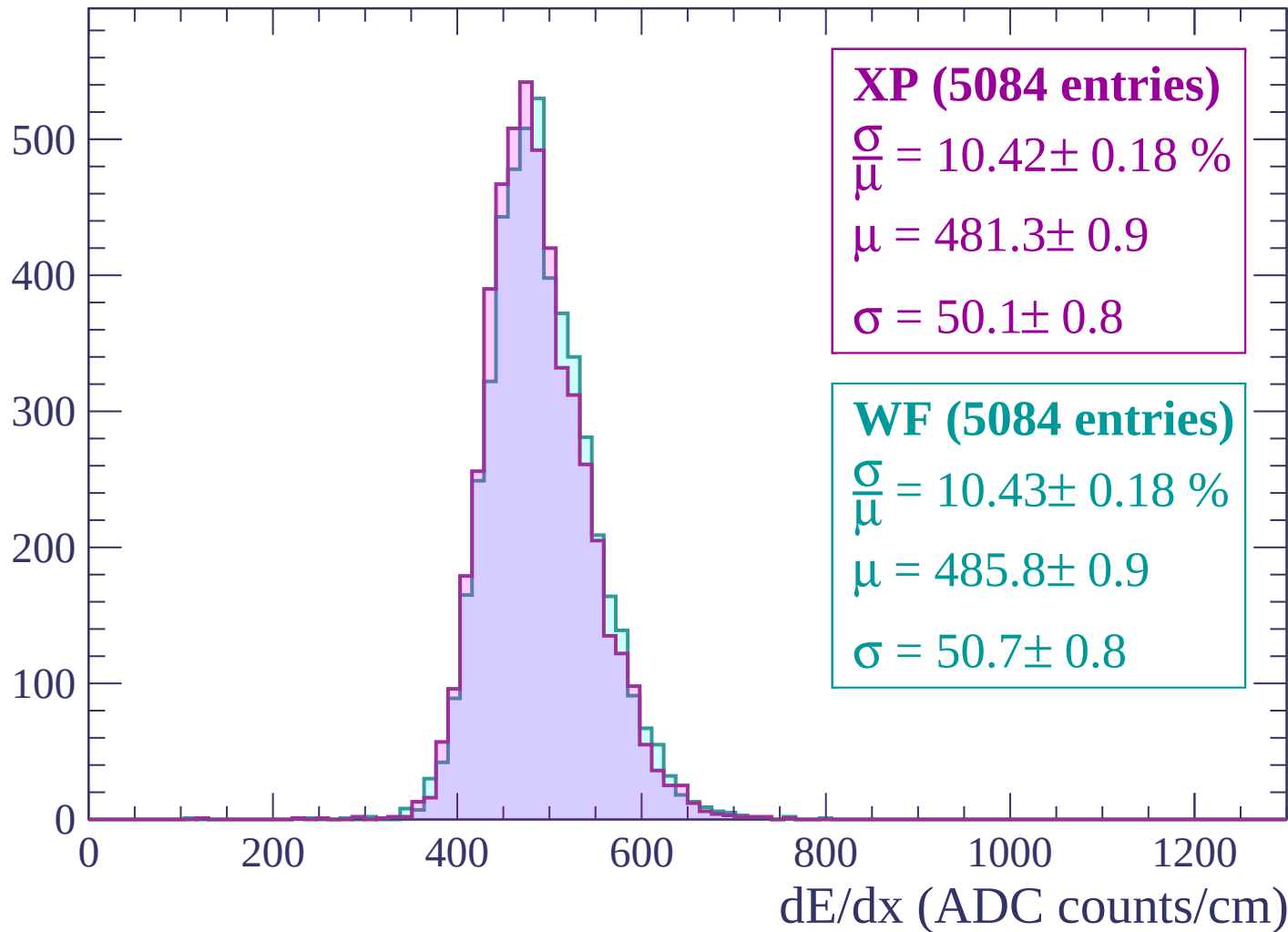
Energy loss | $1800 < p < 1860$

Count



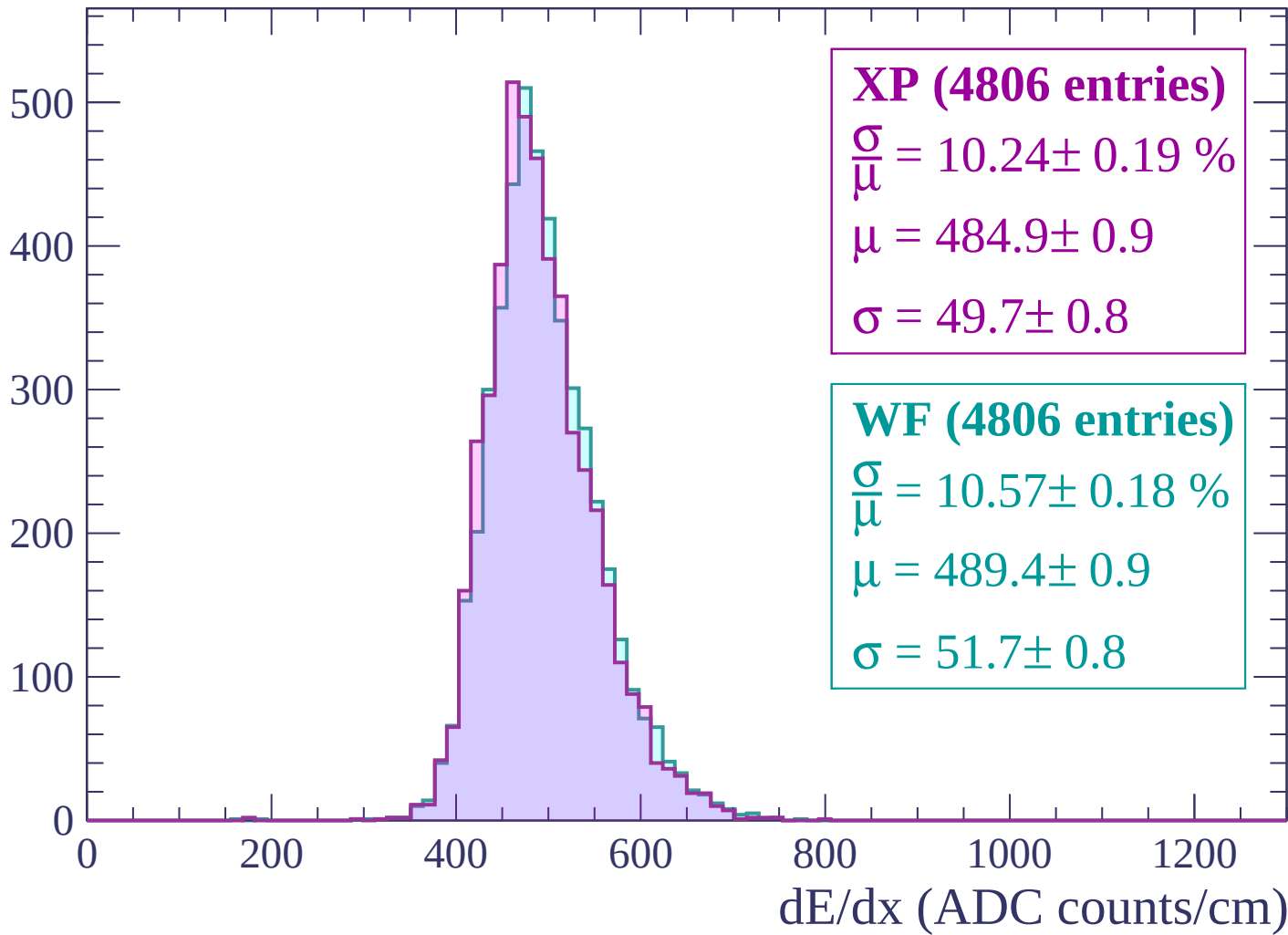
Energy loss | $1860 < p < 1920$

Count



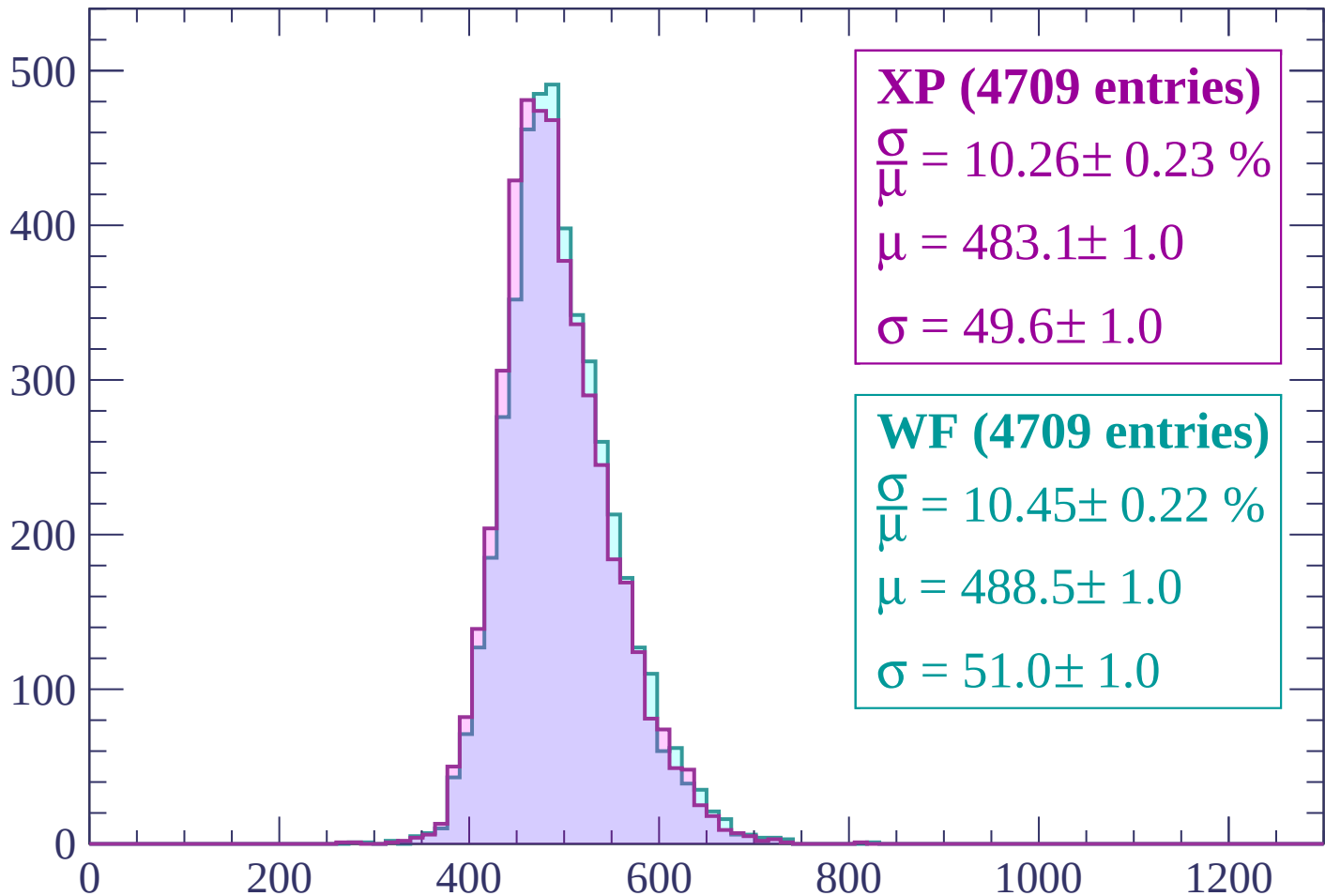
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (4709 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.23 \%$$

$$\mu = 483.1 \pm 1.0$$

$$\sigma = 49.6 \pm 1.0$$

WF (4709 entries)

$$\frac{\sigma}{\mu} = 10.45 \pm 0.22 \%$$

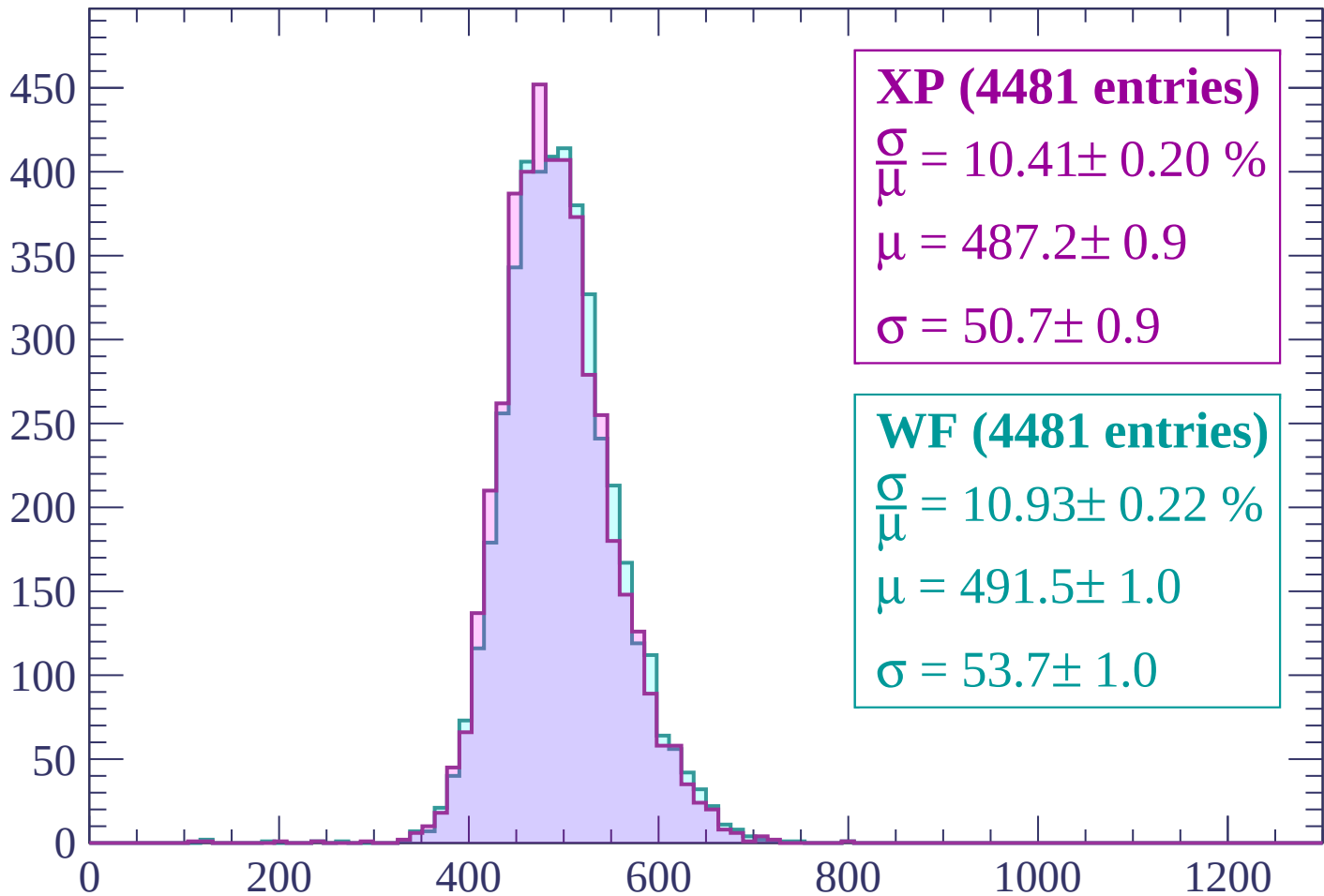
$$\mu = 488.5 \pm 1.0$$

$$\sigma = 51.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (4481 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.20 \%$

$\mu = 487.2 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

WF (4481 entries)

$\frac{\sigma}{\mu} = 10.93 \pm 0.22 \%$

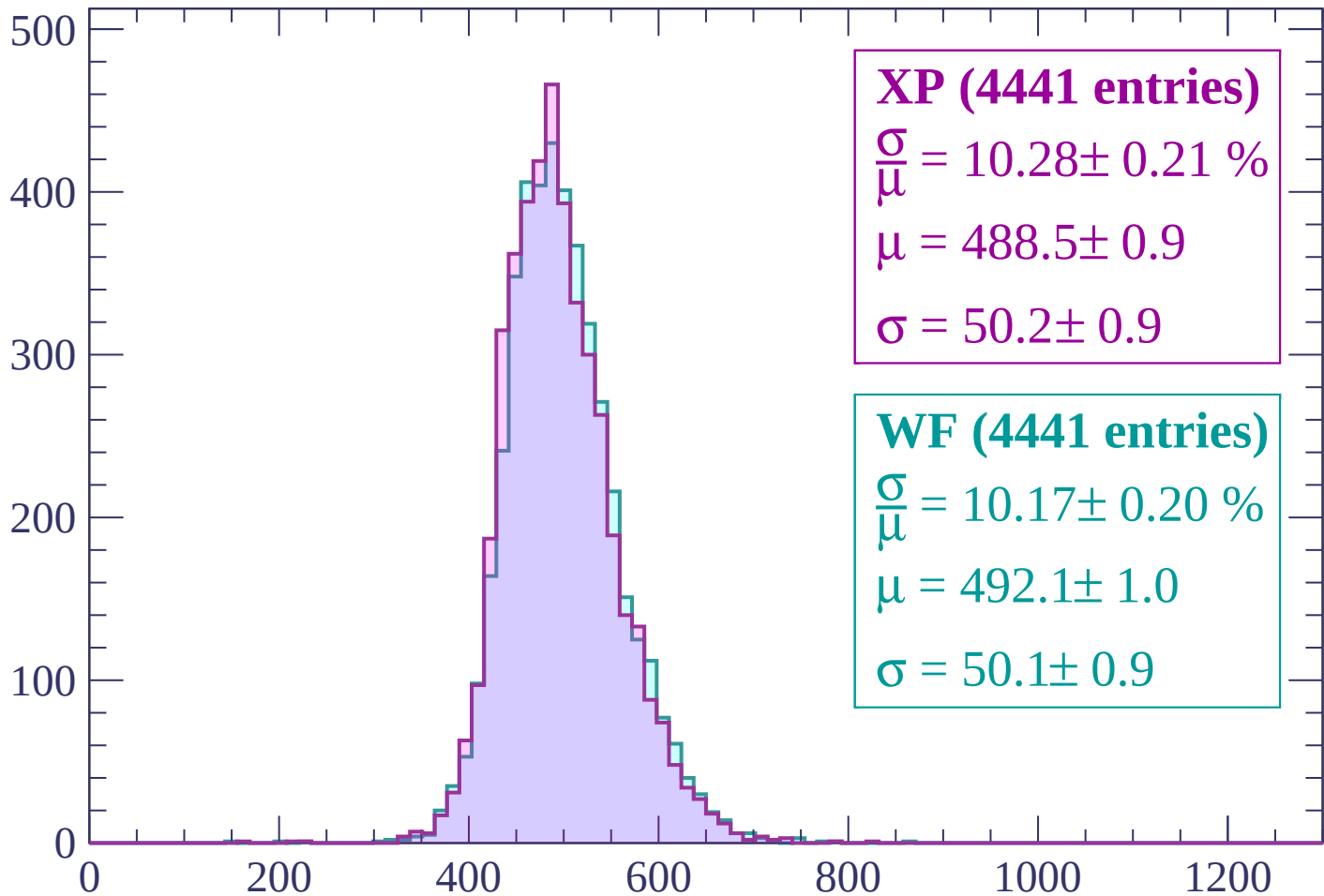
$\mu = 491.5 \pm 1.0$

$\sigma = 53.7 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (4441 entries)

$\frac{\sigma}{\mu} = 10.28 \pm 0.21 \%$

$\mu = 488.5 \pm 0.9$

$\sigma = 50.2 \pm 0.9$

WF (4441 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.20 \%$

$\mu = 492.1 \pm 1.0$

$\sigma = 50.1 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

XP (4295 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.23 \%$

$\mu = 487.9 \pm 1.0$

$\sigma = 51.1 \pm 1.0$

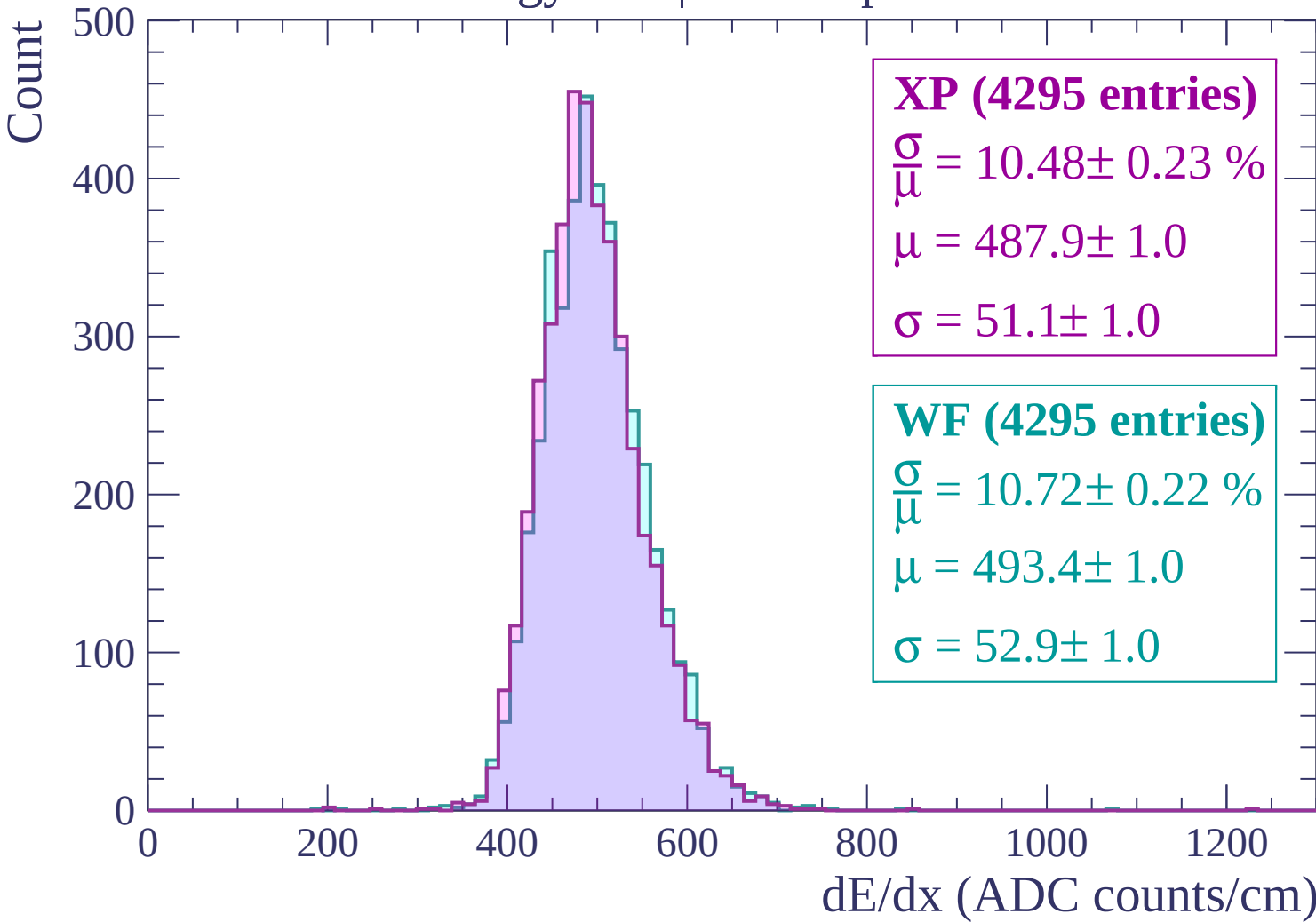
WF (4295 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$

$\mu = 493.4 \pm 1.0$

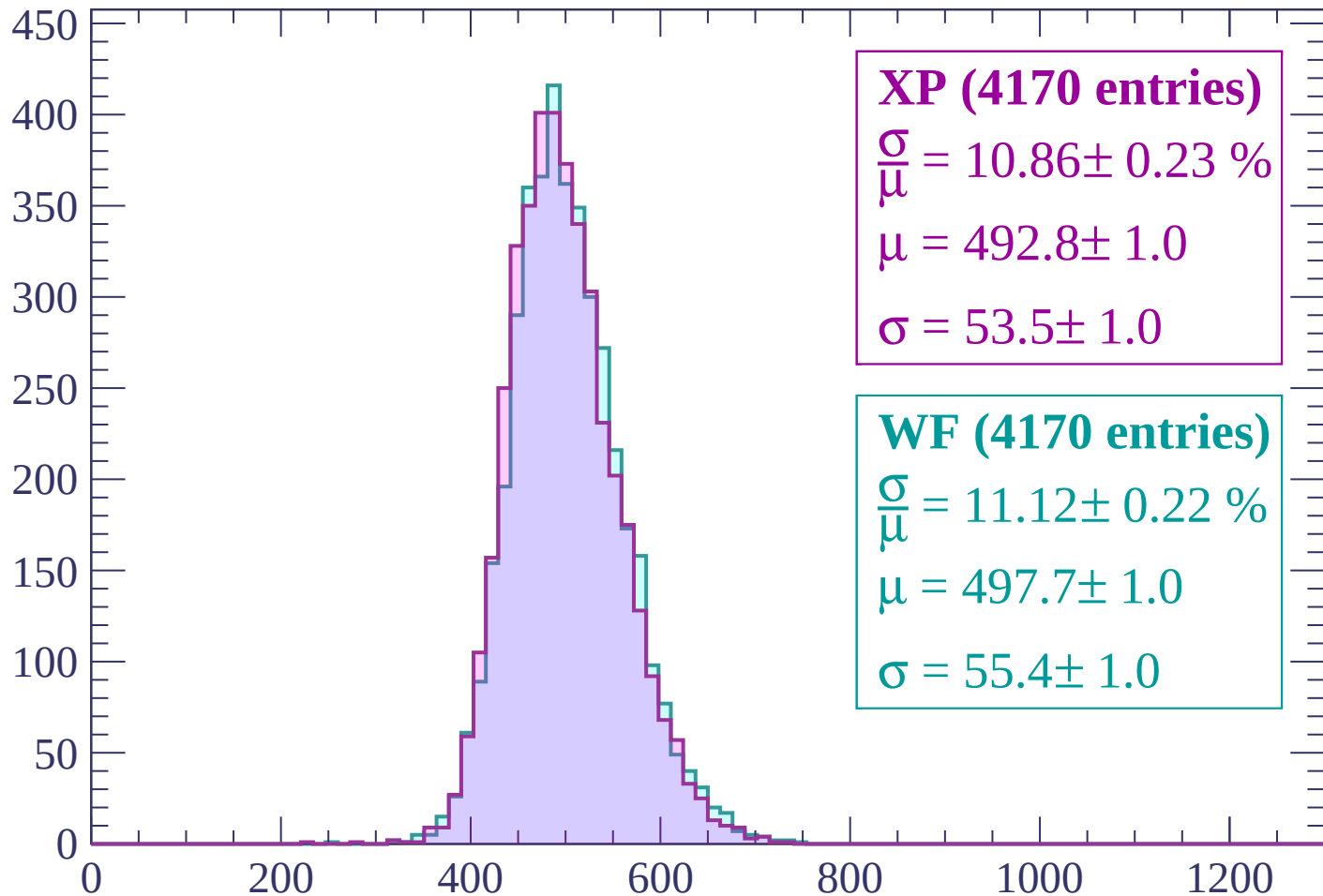
$\sigma = 52.9 \pm 1.0$

dE/dx (ADC counts/cm)



Energy loss | $2220 < p < 2280$

Count



XP (4170 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.23 \%$$

$$\mu = 492.8 \pm 1.0$$

$$\sigma = 53.5 \pm 1.0$$

WF (4170 entries)

$$\frac{\sigma}{\mu} = 11.12 \pm 0.22 \%$$

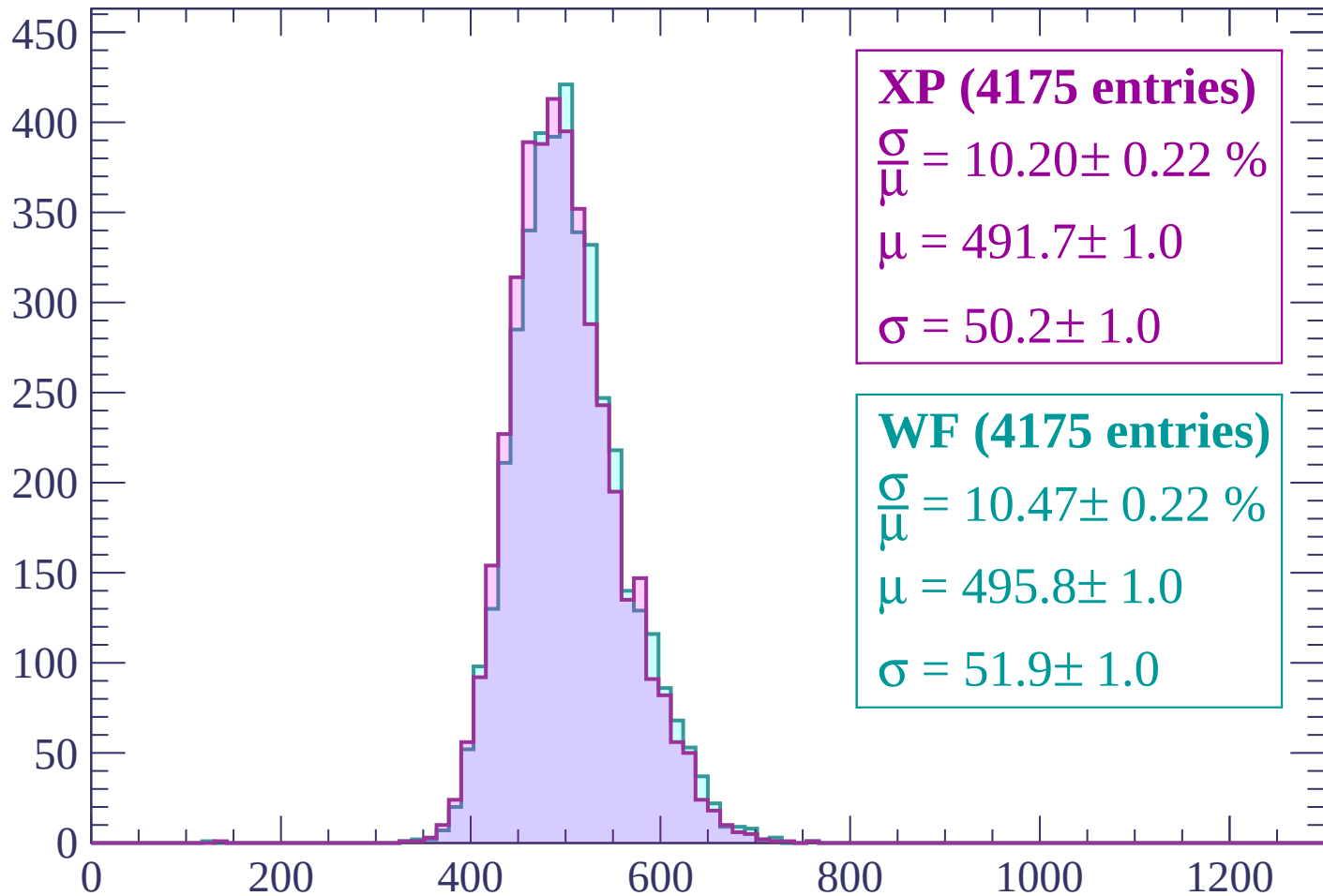
$$\mu = 497.7 \pm 1.0$$

$$\sigma = 55.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (4175 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.22 \%$$

$$\mu = 491.7 \pm 1.0$$

$$\sigma = 50.2 \pm 1.0$$

WF (4175 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.22 \%$$

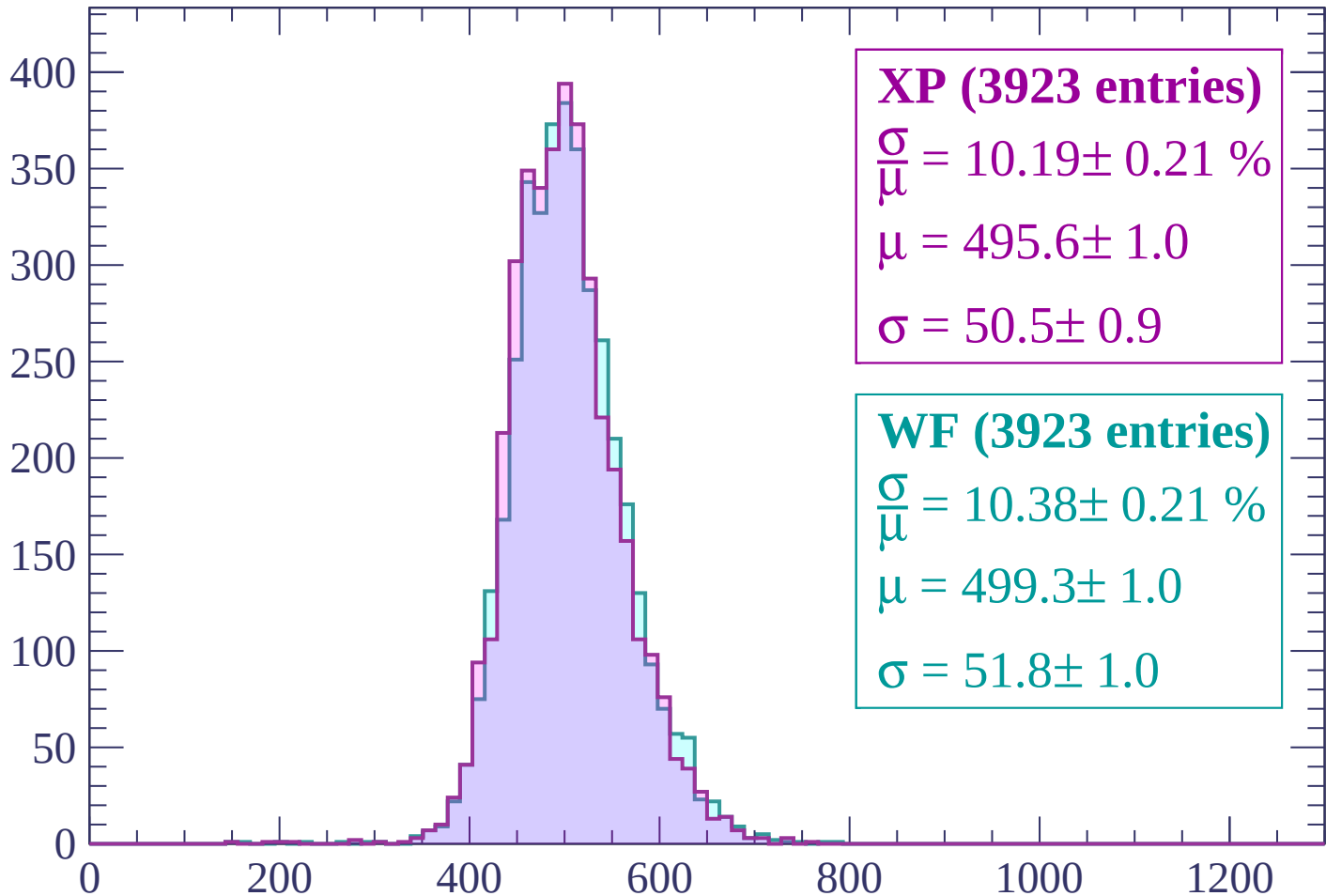
$$\mu = 495.8 \pm 1.0$$

$$\sigma = 51.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (3923 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.21 \%$

$\mu = 495.6 \pm 1.0$

$\sigma = 50.5 \pm 0.9$

WF (3923 entries)

$\frac{\sigma}{\mu} = 10.38 \pm 0.21 \%$

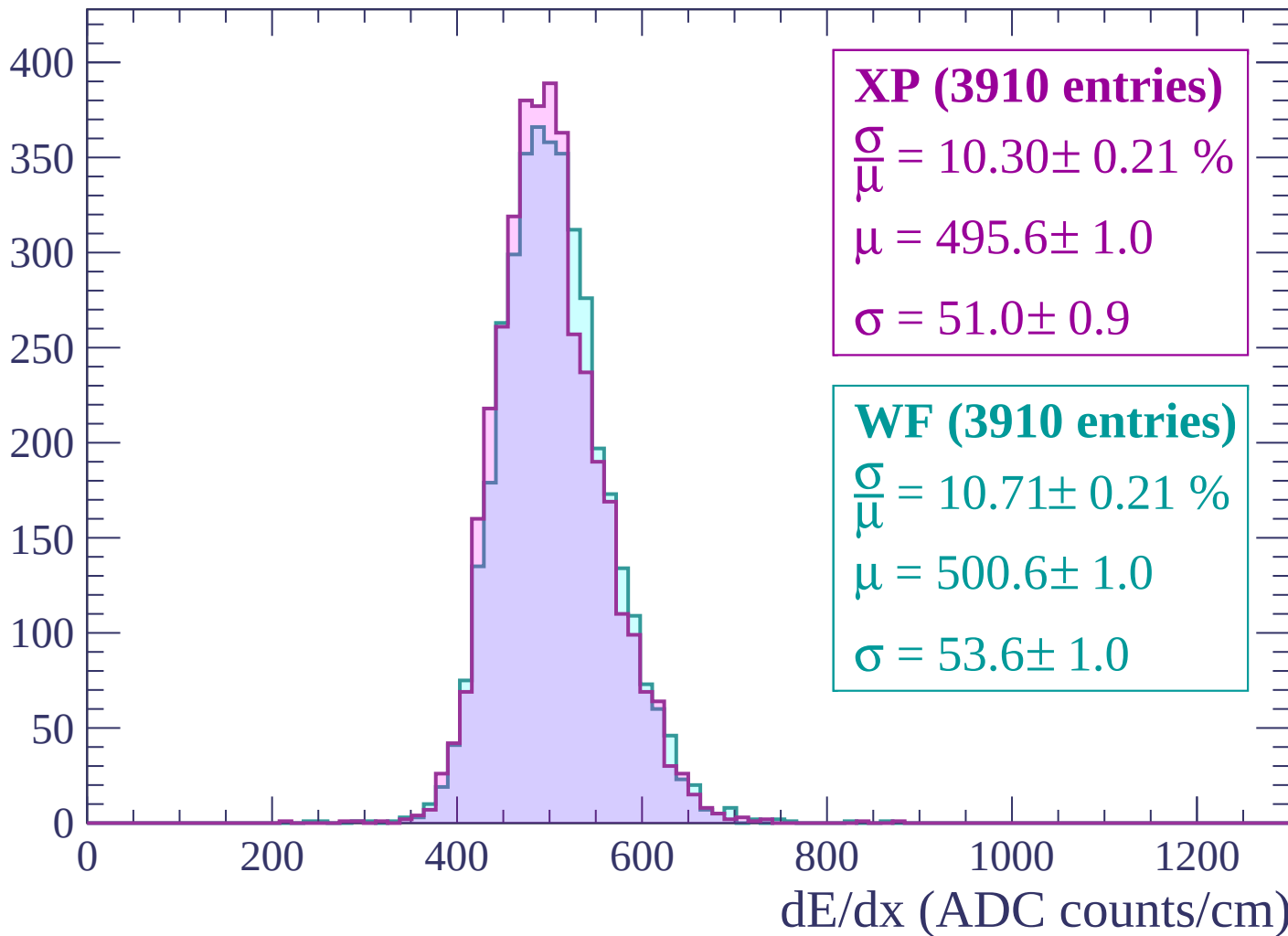
$\mu = 499.3 \pm 1.0$

$\sigma = 51.8 \pm 1.0$

dE/dx (ADC counts/cm)

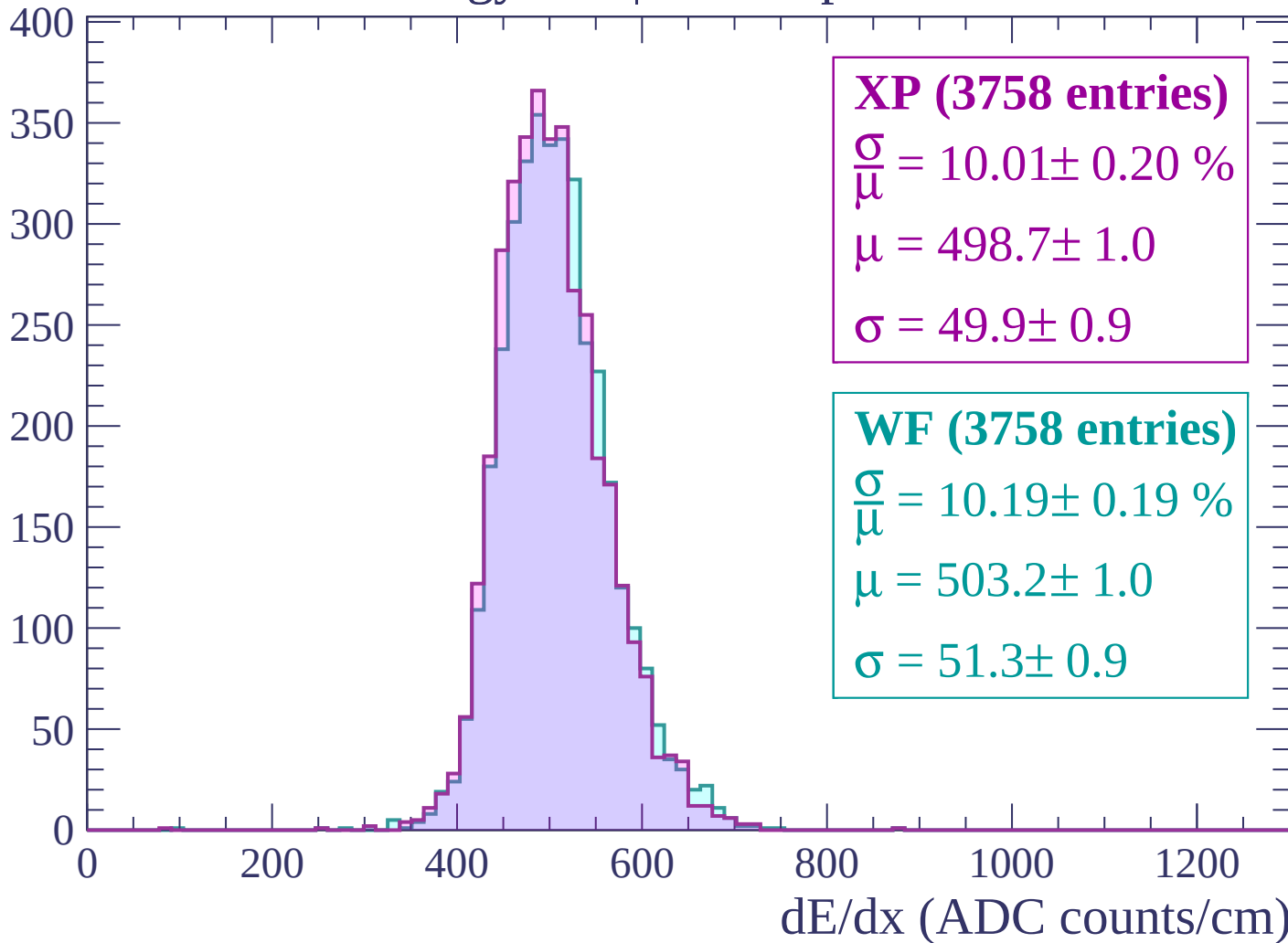
Energy loss | $2400 < p < 2460$

Count



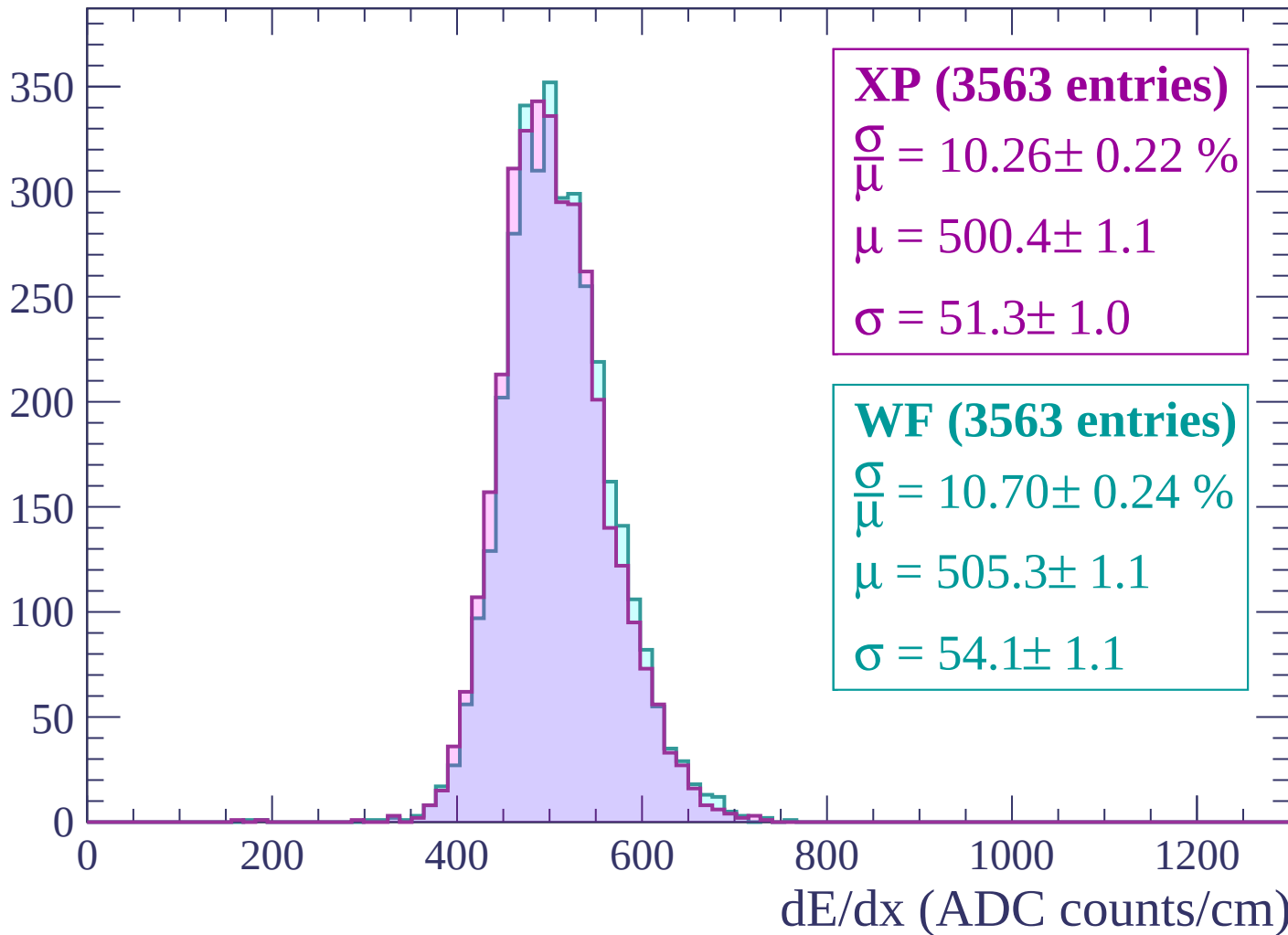
Energy loss | $2460 < p < 2520$

Count



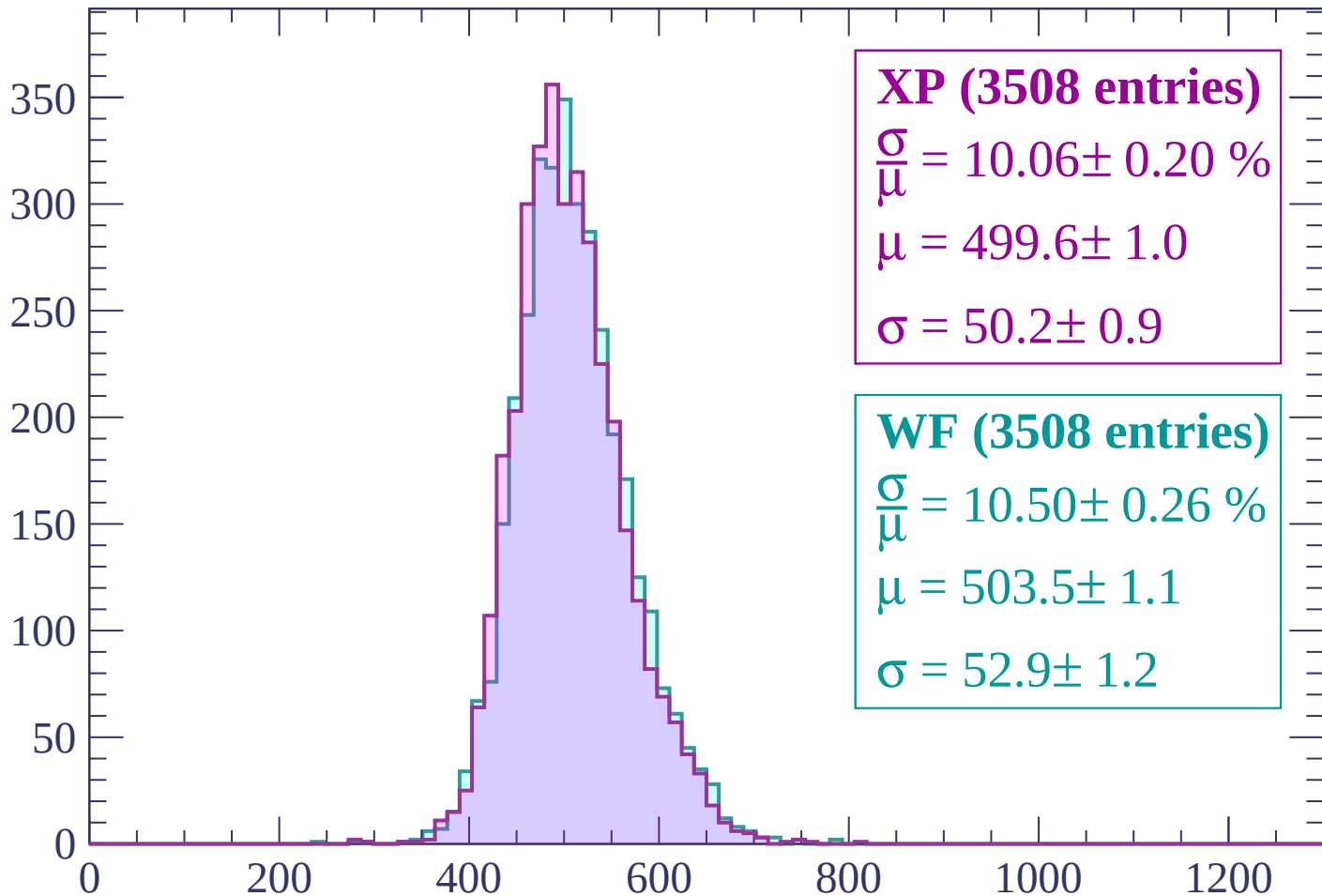
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP (3508 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.20 \%$$

$$\mu = 499.6 \pm 1.0$$

$$\sigma = 50.2 \pm 0.9$$

WF (3508 entries)

$$\frac{\sigma}{\mu} = 10.50 \pm 0.26 \%$$

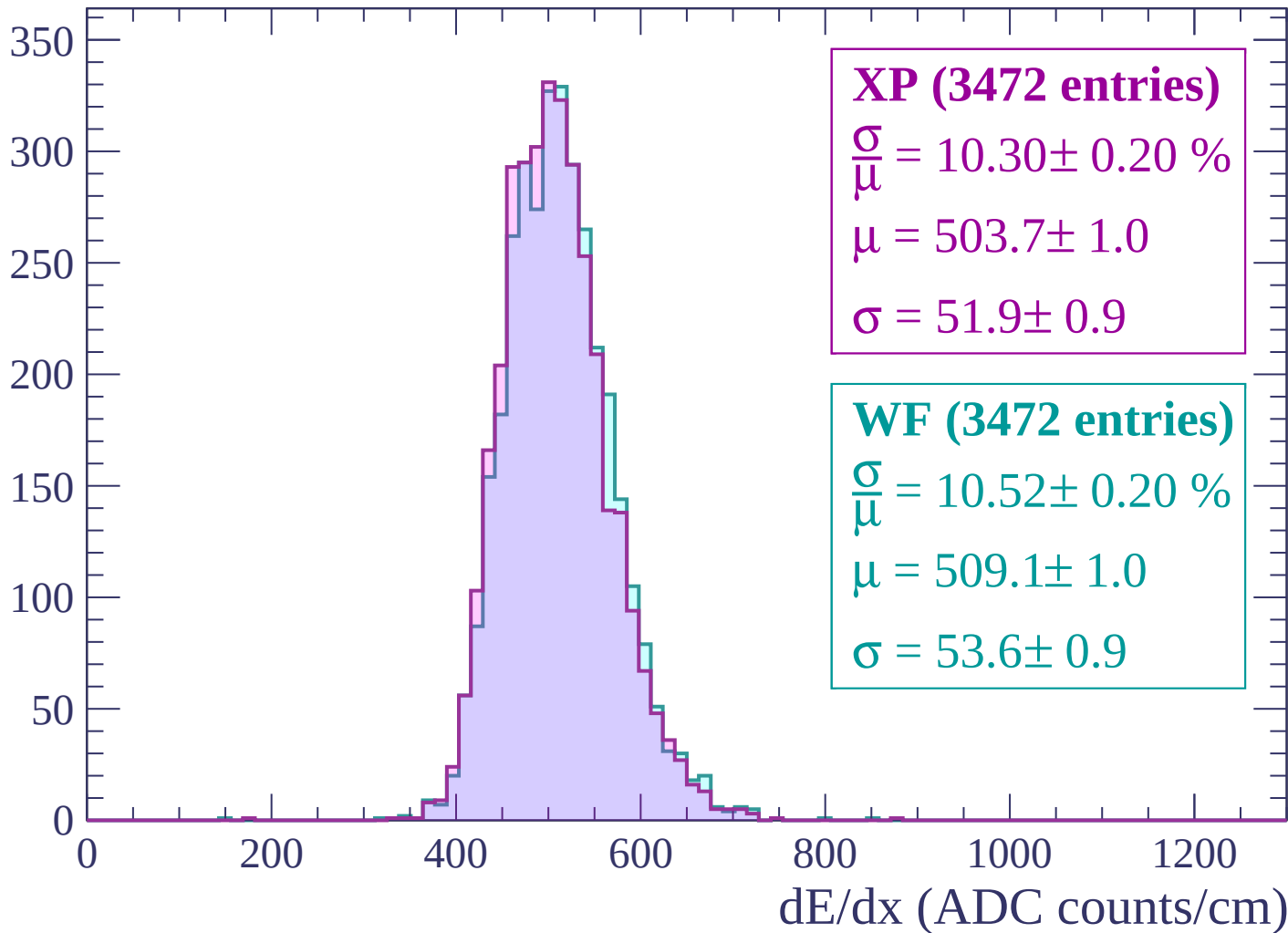
$$\mu = 503.5 \pm 1.1$$

$$\sigma = 52.9 \pm 1.2$$

dE/dx (ADC counts/cm)

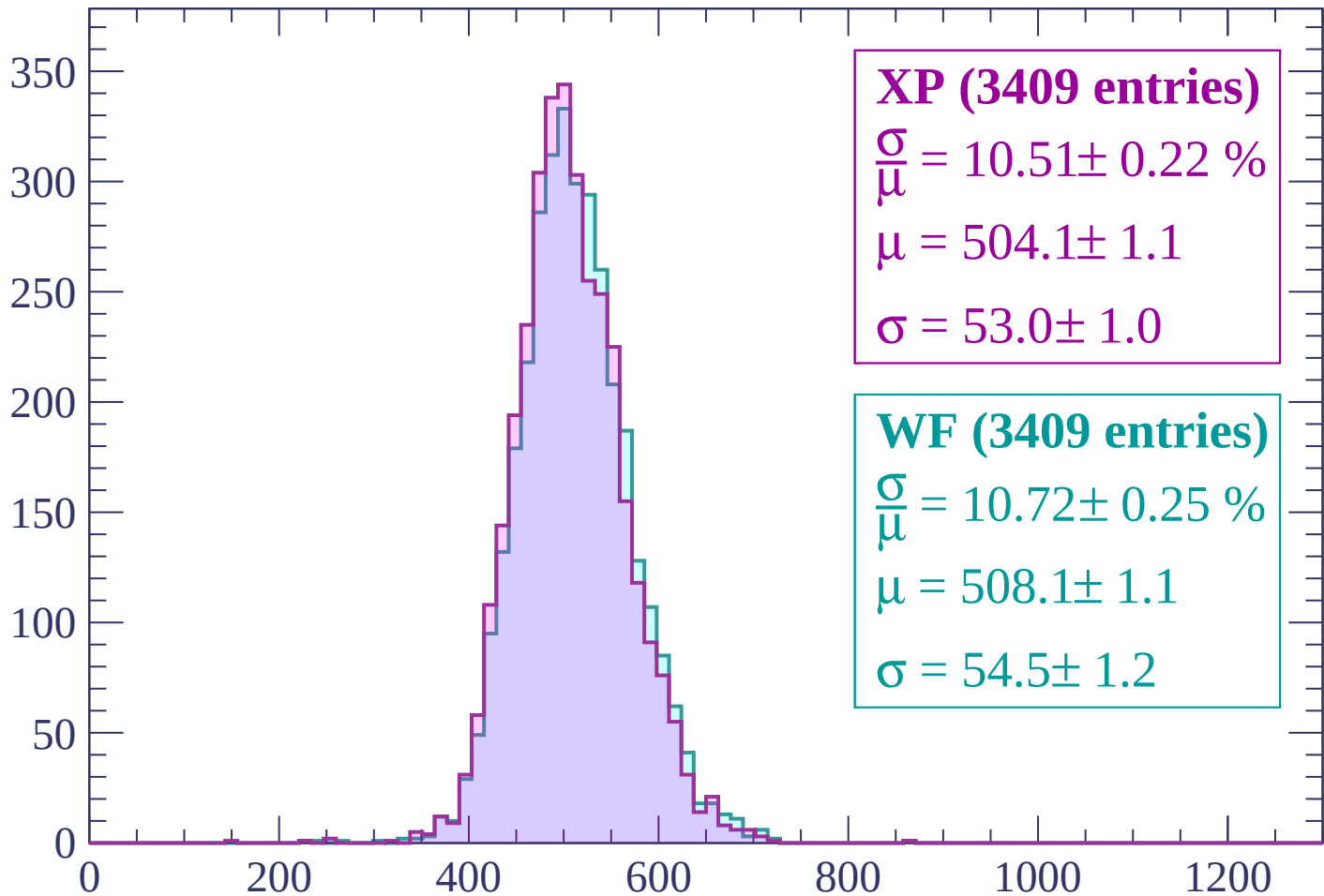
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (3409 entries)

$\frac{\sigma}{\mu} = 10.51 \pm 0.22 \%$

$\mu = 504.1 \pm 1.1$

$\sigma = 53.0 \pm 1.0$

WF (3409 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.25 \%$

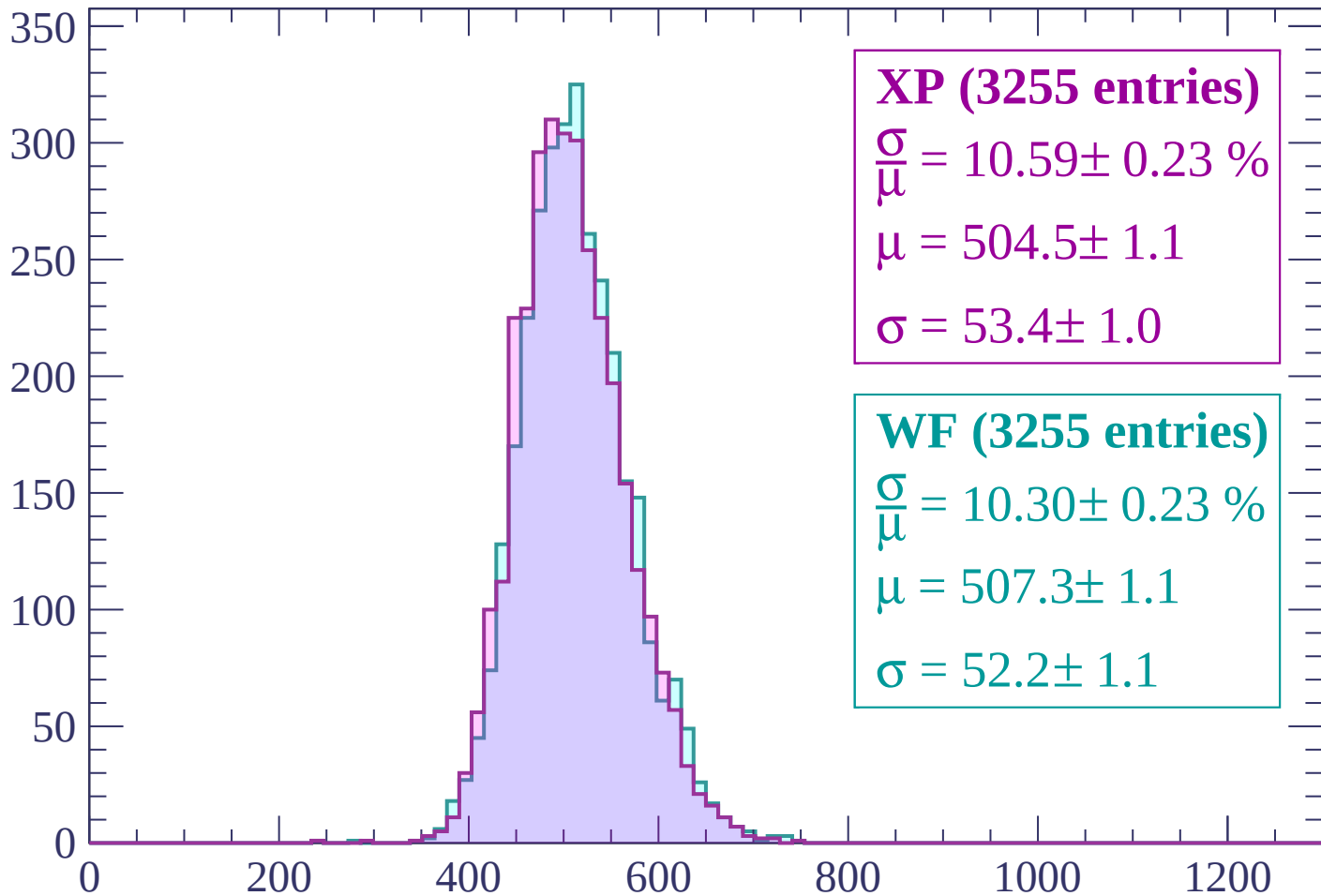
$\mu = 508.1 \pm 1.1$

$\sigma = 54.5 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (3255 entries)

$$\frac{\sigma}{\mu} = 10.59 \pm 0.23 \%$$

$$\mu = 504.5 \pm 1.1$$

$$\sigma = 53.4 \pm 1.0$$

WF (3255 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.23 \%$$

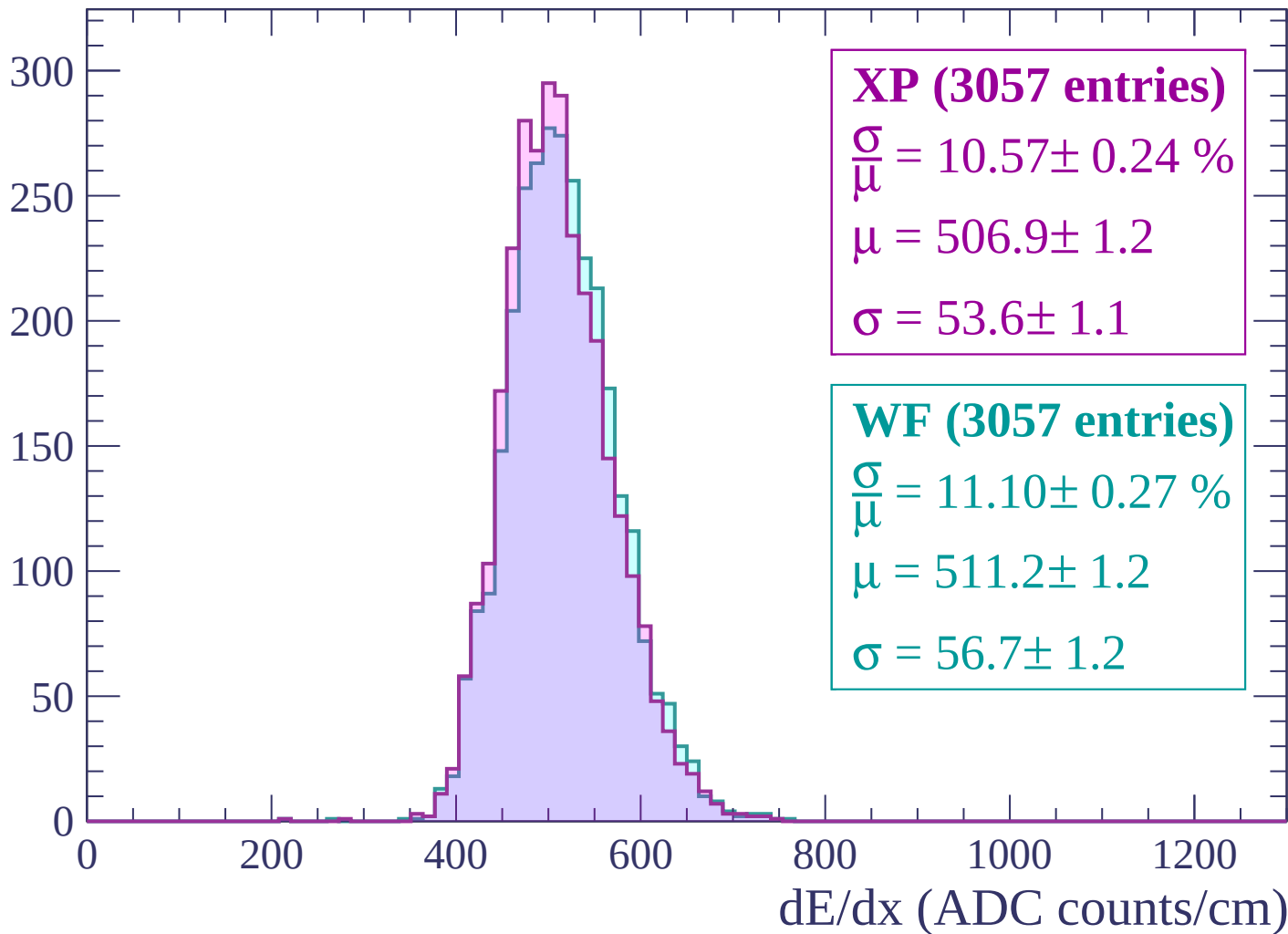
$$\mu = 507.3 \pm 1.1$$

$$\sigma = 52.2 \pm 1.1$$

dE/dx (ADC counts/cm)

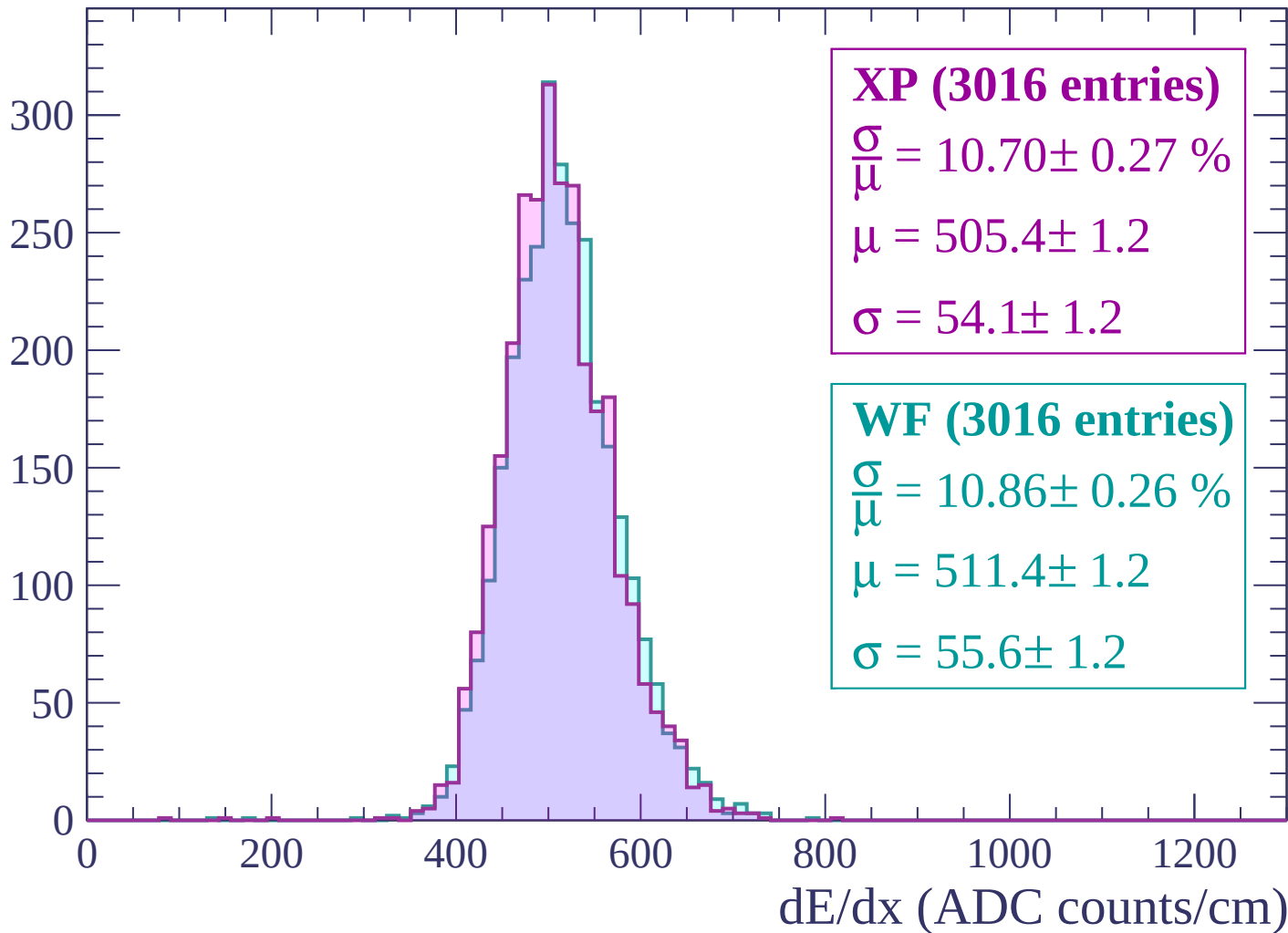
Energy loss | $2820 < p < 2880$

Count



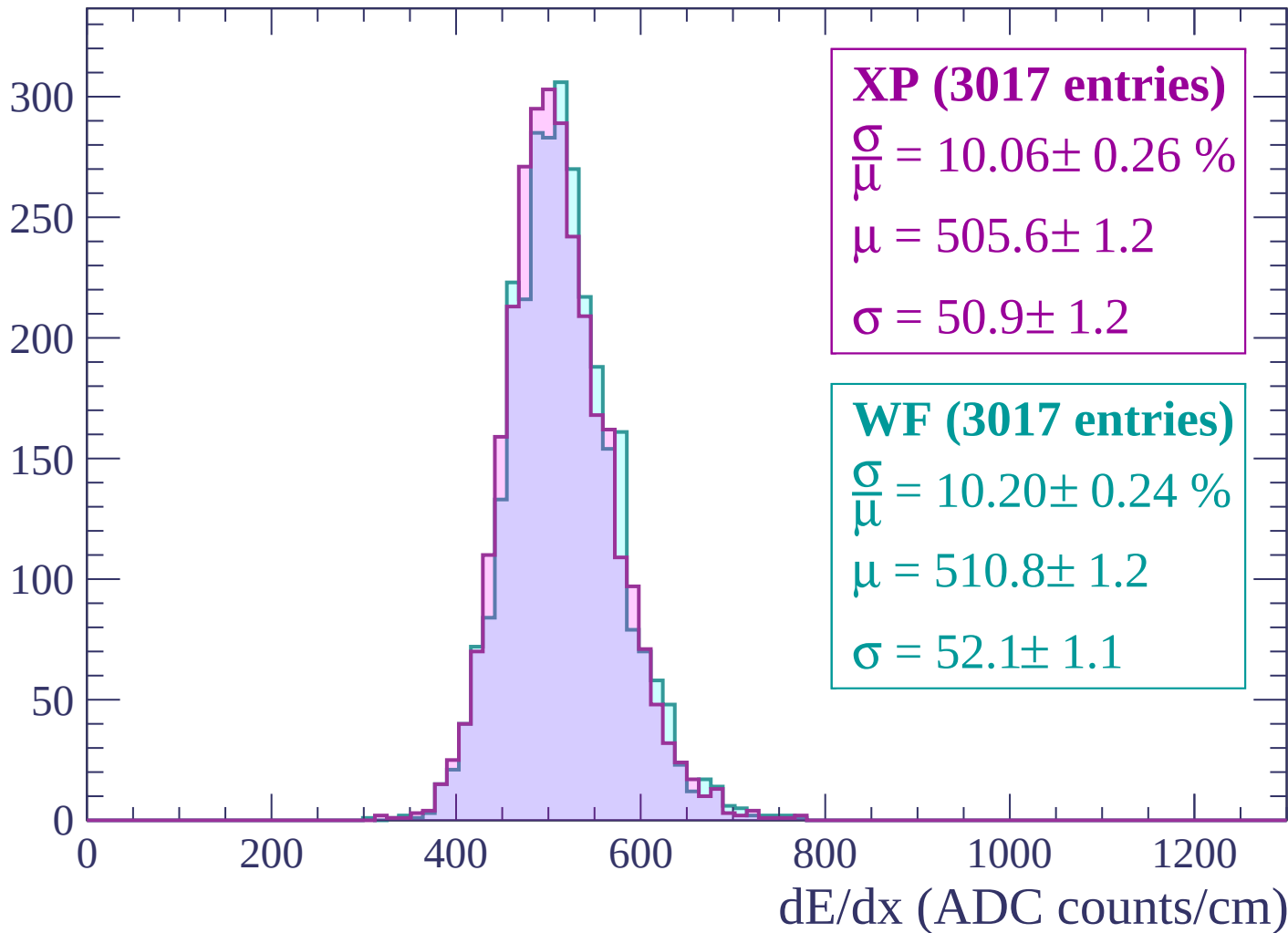
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

